

HERDING, ARMED CONFLICT, AND A CULTURE OF HONOR: GLOBAL EVIDENCE*

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ABSTRACT: We examine the importance of norms of revenge and punishment in perpetuating armed conflicts. Our analysis leverages the ‘culture of honor’ hypothesis from social psychology, which posits that traditional herding practices generate moral systems conducive to revenge-taking. We find that the descendants of herders (i) experience more frequent civil and non-civil conflicts; (ii) are more likely to be involved in conflicts motivated by retaliation; and (iii) exhibit a greater emphasis on revenge-taking in contemporary surveys and historical data. Our evidence suggests that a traditional form of subsistence generated a functional morality that continues to shape conflict across the globe today.

Keywords: Culture of honor, morality, conflict, punishment, revenge, grievance.

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1. Introduction

It is difficult to think of anything more inimical to economic growth than warfare. Within economics, the determinants of conflict and warfare have been extensively studied, and a number of important determinants have been identified. These have predominantly been economic, institutional, political, and geographic in nature.¹ Contrasting these factors, first-hand and ethnographic accounts of the determinants of conflict commonly identify revenge and vengeance as key factors. ‘Grievance,’ in addition to ‘greed,’ is hypothesized to play a key role in explaining the incidence of civil conflicts (e.g., Boehm, 1987; Scheff, 1994; Collier and Hoeffler, 2004). Retribution and revenge are cited as important factors in explaining whether rebel groups can recruit soldiers and gain the local population’s support (Marchais, Mugaruka, Sánchez de la Sierra and Wu, 2022). Similarly, acts of revenge and vengeance can escalate, resulting in sustained large-scale conflicts (Davie, 1929; Chagnon, 1988).

Despite the prominence of the idea that norms of revenge and punishment contribute to conflict, there is no direct empirical evidence that pertains to larger-scale violence like civil wars. Instead, the existing evidence for the importance of punishment and revenge has focused on smaller-scale elements of aggression, such as assaults and homicides. The primary difficulty in assessing the importance of norms of revenge for conflict is that while vengeance may fuel conflict, conflict likely induces vengeance.

To circumvent this problem, we focus on a deeper determinant of the desire for vengeance and punishment that is pre-determined relative to contemporary conflict and has been emphasized by psychologists, known as the ‘culture of honor.’ The main idea, developed by Nisbett (1993) and Nisbett and Cohen (1996), is that economic subsistence in the form of herding (pastoralism) generated a bundle of values, beliefs, and preferences that induce people to respond to threats and wrongdoings with revenge and violence.² This moral code is hypothesized to be especially pronounced among herders because they are particularly vulnerable to exploitation and theft since their livestock is a mobile asset. In such environments, developing a reputation for being violent and willing to take revenge on those who wrong you can be beneficial. As Nisbett and Cohen (1996, p. 5) put it: “a stance of aggressiveness and willingness to kill... is useful in an-

¹For a review of different determinants of conflict, see Fearon and Laitin (2003), Blattman and Miguel (2010), and Burke, Hsiang and Miguel (2015).

²Components of the hypothesis were developed prior in several studies such as Peristiany (1965), Gastil (1971), Black-Michaud (1975), Ayers (1984), Wyatt-Brown (1982), and Fischer (1989).

nouncing a herder's determination to defend his animals."³ This paper studies the relationships between a tradition of pastoralism, cultural and psychological proclivities to seek revenge and punish unfair behavior, and the contemporary incidence of civil and non-civil conflicts.

Our empirical strategy consists of five steps. First, following prior literature, we construct a quantitative measure of the degree to which historical ethnic groups relied on herding practices for economic subsistence. Second, we show that, across the globe, ethnolinguistic groups with a tradition of herding have more frequent and severe conflicts today, including civil conflict. Third, we provide evidence suggesting that the link between ancestral herding and contemporary conflict is largely driven by retaliatory conflicts. Fourth, to better identify psychological mechanisms, we use globally representative survey data to document that the descendants of herders have significantly more pronounced tendencies to seek revenge and punish. Fifth, in historical data, we show that herding societies were more likely to develop a culture of honor and deem violence morally appropriate.

Our measure of traditional pastoralism follows Becker (2025b), who constructs an ethnicity-level measure of the pre-industrial reliance on pastoral production using information contained in the *Ethnographic Atlas*. The measure codes the fraction of subsistence that is obtained from animals that require herding. We then examine whether a tradition of herding is associated with conflict and revenge-taking today.

The possibility that the culture of honor is relevant for larger-scale conflict events, including 'civil conflict' events that include the state on one side, is motivated by a rich body of anecdotal and ethnographic evidence that has emphasized the role of vengeance in fueling violence against the state. Marchais et al. (2022) describe how revenge-taking motives directly contributed to the conflict between the state and rebel groups in the Democratic Republic of the Congo. Revenge-taking motives have also been described as being a key contributor (along with state predation) to popular support for jihadist rebel movements in Western Africa (Benjaminsen and Ba, 2019, 2021) and for the civil conflict in Somalia (Lewis, 1994).

To study the link between a tradition of herding and contemporaneous conflict, we leverage detailed information on the location and incidence of conflicts from the *Uppsala Conflict Data Program (UCDP)*, the world's main provider of data on organized violence. Our main analysis

³Evidence based on agent-based modeling provides some formal support for this presumption (Nowak, Gelfand, Borkowski, Cohen and Hernandez, 2016).

compares ethnolinguistic groups that reside within the same country but potentially differ in their historical reliance on herding. We find that populations that relied on herding to a greater extent historically tend to have more conflicts today. Consistent with existing findings on homicides (Grosjean, 2014), we find that a tradition of herding is associated with a greater prevalence of smaller-scale conflicts that are more localized and do not involve a government. A tradition of herding is also strongly associated with civil conflict events, defined as conflicts that involve a representative of a government on one side. In the *UCDP* data, the median civil conflict involves two deaths (the average is 14). While fairly small, these incidents are important because they can escalate, generating tensions and grievances that can instigate and/or fuel full-scale civil wars.

Given the correlational, reduced-form nature of our analysis, we undertake an extensive set of robustness checks. First, we show that our results are robust to controlling for many economic, social, and historical determinants of conflict emphasized in the literature, including contemporary local development, historical development and institutions, ethnic fractionalization and polarization, terrain ruggedness, and population density. In these analyses, tests of the influence of omitted variables suggest that omitted factors do not drive our results. Second, we show that although herding societies differ from other societies, few of these other characteristics are associated with conflict. Controlling for them does not meaningfully affect our estimates.

While these checks increase our confidence that our estimates reflect a culture-of-honor mechanism rather than a spurious correlation, we acknowledge that unobserved confounders orthogonal to our extensive set of controls, such as unmeasured local institutions or other locality-specific factors, could still contribute to the observed patterns. Our interpretation is, therefore, that historical pastoralism likely represents an important explanatory factor, but not the sole mechanism underlying contemporary conflict in these regions. Indeed, our results directly suggest an important interaction between the culture of honor and institutions. In heterogeneity analyses, we find that the within-country link between herding and conflict is strongly concentrated in countries with a weak rule of law.

From the perspective of the culture of honor hypothesis, herding should be linked specifically to the occurrence of revenge-related conflicts. We thus check whether a tradition of herding is not only linked to the presence of conflicts in general but to violence driven by revenge-taking specifically. We do so in two ways. First, by examining variation in the exact timing of conflict events, we find that the effect of herding on conflict is strongest in the days immediately following

a previous conflict event. The effect declines as time passes, approaching zero after several weeks. In other words, a tradition of herding is uncorrelated with conflict in the absence of recent conflict events, suggesting that our results operate through a retaliation channel.

Second, to provide additional direct evidence on the revenge-taking mechanism, we complement the *UCDP* analysis with an analysis that uses the *ACLED* conflict data, available for the African continent. *ACLED* contains detailed descriptions of conflict events, which allow us to identify conflict events that are described as involving revenge motivations. We find that ancestral herding is strongly associated with conflicts having revenge or retaliation as a motive, yet uncorrelated with conflict events not described as involving revenge.

While this evidence suggests that the link between historical herding and conflict reflects the proclivity to seek revenge, we provide further complementary evidence for this mechanism by leveraging the *Global Preferences Survey*, or GPS (Falk, Becker, Dohmen, Enke, Huffman and Sunde, 2018). This globally representative dataset includes detailed information on respondents' willingness to take revenge and punish others for unfair behavior. Our analysis leverages the within-country variation available from the GPS, linking respondents' revenge-taking and punishment proclivity to historical variation in herding across subnational regions. In all analyses, we find that traditional herding is predictive of individuals' willingness to take revenge and punish others for unfair behavior.

Finally, we deepen our analysis of psychological mechanisms by examining historical data on cultural values. We make use of the recently released dataset on traditional folklore that provides rich information on the beliefs, customs, and stories that were passed through the generations by word of mouth in the form of tales and narratives (Michalopoulos and Xue, 2021). Studying variation across approximately 1,100 ethnic groups, we find that groups that relied more strongly on herding are more likely to have traditional folklore that contains motifs related to vengeance, punishment, retaliation, and, ultimately, violence. We also examine information on the moral beliefs of 60 ethnic groups from across the globe available from the *Standard Cross-Cultural Sample* (SCCS). We find that, historically, a dependence on herding is associated with greater acceptability of violence, particularly outside of one's local community.

In all, the evidence is consistent with a tradition of herding being an important determinant of armed conflicts around the globe. A diverse set of evidence suggests that the underlying mechanism is a 'culture of honor' – psychological and cultural proclivities related to revenge

and punishment. The results for revenge-taking also resolve a potential puzzle that may arise in interpreting conflicts through the lens of a ‘culture of honor.’ One perspective is that, in equilibrium, herding societies with strong revenge norms should not have more conflict due to a deterrence effect on potential attackers. Our results are consistent with such a perspective. Groups with a tradition of herding do not have an increased baseline probability of conflict. Rather, they only see more conflicts conditional on a first conflict event, plausibly driven by retaliatory motives.

Our findings contribute to the existing evidence on honor cultures, which has focused on smaller-scale violence, such as homicides (Black-Michaud, 1975; Nisbett and Cohen, 1996; Nisbett, Polly and Lang, 1995; Cohen, 1998; Hayes and Lee, 2005; Uskul, Cross, Günsoy and Gul, 2019), including in the U.S. South (Grosjean, 2014). They extend our understanding of the consequences of honor cultures by estimating effects on economically important conflict events, as well as the global distribution of tendencies for punishment and revenge-taking.

Our findings also contribute to a deeper understanding of traditionally pastoral groups. Earlier studies have highlighted how a tradition of herding affects female sexuality and child marriage (Becker, 2025a,b). McGuirk and Nunn (2025) study the effects of weather shocks on conflict between transhumant pastoralists and agriculturalists. We provide a series of analyses that suggest that our mechanism is orthogonal to the one analyzed in their paper, and that our results are not driven by transhumant pastoralists.

Finally, by connecting a traditional mode of subsistence to contemporary conflict, our findings also contribute to a better understanding of how historical factors can shape contemporary large-scale conflicts and wars (Jha, 2013; Besley and Reynal-Querol, 2014; Michalopoulos and Papaioannou, 2016) as well as the psychological and social determinants of conflict and warfare (Miguel, Saiegh and Satyanath, 2011; Guarnieri and Tur-Prats, 2023; Tur-Prats, 2021). Related are also studies that have shown that social factors can affect the incidence of conflict, whether they arise due to social obligations (Moscona, Nunn and Robinson, 2020), norms of punishment and cooperation (Fouka and Schläpfer, 2022), status competition (Ager, Bursztyn, Leucht and Voth, 2022), or the influence of leaders or high-status individuals (Dippel and Heblich, 2021; Bai, Jia and Yang, 2023; Jha and Wilkinson, 2023; Cagé, Dagorret, Grosjean and Jha, 2023).

The remainder of the paper proceeds as follows. Section 2 describes the construction of the historical herding index and how it is linked to contemporary data. Section 3 investigates the

link between historical herding and contemporary conflicts. The subsequent sections examine mechanisms. First, by highlighting the role of retaliatory conflicts (Section 4), and then by studying contemporary attitudes and historical folklore on punishment and revenge-taking (Section 5). Lastly, Section 6 offers concluding thoughts.

2. Historical Measure of Economic Dependence on Herding

2.1. Data and Construction of Index

Our analysis uses information on pre-industrial reliance on herding from the *Ethnographic Atlas*, a worldwide database constructed by George Peter Murdock that contains ethnographic information for 1,265 ethnic groups (Murdock, 1967). Information for societies in the sample has been coded for the earliest period for which satisfactory ethnographic data are available or can be reconstructed. The earliest observation dates are for groups in the Old World, where early written evidence is available. For the parts of the world without a written history, the first recorded information tends to follow European contact and can be as late as the 19th or even early 20th centuries. The data capture, to the fullest extent possible, the indigenous characteristics of the group prior to industrialization and European contact.

The *Ethnographic Atlas* has recently seen widespread use in economic history, cultural economics, and cultural psychology (e.g., Alesina, Giuliano and Nunn, 2013; Michalopoulos and Papaioannou, 2013, 2014, 2016; Giuliano and Nunn, 2018; Schulz, Bahrami-Rad, Beauchamp and Henrich, 2019; Enke, 2019). A large-scale validation study recently documented strong correlations between historical ethnic-group level characteristics in the *Ethnographic Atlas* and contemporaneous ethnic-group level measures of those same traits in independent survey data (Bahrami-Rad, Becker and Henrich, 2021).

Herding, or pastoralism, refers to the breeding, care, and use of herd animals. Unlike tending animals such as pigs or chickens, herding involves taking the herds out to natural pasture, which increases the risk of theft. We follow Becker (2025b) to define pre-industrial reliance on herding in the *Ethnographic Atlas* as the product of two parts: (1) the degree to which a society depended on animal husbandry (0–100%), and (2) an indicator taking the value of one if the predominant animal in a society is a herding animal (sheep, cattle, horses, reindeer, alpacas, or camels). As a result, the measure codes the fraction of economic subsistence that is due to herding.

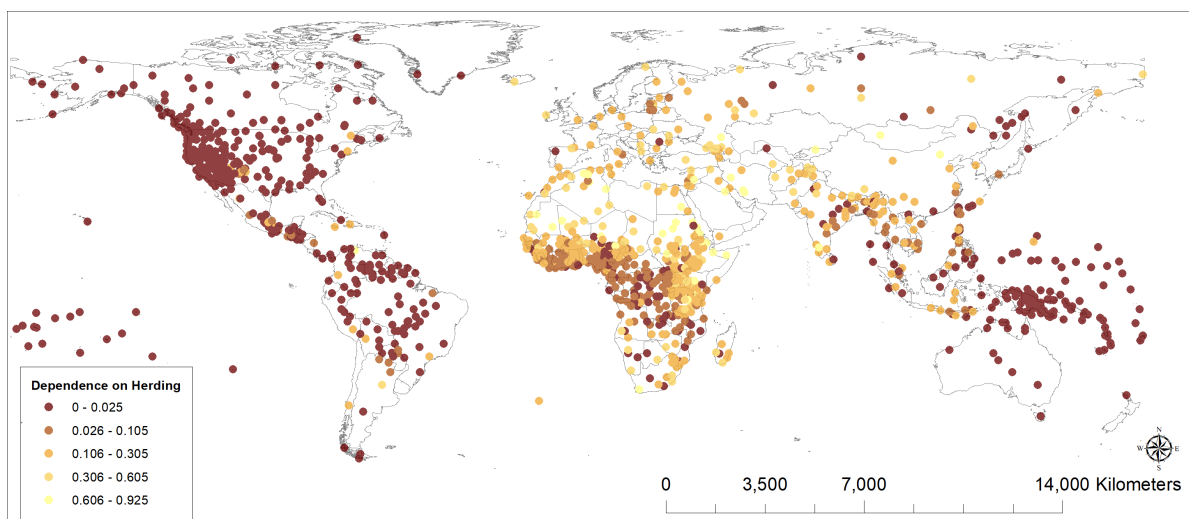


Figure 1: Global distribution of reliance on herding in the *Ethnographic Atlas*. The data apply to the pre-industrial period and capture the traditional subsistence mode of the local population. Source: Becker (2025b).

In total, we have herding data for 1,202 historical ethnic groups. The spatial distribution of ethnic groups' herding dependence is shown in Figure 1, and the histogram of the distribution is reported in Appendix Figure A1. Societies vary substantially in their historical dependence on herding. About one-third of societies traditionally have very little or no herding production (less than 5%). Very few societies depend on herding by more than 50% (about 5%). Most societies have intermediate shares of herding production, with an average dependence of 14%. Societies that depend more on pastoralism tend to be located in Northern Africa, Southern Africa, Northern Europe, the Middle East, and Central Asia.

2.2. Linkage of Herding Index to Conflicts and GPS

Linkage to conflict events. Conflict data provide information on the precise location of contemporary conflict events. We use latitude and longitude to determine the language of the group that currently lives in that location. The location of language groups is taken from the *Ethnologue 16*, a database reporting the dominant language group that is present in each location across the globe. For our analysis, a 'language group' is one of the 7,000+ languages/dialects of the *Ethnologue* spoken in a given country. Therefore, Yoruba speakers in Nigeria are one language group, while Yoruba speakers in Benin are another. We then match each of the *Ethnologue* languages/dialects to one of the approximately 1,300 ethnic groups in the *Ethnographic Atlas*,

which contains information on herding and other historical covariates. As described in Giuliano and Nunn (2018), the matching relies on a combination of linguistic and ethnographic sources.

For each language group, we compute the number of conflict events (and by type) that occur in the territory currently inhabited by that language group. Contemporary conflicts can then be linked to the ancestral dependence on herding of the language group, using their connection to ethnic groups (and information on their reliance on herding). In general, ethnic groups comprise multiple of the more granular language groups. Thus, each language group is connected to one ethnic group, whereas an ethnic group can be connected to multiple language groups.

The matching procedure works best in countries with many ethnic groups that each have their own language or dialect. This is particularly the case in Africa, which is the focus of our auxiliary analyses that use the *ACLED* data. An important aspect of the procedure is that, because it tracks ancestry through language and ethnicity, and not location, it remains valid even when there have been large-scale population movements. This is because some diasporic groups – including the English in New England, French in Canada, Portuguese in Brazil, and Dutch in South Africa – are included in the *Ethnographic Atlas*. In addition, language provides an (admittedly imperfect) proxy for ancestral connections. For example, those in the *Ethnologue* who speak Spanish as their primary language today are linked to the Spanish in the *Ethnographic Atlas*, even those who live in Latin American countries. Similarly, Indigenous groups are identified through their spoken language despite relocation, displacement, and other movements following European contact. The procedure follows the basic logic outlined in Giuliano and Nunn (2018). The variation across space in traditional reliance on herding at the language group level is shown in Figure 2a.

Linkage to GPS. For the analyses that use the *GPS* survey data, we do not know the language or ethnicity of the respondents. We only know their district of residence. Therefore, to connect individuals to ancestral herding practices, we construct measures of the average dependence on herding among the ancestors of those living in each subnational district. By connecting *Ethnologue* languages to *Ethnographic Atlas* ethnic groups, we know the traditional herding measure for all locations on Earth. To compute average ancestral dependence on herding at the district level, we compute population-weighted averages, using gridded population data (at a 30 arc second resolution) taken from the *Landscan* database. The district-level estimate of the average ancestral reliance on herding is shown in Figure 2b.

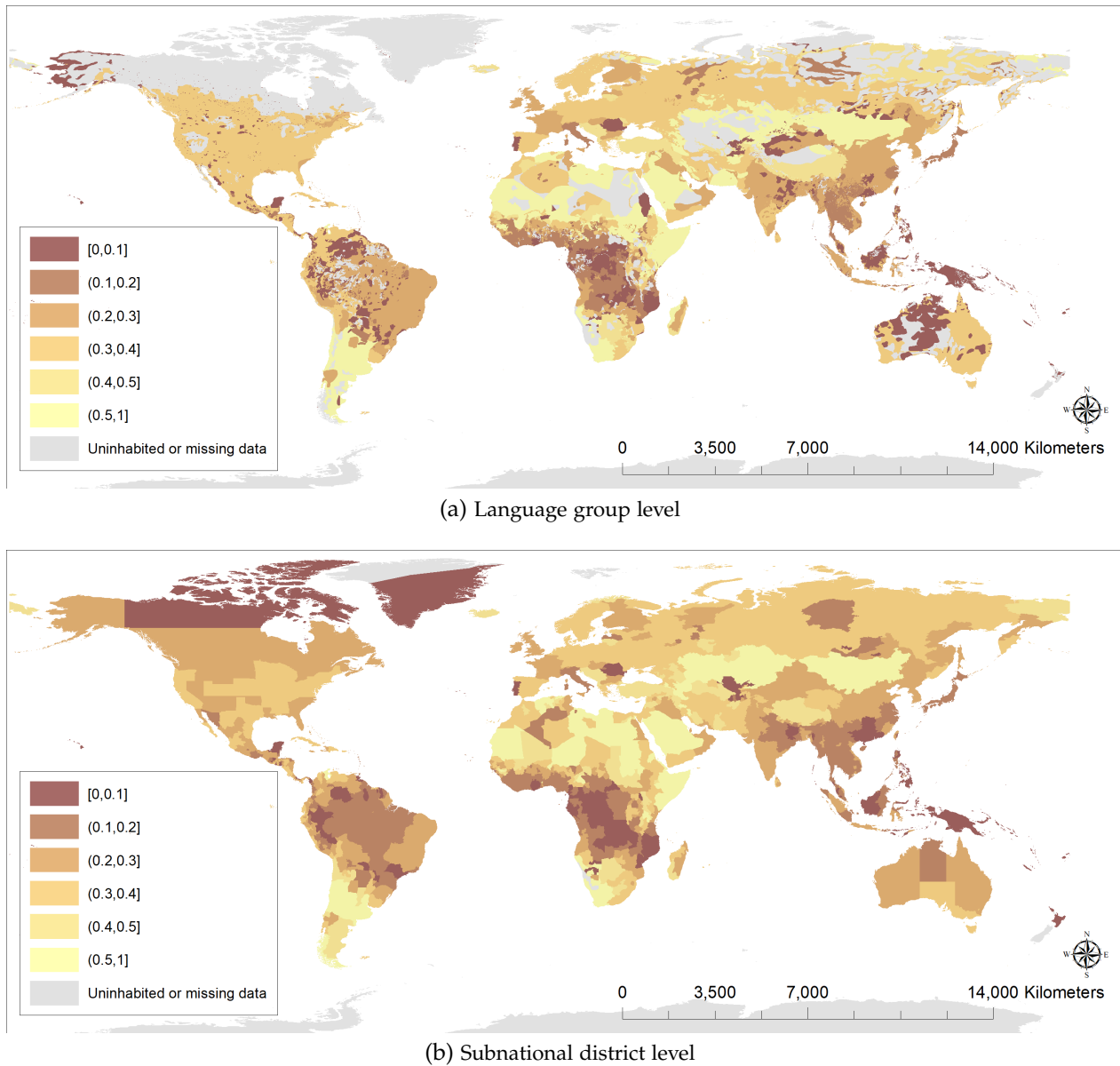


Figure 2: Global distribution of ancestral reliance on herding across language groups and subnational districts.

Several patterns are apparent in Figures 2a and 2b. First, we see a large amount of variation across the globe, with North Africa and the Middle East exhibiting particularly high ancestral dependence on herding. Second, there is substantial variation across language groups and districts within the same country. This allows for more-granular within-country analyses that hold constant all factors that vary at the country level, including national institutions and characteristics of the government, which are particularly important when we examine civil conflicts, where an arm of the national government, like the military or police, is one of the two participants

in the conflict events.

2.3. Ecological Determinants of Herding

A recent study by Becker (2025b) studies the ecological conditions suitable for herding. Building on suitability data constructed by Beck and Sieber (2010) through a maximum entropy model, Becker (2025b) documents that land suitability for herding and traditional subsistence from herding are strongly correlated across ethnic groups in the *Ethnographic Atlas* ($\rho=0.59$), a finding that we reproduce in Appendix Figure A2. Although the measure shows that the environment is an important determinant of herding, we do not use environmental suitability as our primary variable for multiple reasons. First, the land suitability measure for herding (vs. agriculture) is unavailable for the Americas. If we were to use the measure, our analysis would no longer be global, and we would lose over one-third of our sample. Second, taking into account how the measure is constructed, it is unlikely to be more exogenous than the measure we use. As Beck and Sieber (2010) document, their measure is constructed by taking ethnographic data on the prevalence of herding – i.e., the data we use in our analysis – and then predicting these values based on flexible functional forms of a set of geographic characteristics. From the estimated parameters, the model is used to create a measure of predicted pastoral suitability. (The same procedure is used for other suitability measures.) Thus, if one is concerned about other historical factors affecting historical herding, variation from these factors may be present in the constructed herding suitability measure as well.⁴ Instead, our approach, detailed in Section 3, accounts for an extensive set of observable geographic, historical, and contemporaneous characteristics that are potentially correlated with herding and/or conflict.

3. Traditional Herding and Contemporary Conflict

In its original form, the ‘culture of honor’ theory was written to explain relatively small-scale elements of aggression (Nisbett and Cohen, 1996), probably because psychologists tend to focus more on individual behavior than on group-level outcomes. Indeed, the vast majority of the available evidence for the validity of the ‘culture of honor’ theory focuses on within-group violence, such as assaults or homicides. Yet, there is also a considerable body of anecdotal

⁴For the same reason, if we were to use the measure as an instrument, because of its construction, we would not expect the IV estimates to be more reliable than the OLS estimates.

evidence that attributes different types of between-group conflict to a tradition of herding and resulting norms of revenge. For example, between-group conflict among pastoralists has been documented in Northern Cameroon (Moritz, 2008), among the Turkana in East Africa (Mathew and Boyd, 2011), as well as among herders in Kenya and Ethiopia (Abbink, 2009; Galaty, 2016; Beyene, 2017). A common thread that runs through these anthropological, ethnographic, and historical contributions is the importance that grievance and revenge play in the initiation and escalation of conflict, as well as for recruitment.

One question of our study is whether these patterns also extend to larger-scale conflicts, including civil wars. There are numerous examples suggesting this possibility. For instance, the civil conflict in Somalia can be traced back to grievances and revenge-taking between competing clans (Lewis, 1994). Direct evidence from the Congo similarly places grievance and revenge at the center, particularly in explaining why militant groups have been able to successfully recruit and win the support of the local populations by appealing to retaliation (Marchais et al., 2022). Among pastoral groups in Western Africa, revenge and retaliatory motives, often stemming from land-use conflicts, are important motives for their support of jihadist rebel groups aiming to overthrow the government (Benjaminsen and Ba, 2019). For example, a recent study of the determinants of Fulani support for jihadist groups in central Mali identifies the first causes as being past human rights violations and predatory behavior of the army and other state representatives (Benjaminsen and Ba, 2021).

More generally, a desire for revenge plays a critical role in mobilizing non-state actors. As the recent study by Souleimanov, Siroky and Colombo (2023) puts it, “non-material incentives may be far stronger in the recruitment of fighters in societies with a tradition of blood revenge. . . in many war zones, the custom of blood revenge features as the primary motive for violent mobilization.”

3.1. Conflict Data

Our primary data on conflict are taken from the *Uppsala Conflict Data Program* (UCDP), the world’s main provider of data on organized violence. The dataset covers the whole world (with the exception of Syria) for the period 1989–2016. The unit of observation in the dataset is a conflict event, defined as an “incident where armed force was used by an organized actor against another organized actor, or against civilians, resulting in at least one direct death at a specific location and

a specific date.”⁵ For each conflict event, the dataset reports the starting and ending dates, the conflict location’s geographic coordinates, the conflict type, and the number of deaths.

Our analysis requires us to link the frequency of contemporary conflicts to historical economic dependence on herding. As discussed in Section 2 and detailed in Appendix B, we do this by using UCDP’s location information to associate the conflict events with the traditional practices of ethnic groups using the global distribution of languages and dialects mapped in the *Ethnologue*.

For each language group, we aggregate the UCDP data into three types of conflict events over the 1989–2016 period covered by the data:⁶

1. Total conflicts: an aggregate measure that includes all conflicts described as ‘civil’ or ‘non-civil’ conflicts below.
2. Civil conflicts: conflict events that involve the agents of the government (such as the military or police officers) as one of the participants.
3. Non-civil conflicts: conflict events that do not involve government agents as one of the participants.

Our baseline measures are the *number of conflict events* within each category.

Of note is that our conflict measures omit interstate conflicts, where both actors involved are national governments. This is a form of conflict that is not relevant to our hypothesis because the location of military strikes in warfare is primarily determined by strategic objectives (the amount of land being controlled at that point, and the technologies available to both sides) rather than by revenge-taking motives of the local population. Indeed, the location of conflict events between two governments is often entirely unrelated to the local populations and, instead, imposed by potentially geographically distant governments.⁷

For ease of reference, we report descriptive statistics for the dependent and independent variables for all analyses in Appendix Table A1.

⁵The UCDP dataset has a 25-deaths-in-any-calendar-year threshold as an inclusion criterion, and events with at least one death are included once the threshold is passed. If a pair of actors meets the threshold in any year, all events associated with this pair (including those in other years, even when the threshold is not crossed) are included.

⁶The finer details of how each conflict type is identified using the UCDP data are provided in Appendix B.

⁷There are 135,181 conflicts in our dataset. Of these, the majority (103,672) are civil conflicts, 30,726 are non-civil conflicts, whereas there are only 783 interstate events.

3.2. Estimation Strategy and Covariates

Our analysis examines the relationship between a tradition of herding and conflict today by looking across language groups globally. Our estimating equation takes the following form:

$$f(y_{i,e,c}) = \alpha_c + \beta \text{Herding}_e + X_e^H \Gamma + X_i^C \Omega + \varepsilon_{i,e,c}, \quad (1)$$

where the unit of observation is an *Ethnologue* language group i , which is associated with an *Ethnographic Atlas* ethnicity e , and is located in country c . As noted, while each language group i is linked to an ethnicity e , languages and ethnicities are not synonymous nor matched one-to-one. Typically, ethnic groups from the *Ethnographic Atlas* tend to be matched to multiple language groups in the *Ethnologue*. Thus, ethnic groups comprise multiple language groups. We, thus, write $y_{i,e,c}$ for the number of conflicts that take place in the area of language group i , which is part of the ethnic group e and located in country c . Herding_e is our measure of ancestral dependence on herding, calculated at the ethnicity level.

The dependent variable, $f(y_{i,e,c})$, is (a potentially non-linear transformation of) the number of conflict events that occur in the territory of language group i located in country c . We report two types of specifications. First, to take into account that the conflict data have a long right tail (large outliers), our baseline measure is the inverse hyperbolic sine (IHS, arsinh), estimated using OLS. While this has the disadvantage that coefficient estimates are difficult to interpret, an advantage is that the linear estimation facilitates the calculation of Oster’s (2019) δ , which is informative about the sensitivity of the estimates to potential omitted variables. Second, since the number of conflicts is a count variable, we also report results using a Poisson Pseudo-Maximum Likelihood estimation (PPML), which is particularly well suited for count data with many zeros and large dispersion, and does not require a transformation.⁸

Turning to covariates, α_c denotes country fixed effects, which account for country-level determinants of conflict, including cross-country differences in real per capita GDP, the quality of

⁸We test the robustness of our results using a negative binomial specification (Appendix Table A2). We also examine the relationship between herding and conflict at the extensive margin (the probability of having any conflict) and intensive margin (the frequency of conflict conditional on having any conflict), finding a positive effect at both margins. The results are reported in Appendix Tables A3 and A4. In Table A3, we lose significance at the conventional level in some of the specifications and only when the standard errors are clustered at the country level.

domestic institutions, and international geopolitical characteristics.⁹

While the inclusion of country fixed effects captures most of the determinants of conflict that have been examined in the literature, there may be other factors that vary subnationally and that could potentially confound the estimated effects of herding. Thus, our specification also includes a vector of controls for historical characteristics which, like the herding variable, are measured at the ethnic group level, which is matched to language groups. The vector, which we denote X_e^H , includes the absolute value of historical latitude, settlement complexity, and levels of jurisdictional hierarchy beyond the local community from the *Ethnographic Atlas*.

The equation also includes a vector of six contemporary covariates, measured at the language-group level, denoted X_i^C . Three variables measure contemporary social conditions potentially relevant for conflict: linguistic fractionalization and polarization, and log population density.¹⁰ Two variables capture contemporary livelihood and development: the share of land that is used for herding today and nighttime light density. Finally, we also include the terrain ruggedness of the land currently inhabited by language groups since uneven and mountainous terrain has been linked to conflict (Carter, Shaver and Wright, 2019).¹¹

Given the correlational nature of our analysis, we present a wide range of empirical specifications. First, we always report specifications that include only country fixed effects. This is because some of the covariates that plausibly drive conflicts are endogenous to herding (such as settlement complexity or institutional quality), and we wish to ensure that the inclusion of bad controls does not drive our results. Second, we report estimates that include the set of baseline historical ethnicity-level and contemporary language-level covariates discussed above. Third, the Appendix presents a wide range of robustness checks that control for additional variables, which we summarize below.

⁹A question with this within-country strategy is whether there is sufficient variation in a tradition of herding among groups within countries. To provide a sense of the within-country variation, in Appendix Table A5, we provide an overview of countries with variation and their corresponding sample sizes. The table shows a list of countries with variation in herding and, in bold, countries with variation in both herding and conflict.

¹⁰Several studies have suggested a relationship between ethnic/linguistic fractionalization and the likelihood of conflict (Alesina, Devleeschauwer, Easterly, Kurlat and Wacziarg, 2003; Alesina, Baqir and Easterly, 1999; Collier and Hoeffler, 1998). We construct a measure of linguistic fractionalization as follows. For each language group, we first generate a 500km-radius circle centered at its centroid. Letting n_i be the share of the i -th language group (in terms of population) within the circle such that $\sum_i n_i = 1$, fractionalization is defined as $F = 1 - \sum_i n_i^2$. It captures the degree to which the circle is split into distinct groups. The results are robust when we construct the measure using buffers that are 20, 50, 100, or 1000 kilometers. Another important determinant of conflict is polarization, which captures how far the distribution of linguistic groups is away from the bipolar (1/2, 0, ..., 0, 1/2) distribution (Montalvo and Reynal-Querol, 2005; Esteban and Ray, 2008). We construct this variable using an analogous procedure but with the formula: $\sum_i n_i^2 (1 - n_i)$.

¹¹The details of the construction of the historical and contemporaneous controls are provided in Appendix B.

3.3. Main Results

Estimates of equation (1) are reported in Table 1. For each type of conflict, we report three specifications: the first includes country fixed effects only, the second adds the baseline set of covariates discussed above, and the third estimates the same equation using PPML. We report two types of standard errors, either clustered at the *Ethnographic Atlas* ethnic group level (in parentheses) or at the level of the country in which the language group resides (in square brackets).¹² For the OLS specifications, we also report standardized *beta* coefficients, which provide a sense of the magnitude of the estimated effect, and Oster’s (2019) δ , which is informative about the sensitivity of the estimates to possible omitted variables.

The estimated effects are very similar for the different types of conflict. They suggest that an increase in historical dependence on herding by one standard deviation increases the frequency of arsinh armed conflict by about 10% of a standard deviation. This corresponds to about 0.13 conflict events during the period 1989–2016. The PPML coefficient implies that a 0.1 increase in herding is associated with an $\exp(2.883 \times 0.1) - 1 = 33.4$ percent increase in conflict events, equivalent to approximately 7 additional conflict events evaluated at the mean. These relationships are always statistically significant, regardless of how we compute standard errors. Furthermore, the Oster- δ is always around 4 or 5, suggesting that unobservables are unlikely to drive the observed patterns.

Figure 3 reports binned scatterplots of the frequency of each conflict type as a function of historical dependence on herding, controlling for country fixed effects and the other covariates (columns 2, 5, and 8). As shown, the relationships appear quite general and not driven by a small number of influential observations or outliers.

While the effect of a tradition of herding on localized (non-civil) conflicts is expected, especially given the prior evidence on disputes between individuals and homicides, the effects on civil conflicts might be more surprising. An important point is that the category of civil conflicts includes much more than large-scale civil wars. For example, if a police officer wrongs a family (perhaps through attempted extortion that escalates into violence), and a family member takes revenge by killing the police officer, then this is coded as a civil conflict incident. Similarly, if there is a dispute between civilians and a police officer becomes involved in the violence, this

¹²In the civil conflicts analyses, clustering at the country level appears more appropriate because a common actor – the government – is present. Yet the specific ways in which we cluster very rarely impact the statistical significance of our conflict results.

Table 1: Traditional herding and contemporary conflict globally

	Dependent variable: Number of conflict events								
	All conflicts			Civil conflicts			Non-civil conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependence on herding	0.949 (0.295)*** [0.386]**	0.912 (0.266)*** [0.332]**	2.883 (1.096)*** [1.036]**	0.773 (0.256)*** [0.325]**	0.740 (0.233)*** [0.279]**	2.875 (1.327)** [1.200]**	0.735 (0.225)*** [0.291]**	0.785 (0.205)*** [0.251]**	2.899 (1.185)** [1.236]**
<i>Historical ethnicity-level covariates:</i>									
Settlement complexity		-0.007 (0.017) [0.020]	0.203 (0.157) [0.141]		-0.005 (0.016) [0.018]	0.213 (0.177) [0.156]		-0.008 (0.013) [0.016]	0.131 (0.103) [0.097]
Jurisdictional hierarchy		0.057 (0.025)** [0.035]	0.214 (0.161) [0.176]		0.055 (0.021)** [0.026]**	0.159 (0.192) [0.216]		0.053 (0.020)*** [0.025]**	0.379 (0.110)*** [0.096]**
Historical latitude (abs.)		8.194 (7.020) [7.433]	77.702 (26.481)*** [27.732]**		6.323 (6.119) [6.320]	79.927 (31.911)** [33.993]**		6.480 (4.955) [5.485]	73.182 (28.607)** [31.694]**
<i>Contemporary language-level covariates:</i>									
Population density (ln)		0.122 (0.015)*** [0.033]**	0.600 (0.086)*** [0.064]**		0.105 (0.014)*** [0.031]**	0.612 (0.099)*** [0.068]**		0.069 (0.010)*** [0.020]**	0.628 (0.090)*** [0.082]**
Ethnic fractionalization		-0.205 (0.169) [0.251]	-1.924 (1.279) [1.491]		-0.286 (0.154)* [0.223]	-1.822 (1.459) [1.735]		0.047 (0.125) [0.201]	-1.837 (0.835)** [0.960]*
Ethnic polarization		0.336 (0.672) [0.859]	5.134 (3.712) [4.933]		0.693 (0.586) [0.763]	9.279 (3.902)** [5.438]*		-0.779 (0.487) [0.681]	-5.735 (3.278)* [4.316]
Nighttime lights		-0.018 (0.007)*** [0.010]*	-0.151 (0.035)*** [0.032]**		-0.020 (0.006)*** [0.009]**	-0.193 (0.043)*** [0.037]**		-0.008 (0.005) [0.006]	-0.086 (0.022)*** [0.017]**
Share of land for herding		0.163 (0.121) [0.153]	1.564 (0.515)*** [0.445]**		0.101 (0.109) [0.136]	1.632 (0.580)*** [0.520]**		0.077 (0.089) [0.099]	1.542 (0.470)*** [0.615]**
Ruggedness		-0.624 (0.199)*** [0.263]**	-2.217 (0.769)*** [0.856]**		-0.566 (0.180)*** [0.244]**	-2.252 (0.786)*** [0.896]**		-0.541 (0.147)*** [0.210]**	-1.823 (0.607)*** [0.596]**
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.10	0.097		0.092	0.088		0.11	0.12	
Oster δ		4.38			4.37			4.82	
Mean of dependent var	0.63	0.63	20.9	0.51	0.50	16.7	0.35	0.36	5.15
SD of dependent var	1.45	1.45	258.7	1.30	1.30	237.5	1.04	1.05	51.1
Adj. R-squared	0.28	0.31		0.28	0.31		0.22	0.24	
Pseudo R-squared			0.59			0.62			0.55
Number of Obs.	6,979	6,174	5,558	6,979	6,174	5,327	6,979	6,174	5,294
Number of Countries	154	146	106	154	146	100	154	146	84
Number of Ethnic Groups	1,096	976	934	1,096	976	879	1,096	976	906

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

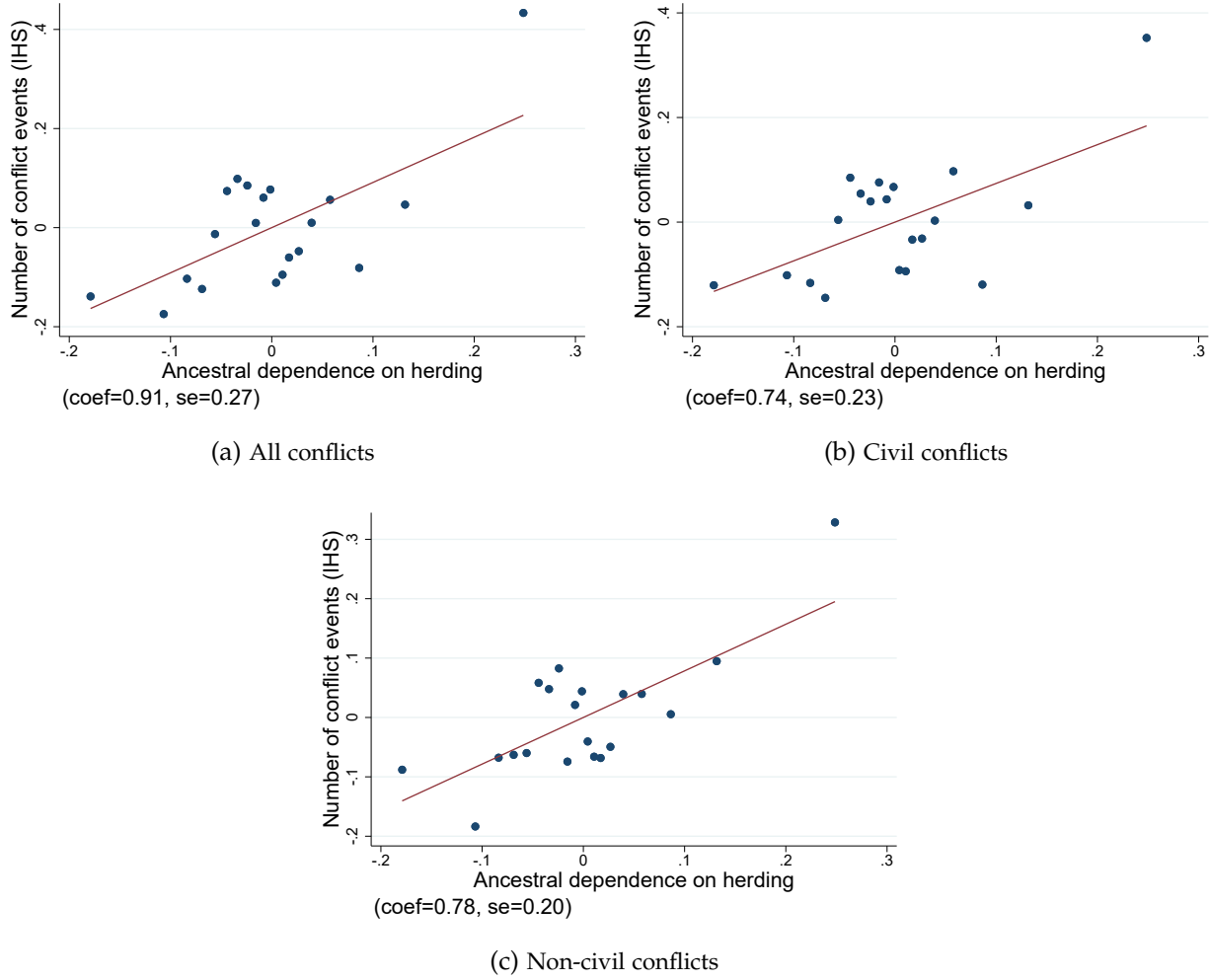


Figure 3: Binscatter partial correlation plots for the relationship between contemporary conflict and a tradition of herding. In each plot, a unit of observation is a language group, defined as a language from *Ethnologue* spoken in a country, $N = 6,174$. Each dot shows the average of (arsinh) conflict events for a given range of values of dependence on herding. Each binscatter is constructed after first partialling out country fixed effects, historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness.

is also coded as civil conflict. In both cases, one of the participants is an agent of the national government. Indeed, in our sample, 82 percent of the civil conflicts last for one day only (with an average duration of 3.8 days). The median number of people killed is two (and the average is 14). These incidents are important because they can escalate, generating tensions and grievances that can either instigate and/or fuel full-scale civil wars. To check whether the culture of honor matters also for the size of conflicts, we re-estimate equation (1), but with the arsinh of the total number of conflict deaths as the dependent variable. As shown in Appendix Table A6, for all conflict types, herding is associated with a greater number of conflict deaths.

Influential observations. Because the herding dependence variable has a skewed distribution, one might be worried that our results are driven by a few language groups with extremely high dependence on herding. To alleviate this concern, we winsorize the herding variable at the 95th percentile (0.405). That is, values greater than this are recoded as 0.405. As reported in Appendix Table A7, this does not meaningfully affect the results.¹³ We also examine whether our findings are driven by a single country by re-estimating our baseline specifications, removing one country at a time. The results remain stable throughout (Appendix Figures A3 and A4.)

3.4. Additional Covariates

In this section, we consider the robustness of our estimates to controlling for additional covariates that are either potential determinants of conflict and/or are potentially correlated with herding. The set of covariates, as well as their correlation with herding, is reported in Appendix Table A8.¹⁴

The robustness of our findings to the inclusion of these covariates is reported in Table 2.¹⁵ In each panel, we begin by first including a large set of geographical and ecological controls (column 1). In column 2, we report estimates that control for historical ethnic characteristics that could be related to conflict. A potential interpretation of our conflict estimates is that they are driven by the fact that herding societies tend to be less sedentary than agricultural societies, which could trigger more conflicts. We control for an indicator for whether the historical settlement pattern is nomadic or semi-nomadic ($v30 = 1$ or 2 in the *Ethnographic Atlas*) and for an indicator for whether it is semi-sedentary or compact but impermanent ($v30 = 3$ or 4 in the *Ethnographic Atlas*). We also control for an indicator capturing whether an ethnic group traditionally had slavery.

In column 3, we control for contemporaneous characteristics of language groups. Motivated by the fact that our findings could be due to pastoral groups having been marginalized in recent history, we include a measure, constructed from the *Ethnic Power Relations (EPR)* dataset, that

¹³The estimates overall remain similar in magnitude with some slight variation, e.g., the PPML coefficient for civil conflict decreases slightly, and the coefficient for non-civil conflict increases. However, the PPML estimates become slightly less precise, and the estimate from one specification – the PPML regression with civil conflicts as the outcome – loses significance at conventional levels.

¹⁴In general, accounting for country fixed effects weakens the correlation between herding and the potential omitted variables. The within-country correlations show that herding societies tended to have less complex settlements, were more likely to be nomadic, had higher levels of jurisdictional hierarchy, and lived further from the equator. Ancestral herding is positively associated with (although with a smaller magnitude) ethnic polarization and segregation, and negatively associated with ethnic fractionalization. Historical herding is also positively correlated with the share of Muslims and negatively correlated with the share of Christians. Several contemporary geographical characteristics are unsurprisingly correlated as well.

¹⁵Appendix Tables A9–A14 report the full specifications, showing the coefficients on all covariates for all conflicts, civil conflicts, and non-civil conflicts, for both the OLS and PPML specifications.

equals one if an ethnic group was “powerless,” “discriminated,” or “self-excluded” at any point in the country from 1989–2016. We also control for a measure of ethnic segregation, which is an important determinant of conflict (Alesina and Zhuravskaya, 2011; Corvalán and Vargas, 2015).¹⁶

In column 4, we include additional variables that capture contemporary development and livelihood (the share of a group’s land that is cropland and the share that is urban). We also account for the potential importance of religion by including the estimated share of a group that belongs to the main religious denominations: Christians, Muslims, Hindus, Buddhists, Jews, and agnostics, with the excluded group being ‘other’ religious denominations (column 5) and for measures of religious fractionalization and religious polarization (column 6).

Lastly, we report two “kitchen sink” specifications, one that includes all covariates (column 7), and another that is more selective and uses a LASSO regression to select predictors of conflict among the controls (column 8). Our coefficients remain reasonably stable across all specifications, losing statistical significance in only three out of the 48 specifications reported in Table 2.¹⁷ For the OLS specifications, the calculated Oster- δ is always between 3.72 and 6.18, suggesting that our estimates are quite robust to the potential presence of omitted variables.

3.5. *Alternative Historical Characteristics*

To further verify that our measure of ancestral dependence on herding does not spuriously pick up the effects of other latent historical group-level characteristics, we examine the sensitivity of our estimates to systematically considering all other characteristics available for a broad cross-section of ethnic groups in the *Ethnographic Atlas*. These measures include local institutional characteristics (types of elections, measures of property rights), different types of social arrangements (patrilineality/matrilineality, patrilocal/matrilocal, cousin marriage, bride price and a measure of kinship tightness), religious characteristics (the presence of moralizing gods), and female participation in agriculture as economic activity.¹⁸ The estimates, reported in Appendix

¹⁶We measure ethnic segregation at the language group level by first generating a 500km radius circle around the centroid of the language group. We then define regions within the circle as 1×1 degree cells and calculate ethnic segregation across cells within the circle. Let T be the total population of the circle, t_j be the population of region j , J be the total number of regions in the circle, M be the total number of language groups in the circle, π_{jm} be the fraction of language group m in region j , and π_m be the fraction of language group m in the circle. The segregation index for the centered language group i is defined as: $S = \frac{1}{M-1} \sum_m \sum_j \frac{t_j}{T} \frac{\pi_{jm} - \pi_m}{\pi_m}$.

¹⁷The estimates lose significance in the PPML specification with geographic controls (column 1).

¹⁸See Enke (2019) for details about the measure of kinship tightness. Other measures of economic activity, such as reliance on gathering or agriculture, are, by construction, mechanically associated with the absence of herding and, for that reason, are not included.

Table 2: Traditional herding and contemporary conflict globally: More controls

<i>Dependent variable: Conflict events</i>								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A: OLS Estimates</i>								
Coef. for Herding:								
All Conflicts	0.806 (0.266)*** [0.329]**	0.952 (0.287)*** [0.358]***	0.971 (0.280)*** [0.347]***	0.915 (0.269)*** [0.339]***	0.971 (0.270)*** [0.320]***	0.918 (0.266)*** [0.333]***	0.943 (0.300)*** [0.322]***	0.894 (0.278)*** [0.304]***
Oster δ	3.99	4.40	4.57	4.38	5.34	4.58	4.59	4.43
Civil Conflicts	0.641 (0.236)*** [0.279]**	0.786 (0.252)*** [0.307]**	0.793 (0.248)*** [0.299]***	0.741 (0.235)*** [0.284]***	0.776 (0.235)*** [0.267]***	0.744 (0.233)*** [0.281]***	0.750 (0.271)*** [0.290]**	0.687 (0.252)*** [0.273]**
Oster δ	3.72	4.33	4.55	4.34	5.01	4.53	4.09	3.72
Non-civil Conflicts	0.708 (0.201)*** [0.251]***	0.832 (0.220)*** [0.263]***	0.830 (0.219)*** [0.258]***	0.787 (0.207)*** [0.256]***	0.828 (0.210)*** [0.257]***	0.791 (0.206)*** [0.254]***	0.827 (0.231)*** [0.245]***	0.833 (0.228)*** [0.240]***
Oster δ	4.84	4.79	4.86	4.80	6.18	5.13	5.33	5.60
<i>Panel B: PPML Estimates</i>								
Coef. for Herding:								
All Conflicts	1.784 (1.154) [1.225]	3.483 (0.887)*** [0.863]***	2.858 (1.177)** [1.159]**	2.852 (1.157)** [1.150]**	3.248 (1.006)*** [0.922]***	2.967 (1.140)*** [1.079]***	3.028 (1.125)*** [1.287]**	2.188 (1.123)* [1.170]*
Civil Conflicts	1.753 (1.254) [1.250]	3.632 (0.971)*** [0.905]***	2.828 (1.446)* [1.362]**	2.833 (1.401)** [1.336]**	3.433 (1.239)*** [1.106]***	2.989 (1.401)** [1.269]**	3.876 (1.057)*** [1.218]***	2.403 (1.092)** [1.106]**
Non-civil Conflicts	2.313 (1.389)* [1.471]	3.609 (1.258)*** [1.238]***	2.949 (1.249)** [1.355]**	2.928 (1.253)** [1.356]**	3.037 (1.174)*** [1.183]**	2.937 (1.191)** [1.243]**	3.097 (1.411)** [1.442]**	3.130 (1.452)** [1.477]**
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographical and Ecological Controls	Yes						Yes	Yes
Historical Controls		Yes					Yes	Yes
Ethnic Group Controls			Yes				Yes	Yes
Contemporary Landuse Controls				Yes			Yes	Yes
Religious Denomination Controls					Yes		Yes	Yes
Religious Fractionalization & Polarization						Yes	Yes	Yes
LASSO Selected								Yes

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Panel A applies the inverse hyperbolic sine (IHS) transformation to the value. Baseline controls include historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A15, show the relationship between each historical characteristic and contemporary conflict, while accounting for country fixed effects (Specification A). We then add the measure of reliance on herding to check whether its estimated relationship with conflict is significantly reduced (or even eliminated) after controlling for the other ethnic characteristics (Specification B). To facilitate the comparisons of coefficients, we report standardized ‘beta coefficients.’ We find that most other ethnic historical characteristics are not correlated with conflict, and controlling for them barely affects the coefficient of the herding variable. The results are similar when we use a PPML specification (Appendix Table A16).

3.6. *Within-Africa ACLED Estimates*

For groups within the African continent, we are able to use the *Armed Conflict Location and Event Data project (ACLED)*, which, although limited in geographic coverage, is much richer than the *UCDP* data. The criterion for a conflict’s inclusion in the database is considerably lower for *ACLED* than for the *UCDP* dataset. Thus, we are able to estimate effects on smaller-scale localized conflicts that only involve local actors and result in fewer deaths.¹⁹ These conflicts, if they do not surpass the 25-death-in-a-calendar-year threshold, are not included in the *UCDP* dataset. Most importantly, as we explain below, the *ACLED* data allow us to test explicitly for revenge-taking as a channel behind our finding of a relationship between herding and conflict.

We undertake our analysis in the same manner as with the *UCDP* data. We connect the location of a conflict event provided in the *ACLED* data to a tradition of herding by location, using the distribution of languages and dialects from the *Ethnologue*, whose ethnographic characteristics we know from the linked *Ethnographic Atlas*. We then again examine the relationship between a tradition of herding and conflict when looking across language groups in the *Ethnologue*. Figure 4 shows a map of all conflict events in the *ACLED* data, overlaid on top of color coding that shows the local degree of ancestral dependence on herding.

More formally, as we did with the *UCDP* data, we estimate equation (1), which includes country fixed effects. The estimates are reported in Table 3 and partial correlation binned scatterplots (for the specification with all covariates) are reported in Figure 5. We report estimates for the

¹⁹There are 140,746 conflicts in the *ACLED* dataset. 137,140 events are within Africa. Of these, 68,044 are civil conflicts, and 61,213 are non-civil conflicts (among which 35,837 are “localized” conflicts). We exclude 29 events that are peaceful interactions between civilians and 7,854 events that involve international organizations or forces active outside of their main country of operation, of which 603 are interstate conflicts between military forces of two countries.

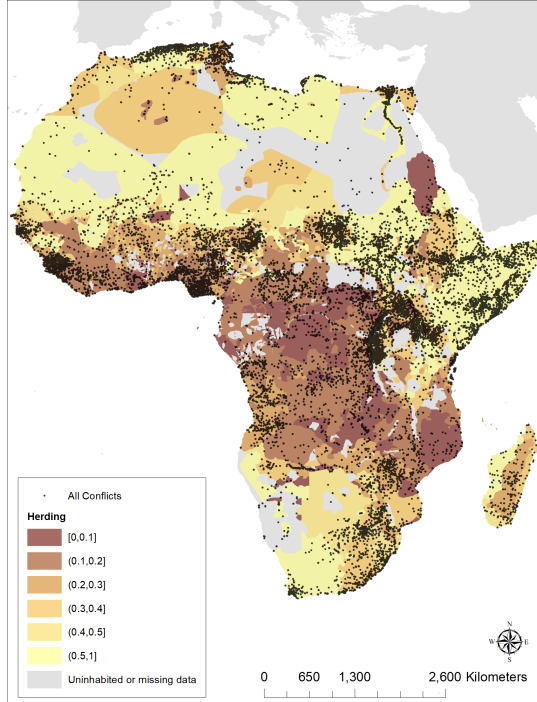


Figure 4: Map of conflict events in *ACLED* dataset, shown along with local ancestral dependence on herding.

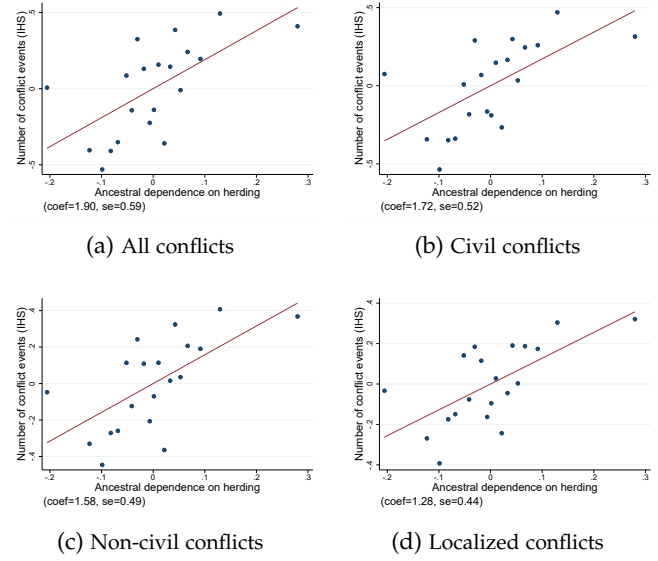


Figure 5: Binscatter partial correlation plots for the relationship between contemporary conflict and a tradition of herding. In each plot, a unit of observation is a language group, defined as a language from *Ethnologue* spoken in a country, $N = 2,133$. Each dot shows the average of (arsinh) conflict events for a given range of values of dependence on herding. Each binscatter is constructed after first partialling out country fixed effects, historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness.

number of all conflicts, civil conflicts, non-civil conflicts, and localized conflicts as outcomes, without and with our baseline set of covariates. Consistent with the *UCDP* estimates, when using the *ACLED* data to look within Africa, we also find that a history of herding is associated with more conflict of all types.

As mentioned above, a benefit of the *ACLED* data is that they have a lower threshold for the inclusion of conflicts, which allows us to also measure ‘localized conflicts,’ smaller-scale conflicts within the same community. The effect of herding on localized conflicts is shown in the final three columns of Table 3.²⁰ We find that a history of herding is also associated with more localized conflicts. In comparing the magnitude of all conflicts, civil, non-civil, and localized, we find similar effects across all types of conflicts: an increase in dependence on herding by one standard deviation increases the frequency of arsinh armed conflict by 11–14% of a standard deviation.²¹

²⁰Localized conflicts are conflict events for which both actors are geographically and/or ethnically local groups. These are identified using the “Interaction” variable from *ACLED*. See Appendix B for details.

²¹Analogous to the *UCDP* analysis, with the *ACLED* data, we report the correlations between historical herding and the regression covariates (Appendix Table A17), we test the robustness of our results to the inclusion of a larger set of controls (Appendix Tables A18–A25), and we check the correlations between conflict in Africa and other historical ethnicity characteristics (Appendix Tables A26 and A27.) The coefficient on herding remains similar in magnitude and significance. In a few specifications, we lose significance at the conventional level; in most cases, this is when the standard errors are clustered at the country level.

Table 3: Traditional herding and contemporary conflict in Africa

	Dependent variable: Number of conflict events											
	All conflicts			Civil conflicts			Non-civil conflicts			Localized conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Dependence on herding	1.706 (0.512)*** [0.420]***	1.904 (0.591)*** [0.491]***	2.085 (0.739)*** [0.760]***	1.687 (0.456)*** [0.383]***	1.716 (0.519)*** [0.463]***	2.418 (0.794)*** [0.877]***	1.378 (0.436)*** [0.374]***	1.577 (0.494)*** [0.383]***	1.813 (0.794)** [0.760]**	1.113 (0.383)*** [0.367]***	1.276 (0.435)*** [0.354]***	2.300 (0.811)*** [0.841]***
<i>Historical ethnicity-level covariates:</i>												
Settlement complexity		-0.006 (0.049) [0.045]	-0.043 (0.081) [0.077]		-0.011 (0.044) [0.044]	-0.018 (0.091) [0.090]		-0.013 (0.043) [0.042]	-0.073 (0.083) [0.080]		-0.032 (0.037) [0.039]	-0.015 (0.078) [0.090]
Jurisdictional hierarchy		0.156 (0.062)** [0.051]***	0.645 (0.137)*** [0.145]***		0.159 (0.053)*** [0.053]***	0.612 (0.141)*** [0.141]***		0.149 (0.055)*** [0.049]***	0.678 (0.150)*** [0.169]***		0.125 (0.047)*** [0.045]***	0.651 (0.123)*** [0.136]***
Historical latitude (abs.)		6.524 (15.027) [15.253]	-10.655 (22.961) [27.983]		11.520 (14.218) [15.711]	-32.255 (28.974) [33.983]		-1.605 (12.682) [14.543]	-3.635 (18.954) [22.166]		-6.990 (11.515) [14.153]	16.118 (20.023) [24.260]
<i>Contemporary language-level covariates:</i>												
Population density (ln)		0.390 (0.045)*** [0.071]***	0.647 (0.084)*** [0.062]***		0.287 (0.040)*** [0.062]***	0.499 (0.089)*** [0.074]***		0.337 (0.038)*** [0.061]***	0.829 (0.096)*** [0.080]***		0.305 (0.033)*** [0.051]***	0.882 (0.096)*** [0.087]***
Ethnic fractionalization		0.445 (1.026) [0.826]	-1.799 (1.027)* [1.178]		0.225 (0.900) [0.825]	-1.526 (1.115) [1.187]		0.326 (1.019) [0.700]	-2.410 (1.130)** [1.232]*		0.175 (0.939) [0.666]	-2.167 (1.102)** [0.987]**
Ethnic polarization		1.893 (2.592) [3.704]	8.330 (3.444)** [3.452]**		2.002 (2.275) [3.140]	7.543 (3.751)** [3.786]**		0.847 (2.423) [3.351]	10.000 (3.556)*** [3.548]***		-0.159 (2.146) [2.868]	8.777 (3.036)*** [3.775]**
Nighttime lights		0.004 (0.021) [0.039]	-0.052 (0.015)*** [0.011]***		-0.003 (0.019) [0.032]	-0.053 (0.017)*** [0.012]***		0.007 (0.019) [0.035]	-0.058 (0.014)*** [0.013]***		0.000 (0.017) [0.030]	-0.067 (0.014)*** [0.013]***
Share of land for herding		0.335 (0.260) [0.299]	1.376 (0.843) [0.862]		0.227 (0.228) [0.270]	1.497 (0.913) [0.931]		0.401 (0.214)* [0.252]	1.351 (0.762)* [0.838]		0.472 (0.189)** [0.221]**	1.725 (0.623)*** [0.731]**
Ruggedness		-0.810 (0.561) [0.648]	0.376 (1.094) [1.556]		-0.179 (0.518) [0.571]	0.536 (1.038) [1.535]		-0.806 (0.454)* [0.738]	0.175 (1.274) [1.669]		-0.949 (0.376)** [0.601]	-1.311 (1.178) [1.376]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.12	0.13		0.14	0.14		0.12	0.13		0.11	0.12	
Oster δ	8.69	8.69		4.95	4.95		9.51	9.51		8.32	8.32	
Mean of dependent var	1.63	1.65	55.9	1.23	1.24	28.9	1.20	1.21	27.0	0.92	0.93	16.2
SD of dependent var	2.07	2.08	419.3	1.83	1.84	227.4	1.77	1.78	208.7	1.57	1.58	121.6
Adj. R-squared	0.28	0.33		0.29	0.33		0.25	0.31		0.25	0.31	
Pseudo R-squared			0.62			0.61			0.64			0.67
Number of Obs.	2,285	2,133	2,119	2,285	2,133	2,119	2,285	2,133	2,119	2,285	2,133	2,119
Number of Countries	56	56	45	56	56	45	56	56	45	56	56	45
Number of Ethnic Groups	497	449	448	497	449	448	497	449	448	497	449	448

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

4. The Culture of Honor Mechanism I: Revenge-Taking Motives in Conflicts

We hypothesize that a tradition of herding is associated with conflict because these groups developed moral systems that support punishment and revenge-taking when one experiences wrongdoings from others. In this section, we examine evidence for this mechanism. Our analysis proceeds in two steps. First, we directly test for the culture of honor mechanism by examining whether the link between herding and conflict arises due to revenge-taking motives and retaliatory conflicts. Second, we also test and rule out an alternative possible explanation. Since pastoral populations are often transhumant (i.e., seasonally nomadic), this characteristic of herding may drive our results (McGuirk and Nunn, 2025).

4.1. The Role of Revenge-Taking in Conflict

a. Interaction With Previous Conflicts

The first exercise we undertake is to check whether the effect of herding on contemporary conflicts is greater when there is potential for retaliation. In other words, we examine whether herding leads to more conflict when there is a recent history of conflict. In these settings, it is more likely that any future conflict is in retaliation or to avenge wrongdoings during past conflicts.

We examine this possibility with the following panel regression equation that varies by language group and year:

$$f(y_{i,e,c,t}) = \alpha_c + \lambda_t + \theta \text{Herding}_e + \beta \text{Herding}_e \times \mathbb{I}_{i,e,c,t-1}^{y>0} + \eta \mathbb{I}_{i,e,c,t-1}^{y>0} + X_e^H \Gamma + X_i^C \Omega + \varepsilon_{i,e,c,t}, \quad (2)$$

where i indexes language groups, e the ethnic group to which the group belongs, c the country in which the group is located, and t indexes the year. α_c denotes country fixed effects, λ_t denotes year fixed effects, and X_e^H and X_i^C denote our vectors of (ethnicity-level) historical and (language-group-level) contemporary group-level covariates. $y_{i,e,c,t}$ is one of our measures of the number of conflict events (all, civil, and non-civil conflicts) that occur in the territory of a language group i , located in country c , at time t . The variable $\mathbb{I}_{i,e,c,t-1}^{y>0}$ is an indicator that equals one if there was a conflict in language group i 's territory in the recent past.

The first measure of past conflict that we consider codes whether there was a conflict during the previous calendar year. The estimates for the UCDP data are reported in Table 4. In contrast to the analyses above, here we report only OLS arsinh estimates because interaction coefficients

Table 4: Traditional herding and contemporary conflict globally: Interaction with previous conflicts

	<i>Dependent variable: Number of conflict events (IHS)</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	-0.001 (0.034) [0.037]	-0.011 (0.025) [0.036]	-0.009 (0.028) [0.032]	-0.012 (0.022) [0.030]	0.022 (0.019) [0.019]	0.011 (0.015) [0.020]
Herding $\times$ Previous Conflict (1y)	1.538 (0.229)*** [0.229]***	1.655 (0.245)*** [0.273]***	1.695 (0.234)*** [0.224]***	1.777 (0.253)*** [0.275]***	0.771 (0.195)*** [0.292]***	0.930 (0.194)*** [0.294]***
Previous conflict indicator (1y)	1.060 (0.066)*** [0.080]***	0.996 (0.070)*** [0.074]***	0.971 (0.068)*** [0.084]***	0.918 (0.075)*** [0.086]***	0.944 (0.074)*** [0.108]***	0.896 (0.077)*** [0.095]***
Baseline controls	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	0.092	0.092	0.072	0.072	0.039	0.040
SD of dependent var	0.50	0.51	0.45	0.45	0.31	0.31
Adj. R-squared	0.41	0.42	0.41	0.41	0.34	0.34
Number of Obs.	197,008	174,692	197,008	174,692	197,008	174,692
Number of Countries	211	211	211	211	211	211
Number of Ethnic Groups	1,104	985	1,104	985	1,104	985

Note. The unit of observation is a language group and a year. A language group is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for each year during the period 1989-2016. They are measured as the inverse hyperbolic sine (IHS) of the value. the Previous conflict indicator is based on the specification's dependent variable measured in the previous year. Baseline controls include historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

in nonlinear PPML specifications are not directly interpretable as interaction effects. In each specification, the 'Previous conflict' variable is the specification's dependent variable measured in the year prior. We find that herding only affects the incidence of conflict when there was a conflict in the previous year. By contrast, the raw herding coefficient, which captures the estimated effect following a year of peace, is very small and not statistically different from zero. In other words, the estimates in Table 4 suggest that if there is no prior conflict event, then herding and non-herding societies are equally likely to see conflict. Differences between herding and non-herding societies only emerge in the presence of a prior conflict event, with herding societies being more likely to see additional conflicts.²²

To better understand the precise timing that underlies these patterns, we estimate a variant of

²²We also test that the interaction effect is not driven by fractionalization or polarization, by interacting each of these two terms with previous conflicts. As reported in Appendix Tables A28 and A29, the results remain robust.

equation (2), where the time dimension varies at the month level (i.e., the unit of observation is a language group and month) and we allow the effect of herding to differ depending on whether there was a conflict in each of the previous twelve months. Figure 6a summarizes the estimates. We find that a tradition of herding has a large positive effect on conflict in the month immediately following a previous episode of conflict. The effect declines as one moves further in time from the conflict and approaches zero starting at four months following the conflict. Given that the duration of conflict events is fairly short – 3.8 days on average – we undertake the same exercise but using a more granular time dimension that varies at the level of 10-day intervals. As shown in Figure 6b, a very similar pattern emerges. We estimate a large effect of herding on the probability of conflict following the first 10 days after a conflict event. As one moves further in time from the event, the effect declines and approaches zero at about 40 days after the conflict. Our findings of quick retaliation among herding societies are consistent with ethnographic accounts showing that counterattacks are implemented quickly so that it is clear to the enemy that the attack is a retaliation for a particular wrongdoing (e.g., Mathew, 2022).

Overall, the estimates document a clear pattern. A tradition of herding is strongly associated with the occurrence of conflict events in the immediate aftermath of previous conflicts, but is uncorrelated with conflicts when no prior conflict event took place. These results are consistent with the hypothesis that herding is linked to conflict through the mechanisms posited by the culture of honor theory: retaliation in response to prior wrongdoings.

Replication in the ACLED dataset. We also undertake the same exercises using the *ACLED* data for Africa. The estimates, which are reported in Appendix Tables A30–A32 and Figures 6c and 6d, show that we find very similar patterns. Herding is strongly linked to conflict in the immediate aftermath of prior conflicts and much more weakly correlated with conflicts if no prior conflict took place.

b. Duration and prolonged conflicts

Given that retaliation increases the probability that a conflict continues, a natural implication is that conflicts related to historical reliance on herding should also last longer. We test for this possibility in two ways. First, we use duration in months as an alternative dependent variable. The estimates, reported in Appendix Table A33 for *UCDP* conflicts and Appendix Table A34 for

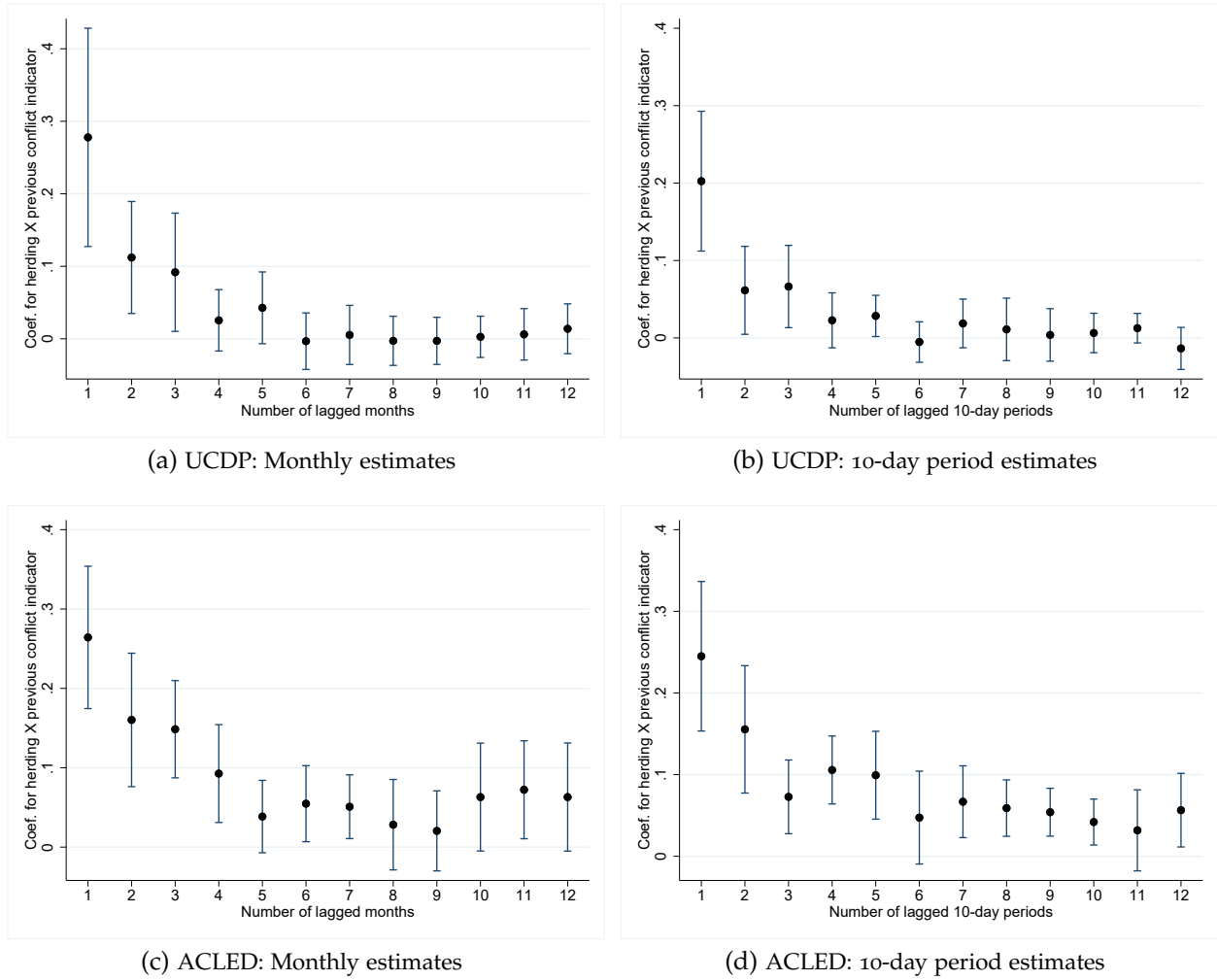


Figure 6: Coefficient plot for the interaction terms between herding and conflict indicators for the past 12 months (panels (a) and (c)) and for the past 120 days (panels (b) and (d)). A unit of observation is a country-language group in the *Ethnologue* and a year-month for panels (a) and (c), and a 10-day period for panels (b) and (d). In the top panels, the dependent variable is the arsinh number of conflicts from *UCDP*. In the bottom panels, the dependent variable is the arsinh number of conflicts in the *ACLED* data. Control variables include country fixed effects, time period fixed effects (either year-month or 10-day period), historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, share of land used for herding, and terrain ruggedness. Error bars show 95% confidence intervals, computed based on clustering at the ethnic-group level.

ACLED conflicts within Africa, show that historical reliance on herding is associated with conflict when we measure it as the number of months. The standardized beta coefficients range from 0.09 to 0.11 for *UCDP* and 0.10 to 0.14 for *ACLED*.

The second strategy is to use a multinomial logit estimator that distinguishes between (1) the absence of conflict during the full time period (e.g., 1989–2016 in the *UCDP* data), (2) at least one conflict but none of them were ‘prolonged,’ and (3) at least one conflict and at least one of

Table 5: Traditional herding and contemporary conflict globally: Prolonged events

	<i>Dependent variable: Incidence of conflict events (multinomial)</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)
Cat. 1 (omitted): no incidence						
Cat. 2: no prolonged Dependence on herding	0.226 (0.412) [0.598]	0.068 (0.522) [0.592]	0.166 (0.423) [0.660]	0.136 (0.569) [0.665]	0.760 (0.481) [0.645]	0.617 (0.567) [0.664]
Cat. 3: prolonged Dependence on herding	2.201 (0.576)*** [0.921]**	2.496 (0.659)*** [0.830]***	1.942 (0.610)*** [0.931]**	2.137 (0.681)*** [0.799]***	2.600 (0.687)*** [1.107]**	3.412 (0.773)*** [0.841]***
Baseline controls	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Elasticity of herding on:						
Cat. 1	-0.054	-0.061	-0.041	-0.047	-0.047	-0.060
Cat. 2	-0.025	-0.052	-0.020	-0.028	0.051	0.024
Cat. 3	0.23	0.28	0.21	0.25	0.29	0.41
Share of Cat.2	0.121	0.123	0.101	0.103	0.083	0.084
Share of Cat.3	0.099	0.099	0.085	0.084	0.057	0.058
Equality of Coefficients (p-value)	0.000	0.000	0.001	0.003	0.001	0.000
Pseudo R-squared	0.28	0.31	0.29	0.34	0.27	0.31
Number of Obs.	7,036	6,239	7,036	6,239	7,036	6,239
Number of Countries	211	211	211	211	211	211
Number of Ethnic Groups	1,104	985	1,104	985	1,104	985

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables take the value of 0 if no conflicts occurred according to the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016, 1 if at least one conflict occurred but those between the same actors never spread over 3 months, and 2 if conflicts between the same actors spread over more than 3 months. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

them was ‘prolonged,’ where ‘prolonged’ is defined as follows. For all groups that experienced a conflict, we calculate the longest duration among all conflict episodes (i.e., sequence of conflict events between the same pair of actors). We then use the median length as our cutoff to divide groups who experienced conflict into those who experienced a ‘prolonged’ conflict or only non-prolonged (i.e., temporary) conflicts. In the UCDP data, this cutoff is three months, and in the *ACLED* data, it is 10 months. The estimates, reported in Table 5 for the world and Appendix Table A35 for Africa, show that the association between herding and conflict is stronger for prolonged conflicts than for temporary ones.

c. *Text Analysis of Revenge-Taking in ACLED*

To test more directly for revenge-taking motives, we leverage the description of conflict events provided in the *ACLED* data. An example of a description is:

Clashes between military and pastoralist youth group in Akot when 1 youth was killed by military forces for looting in the area. Youth group then killed 6 policeman and 4 soldiers in retaliation. 2 civilians also killed.

We investigate whether traditional herding is more strongly associated with the frequency of conflict events that are described as involving revenge-taking actions. If the culture of honor hypothesis explains the observed relationship between ancestral herding and conflict, then we should see that this is primarily driven by herding and its association with revenge-motivated conflicts rather than non-revenge-motivated conflicts.

To classify conflict events as being revenge-related, we first retrieve from the knowledge representation project *ConceptNet* the list of top-50 terms relevant for each of the following seed words: *punish*, *retaliation*, *revenge*.²³ We then classify a conflict as revenge-related if at least one of the terms in this bag-of-words appears in the textual description. In the example above, because the word *retaliation* appears, it would be classified as a revenge-motivated conflict event. Among the 127,668 events in Africa in the *ACLED* database for which descriptions are available, 1,936 events report a revenge-taking action.²⁴ This is certainly an underestimate because, for many events, the descriptions are quite sparse. However, we expect that the events described as being retaliatory are almost certainly revenge-motivated conflicts.

We jointly estimate the effect of herding on both revenge-motivated and non-revenge-motivated conflict events within a multinomial logit framework. We construct a categorical variable, $Incidence_{e,c}$, which, for each group and country, takes the value of 0 if no conflicts occurred on the territory from 1997–2016, the value of 1 if at least one conflict occurred but none of them were described as revenge-motivated, and 2 if at least one event occurred that was described as revenge-motivated. We then use a multinomial logistic regression to investigate the effect of traditional economic dependence on herding on this variable.

Our hypothesis is *not* simply that herding is linked to the occurrence of conflicts that feature a revenge-taking motive (such a correlation could effectively be mechanical and driven by the fact

²³See Appendix B for the complete lists of words.

²⁴Of these, 632 are civil conflicts, and 1,205 are non-civil conflicts (of which 919 are “localized” ones). The other 99 conflicts involve international actors, of which 7 are interstate conflicts. 9,472 of the events do not provide descriptions.

Table 6: Traditional herding and contemporary conflict in Africa: Multinomial logit analysis

	<i>Dependent variable: Incidence of conflict events (multinomial)</i>							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cat. 1 (omitted): no incidence								
Cat. 2: no revenge-taking								
Dependence on herding	0.204 (0.567) [0.530]	0.794 (0.729) [0.685]	0.661 (0.494) [0.541]	0.886 (0.653) [0.700]	0.364 (0.537) [0.543]	0.961 (0.694) [0.583]*	0.303 (0.511) [0.496]	0.618 (0.655) [0.513]
Cat. 3: revenge-taking								
Dependence on herding	2.590 (0.720)*** [0.625]***	3.159 (0.952)*** [0.828]***	3.346 (0.807)*** [0.731]***	3.672 (1.063)*** [1.022]***	2.882 (0.748)*** [0.640]***	3.211 (0.995)*** [0.814]***	2.647 (0.736)*** [0.601]***	2.651 (0.964)*** [0.773]***
Baseline controls	No	Yes	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Elasticity of herding on:								
Cat. 1	-0.096	-0.16	-0.11	-0.14	-0.090	-0.14	-0.070	-0.090
Cat. 2	-0.058	-0.012	0.0070	0.025	-0.024	0.039	-0.014	0.026
Cat. 3	0.38	0.43	0.49	0.54	0.43	0.46	0.40	0.39
Share of Cat.2	0.393	0.399	0.359	0.364	0.344	0.347	0.281	0.284
Share of Cat.3	0.125	0.125	0.074	0.074	0.090	0.091	0.073	0.074
Equality of Coefficients (p-value)	0.000	0.002	0.000	0.003	0.000	0.007	0.000	0.019
Pseudo R-squared	0.14	0.19	0.16	0.20	0.14	0.19	0.14	0.20
Number of Obs.	2,285	2,133	2,285	2,133	2,285	2,133	2,285	2,133
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	497	449	497	449	497	449	497	449

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables take the value of 0 if no conflicts occurred according to the *Armed Conflict Location and Event Data Project* (ACLED) during 1997 to 2016, 1 if at least one conflict occurred but none of them were described as revenge-motivated, and 2 if at least one event that occurred during this time was described as revenge-motivated. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

that herding societies have a higher base rate of conflicts). Rather, our more specific hypothesis is that the link between herding and conflict is stronger for revenge conflicts than for non-revenge ones.

The estimated coefficients for the latent variables and the elasticities of herding on each of the conflict categories, with “no conflict” as the omitted category, are reported in Table 6. We find that traditional dependence on herding strongly increases the incidence of revenge-related conflicts, but has almost no effect on non-revenge conflicts (coefficients of 2.590 ($p < 0.01$) and 0.204 ($p = 0.72$), respectively). As noted, the effect of interest is not whether herding is linked to revenge-taking conflicts, but whether the coefficient for revenge-related conflicts is substantially larger than for non-revenge conflicts, a difference that is highly statistically significant ($p < 0.01$). Moreover, the estimated coefficient is effectively zero for non-revenge conflicts.

We check the robustness of our findings to alternative classification methods that leverage large language models (LLMs). We first use a zero-shot prompting strategy, where we ask the model to

classify each event as revenge-related or not based on its description.²⁵ We also adopt a synthetic-text approach following Halterman (2025), which combines LLM-generated synthetic text with supervised machine learning to classify events as revenge-related or not.²⁶ The results, reported in Appendix Tables A36–A38, are in line with the bag-of-words approach. Traditional herding is linked to a higher incidence of conflicts primarily (if not exclusively) because it increases the probability of conflicts with a revenge-taking motive.

Contemporary herding and revenge-motivated conflicts. Our research hypothesis of cultural persistence does not require that traditionally pastoral societies are still herders today. However, in the spirit of the ‘culture of honor’ theory, one may ask whether contemporary herders are more likely to be involved in conflicts with a revenge-taking motive. At the language-group level, we do not know to what degree the respective group depends on herding today. However, the descriptions of conflict events in the *ACLED* data sometimes allow an inference about the subsistence mode of the parties involved. We retrieve from *ConceptNet* the top-50 list of terms relevant for each seed of *herding* and *herder* and then classify a conflict event as having herders involved if at least one term in this bag-of-words appears in the description. We then analyze the relationship between the mention of herders and revenge-taking actions across approximately 130,000 conflict events.²⁷

The dependent variable is an indicator that equals one if the conflict event description mentions a revenge-related term. The independent variable is an indicator variable that equals one if the description includes a herding-related term. Consistent with a ‘culture of honor’ (and herding) being particularly relevant for revenge-taking, Appendix Table A39 shows that if a description mentions herding, the event is more likely to be described as an act of revenge or retaliation. Interestingly, this appears to be driven by non-civil and localized conflicts. The estimate looking at civil conflicts specifically is positive but smaller in magnitude and underpowered. This is true both in the raw data and if we control for description length, country fixed effects, year fixed effects, and information source fixed effects. We obtain similar results when we use GPT and

²⁵Each event was individually processed using a specifically crafted prompt (reported in Appendix B) that was tested and refined to ensure that it would only label an event as true when the reference to revenge was explicit and direct. We run our prompt, setting the temperature to zero.

²⁶We use GPT to generate 500 short event descriptions designed to mimic ACLED event descriptions. We then train a logistic regression classifier on these synthetic texts using both TF-IDF and sentence embeddings as features and apply the resulting classifier to all ACLED events. Importantly, the LLM is used only to generate the synthetic training corpus. Thus, no real ACLED event is labeled by GPT or by a human in the exercise.

²⁷Among all events, 2,979 have herding-related words in their descriptions.

the two previously described LLM-based strategies to identify revenge-motivated conflicts and conflicts involving herding activities (Appendix Table A40).²⁸

The role of omitted variables. A correlational reduced-form approach such as ours is invariably subject to concerns over potential omitted variables. Almost by construction, a study like ours can never perfectly rule out that unobserved or unmeasured differences in, for example, local institutional quality are correlated with a tradition of herding and produce a higher frequency of conflicts. However, our results on the role of retaliatory conflicts – which are entirely natural and even predicted from the perspective of a ‘culture of honor’ – are harder to explain from the perspective of economic factors such as institutions or local development. In particular, for such an unobserved factor to explain our results, it would have to be that this factor (i) is correlated with a tradition of herding; (ii) increases the probability of conflict only if there was prior conflict; (iii) does not produce conflicts without a revenge motive; (iv) causes perpetuated conflicts with a retaliatory motive after an initial conflict has taken place; and (v) is sufficiently uncorrelated with all of our covariates that the Oster- δ tests incorrectly indicate that unobservables are unlikely to drive the results. We cannot rule out that such a factor exists, but we leave it to the reader to assess the plausibility of this combination of patterns.

Potential deterrence effects. A game-theoretic, rational analysis of conflict might predict that societies with strong revenge norms do not see more conflict due to a deterrence effect: potential aggressors realize that traditional herding societies are more likely to retaliate aggressively, increasing the cost of attacking. Consistent with this, in the seminal experiments of Cohen, Nisbett, Bowdle and Schwarz (1996) and Nisbett and Cohen (1996), participants from herding backgrounds (the Southern U.S.) were more polite and deferential at baseline (relative to participants from non-herding backgrounds) but became more aggressive and hostile when insulted. This pattern suggests that a culture of honor may have benefits. By cultivating a reputation for swift and disproportionate retaliation, herding societies signal high costs to would-be aggressors, potentially reducing the incidence of initial attacks.

The effectiveness of such deterrence likely depends critically on several factors. First, it requires common knowledge of the herding society’s revenge norms and a shared belief that these norms will be enacted. Second, it assumes rational actors who can accurately assess costs

²⁸Herding correlates positively with civil conflicts when we use these alternative text analysis methods.

and benefits and update their beliefs based on observable patterns. Third, and perhaps most importantly, it presumes that conflicts arise from deliberate cost-benefit calculations rather than accidents, misperceptions, emotions, or actions by agents with misaligned incentives. If any of these conditions is not satisfied, even temporarily, then we may observe off-equilibrium ‘trembles’ that trigger culture-of-honor-related revenge conflicts.

The estimates reported in this section provide empirical evidence relevant to this issue. Indeed, herding societies do not have more conflict at baseline. Rather, they only have more conflict after a prior conflict event takes place. While in a theoretical world characterized by a subgame-perfect equilibrium, such initial conflict should not arise, the real world is much messier. Initial attacks occur because a situation is misunderstood, because emotions drive decisions, or because a state- or non-state-actor views it as in their short-run interest to start a conflict even though it isn’t in the group’s long-run interest, and so on. Once such actions (which in theory should be off-equilibrium) have occurred, the higher propensity for retaliatory conflicts among herders produces a correlation between herding and conflict.

The importance of the institutional environment. A key prediction of Nisbett and Cohen (1996) and also Grosjean (2014) is that cultures of honor are more likely to emerge and persist in weak institutional environments where formal legal systems cannot reliably protect property or punish aggressors. In such contexts, cultivating a reputation for violent retaliation serves as a substitute for state-provided security. Thus, the relationship between a tradition of herding and conflict should be strongest where state capacity is weakest, a prediction that can be considered empirically.

We examine whether the relationship between ancestral herding and conflict is moderated by institutional quality, proxied by a country’s measured rule of law, by splitting our sample into two groups: those in countries above and those in countries below the median value of the rule of law.²⁹ The estimates, which are reported in Table 7, show that the estimated relationship between historical dependence on herding and conflict is driven by countries with lower quality

²⁹We use the World Bank’s measure of Rule of Law, measured in 1996, the earliest year for which data are available. The variable, which is from the Worldwide Governance Indicators (WGI) dataset, captures perceptions of the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence. The variable is measured on a scale from -2.5 to +2.5, with higher values indicating a stronger rule of law.

of institutions.³⁰ This pattern is consistent with the culture of honor theory as initially posited by Nisbett and Cohen (1996).

4.2. *Checking for the Role of Transhumance*

A potential alternative to the ‘culture of honor’ hypothesis in explaining the estimates is the role of transhumance – i.e., seasonal migration – among some pastoralists. As recently shown, transhumance can lead to conflict in neighboring territories of agricultural ethnic groups if the timing of seasonal migrations is disturbed by adverse rainfall shocks (McGuirk and Nunn, 2025).

Conceptually, the findings we document here and those found in McGuirk and Nunn are expected to be orthogonal. Our estimates show more conflict *within* the territories of traditional pastoral groups. By contrast, McGuirk and Nunn show more conflict *outside* of the territories of traditional pastoral groups. To more formally assess whether this alternative channel can explain our findings, we disaggregate herding populations into two groups: those who are mobile (i.e., transhumant) and those who are not. We then re-estimate equation (1) using two herding variables, one that captures transhumant herding and another that captures non-transhumant herding.³¹

The estimates for both types of herding are reported in Table 8 for the global *UCDP* sample and Appendix Table A42 for the Africa *ACLED* sample. We find that both transhumant and non-transhumant herding are strongly correlated with conflict, suggesting that our findings are not solely explained by transhumance. The estimated coefficient for transhumant herding is often larger than for non-transhumant herding. However, in none of the specifications is the difference statistically significant at conventional levels.³²

We also pursue a complementary strategy where we directly control for potential spillovers that could arise from transhumant pastoralism. McGuirk and Nunn (2025) document how transhumant pastoralism causes conflict in the nearby territories that are suitable for agriculture when adverse rainfall shocks during the wet season cause herds to migrate before farmers are

³⁰The pattern is very similar if we split countries into four groups. The relationship is driven by countries in the low and low-middle category (Appendix Table A41).

³¹The first variable is equal to the traditional herding index for groups that are identified as being migratory, and the second is the traditional herding index for groups that are not migratory. The classification of whether a group is migratory or not follows McGuirk and Nunn (2025) and is detailed in Appendix B.

³²The coefficient on non-transhumant herding loses statistical significance in the specification for civil conflicts in the global sample. In the African sample, it loses significance when we use the PPML specification for all, civil and non-civil conflicts (Appendix Table A42).

Table 7: Traditional herding and contemporary conflict by rule of law

	<i>Dependent variable: Number of all conflict events</i>					
	By rule of law					
	Below Median			Above Median		
	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	1.888 (0.410)*** [0.346]***	1.736 (0.395)*** [0.327]***	4.547 (1.139)*** [0.721]***	-0.294 (0.326) [0.539]	-0.046 (0.273) [0.372]	1.024 (1.750) [1.995]
<i>Historical ethnicity-level covariates:</i>						
Settlement complexity		-0.035 (0.031) [0.034]	0.364 (0.144)** [0.090]***		0.021 (0.017) [0.023]	-0.013 (0.166) [0.173]
Jurisdictional hierarchy		0.056 (0.038) [0.035]	0.157 (0.221) [0.346]		0.059 (0.032)* [0.061]	0.323 (0.139)** [0.142]**
Historical latitude (abs.)		7.046 (14.837) [10.802]	40.184 (19.250)** [18.714]**		7.510 (5.368) [10.361]	100.389 (79.155) [79.519]
<i>Contemporary language-level covariates:</i>						
Population density (ln)		0.237 (0.029)*** [0.041]***	0.637 (0.122)*** [0.116]***		0.062 (0.013)*** [0.024]**	0.662 (0.225)*** [0.165]***
Ethnic fractionalization		-0.289 (0.292) [0.439]	-5.464 (1.647)*** [2.597]**		-0.034 (0.227) [0.252]	0.752 (1.157) [0.694]
Ethnic polarization		0.515 (1.400) [1.542]	6.832 (4.190) [5.862]		0.419 (0.683) [0.814]	-0.148 (3.456) [4.152]
Nighttime lights		-0.011 (0.011) [0.017]	-0.152 (0.033)*** [0.034]***		-0.019 (0.009)** [0.010]*	-0.158 (0.063)** [0.051]***
Share of land for herding		0.372 (0.171)** [0.217]*	2.047 (0.562)*** [0.570]***		-0.177 (0.157) [0.175]	0.903 (0.879) [0.899]
Ruggedness		-0.118 (0.278) [0.356]	-4.114 (1.235)*** [0.812]***		-0.822 (0.216)*** [0.334]**	-1.162 (0.798) [1.016]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.18	0.16		-0.034	-0.0056	
Oster δ		3.01			-2.41	
Mean of dependent var	0.88	0.91	27.3	0.43	0.41	15.6
SD of dependent var	1.62	1.64	337.1	1.26	1.24	166.7
Adj. R-squared	0.25	0.30		0.31	0.32	
Pseudo R-squared			0.69			0.53
Number of Obs.	3,097	2,677	2,537	3,818	3,436	3,002
Number of Countries	59	59	53	89	82	51
Number of Ethnic Groups	617	541	539	689	631	576

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table 8: Traditional herding and contemporary conflict globally: Transhumant and non-transhumant herding

	Dependent variable: Number of conflict events								
	All conflicts			Civil conflicts			Non-civil conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Transhumant herding	1.368 (0.364)*** [0.380]***	1.254 (0.382)*** [0.382]***	3.332 (1.379)** [1.131]***	1.139 (0.336)*** [0.335]***	1.059 (0.353)*** [0.341]***	3.227 (1.540)** [1.276]**	1.135 (0.282)*** [0.301]***	1.038 (0.291)*** [0.292]***	3.110 (1.250)** [1.126]***
Non-transhumant herding	1.083 (0.376)*** [0.472]**	0.578 (0.336)* [0.417]	2.167 (1.157)* [1.170]*	0.879 (0.315)*** [0.414]**	0.429 (0.285) [0.339]	1.863 (1.323) [1.315]	0.754 (0.282)*** [0.311]**	0.537 (0.258)** [0.315]*	2.779 (1.318)** [1.430]*
<i>Historical ethnicity-level covariates:</i>									
Settlement complexity		0.007 (0.018) [0.022]	0.241 (0.184) [0.150]		0.008 (0.016) [0.019]	0.248 (0.199) [0.161]		0.002 (0.013) [0.016]	0.147 (0.124) [0.109]
Jurisdictional hierarchy		0.057 (0.024)** [0.035]	0.206 (0.165) [0.178]		0.055 (0.021)*** [0.026]**	0.154 (0.196) [0.220]		0.053 (0.020)*** [0.025]**	0.374 (0.116)*** [0.103]***
Historical latitude (abs.)		8.987 (7.066) [7.690]	79.996 (27.403)*** [27.884]***		7.063 (6.174) [6.515]	83.298 (33.326)** [34.360]**		7.069 (4.973) [5.640]	73.761 (29.086)** [31.749]**
<i>Contemporary language-level covariates:</i>									
Population density (ln)		0.123 (0.015)*** [0.033]***	0.619 (0.097)*** [0.065]***		0.107 (0.014)*** [0.032]***	0.632 (0.111)*** [0.071]***		0.070 (0.010)*** [0.020]***	0.634 (0.093)*** [0.082]***
Ethnic fractionalization		-0.218 (0.171) [0.253]	-1.964 (1.276) [1.482]		-0.297 (0.156)* [0.224]	-1.868 (1.453) [1.725]		0.037 (0.126) [0.201]	-1.831 (0.825)** [0.941]*
Ethnic polarization		0.361 (0.676) [0.861]	5.275 (3.796) [5.008]		0.716 (0.588) [0.759]	9.425 (3.959)** [5.477]*		-0.760 (0.490) [0.684]	-5.706 (3.291)* [4.352]
Nighttime lights		-0.018 (0.007)*** [0.010]*	-0.156 (0.036)*** [0.029]***		-0.020 (0.006)*** [0.009]**	-0.198 (0.045)*** [0.036]***		-0.008 (0.005) [0.006]	-0.087 (0.023)*** [0.019]**
Share of land for herding		0.173 (0.122) [0.157]	1.506 (0.530)*** [0.439]***		0.110 (0.109) [0.139]	1.537 (0.607)** [0.525]***		0.084 (0.089) [0.101]	1.550 (0.472)*** [0.622]**
Ruggedness		-0.592 (0.194)*** [0.253]**	-2.099 (0.804)*** [0.932]**		-0.536 (0.176)*** [0.236]**	-2.112 (0.834)** [1.002]**		-0.518 (0.145)*** [0.202]**	-1.786 (0.629)*** [0.602]***
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Diff(Transhumant vs. non-transhumant) p-value	0.513	0.147	0.374	0.513	0.139	0.343	0.239	0.149	0.745
Mean of dependent var	0.63	0.63	20.9	0.51	0.51	16.7	0.35	0.36	5.15
SD of dependent var	1.46	1.46	258.7	1.31	1.31	237.5	1.05	1.06	51.1
Adj. R-squared	0.29	0.32		0.29	0.32		0.23	0.26	
Pseudo R-squared			0.59			0.62			0.55
Number of Obs.	6,502	6,239	5,558	6,502	6,239	5,327	6,502	6,239	5,294
Number of Countries	211	211	106	211	211	100	211	211	84
Number of Ethnic Groups	1,017	985	934	1,017	985	879	1,017	985	906

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

finished farming the land. This can result in herds damaging crops, which causes disputes that often escalate into conflict. To capture these and similar spillover mechanisms, we control for variables that capture the prevalence of transhumant pastoralism of all (contiguous) neighbors of a language group. We also allow the effect of the variables to differ depending on the average agricultural suitability of the language group itself.³³

The estimates are reported in Table 9 for the UCDP global sample, and Appendix Table A43 for the ACLED Africa sample. The top rows report the baseline estimates of a group's ancestral dependence on herding for comparison. The next rows report the same estimates from a specification that controls for the transhumant pastoralism of neighbors, the group's own dependence on agriculture, and the interaction between the two variables. The transhumant pastoralism measures are taken from McGuirk and Nunn (2025).³⁴ We report results using various strategies for measuring the transhumant pastoralism of a group's neighbors: the equally weighted mean, maximum value, an indicator if at least one neighbor has a transhumant pastoralism measure above 0.15, and the same measure but using either 0.35, 0.55, or 0.75 as cutoffs. In total, we report estimates using six neighbor transhumant pastoralism measures. In all specifications, our coefficients of interest remain positive, of similar magnitude to the baseline estimate, and almost always statistically significant.³⁵

The final check we perform is to disaggregate conflicts between those occurring during the growing and non-growing seasons. The mechanism examined in McGuirk and Nunn (2025) is one where transhumant pastoral conflict occurs primarily in the growing season when potential grazing land is not yet fallow and is still being used to cultivate crops. We identify the growing season as April to September north of the equator and October to March south of the equator and estimate equation (1) separately for conflicts occurring during the two seasons. OLS and PPML estimates are reported in Appendix Tables A44 and A45 globally, and Appendix Tables

³³McGuirk and Nunn (2025) predict an increasing conflict in a location (in our case, a language group) if the location relies on agriculture and has neighbors who are transhumant pastoralists. Therefore, our specification includes the own suitability of a language group's polygon for agriculture, measures of the transhumant pastoralism of neighboring language groups, and the interaction between the two.

³⁴Transhumant groups are those that, according to the *Ethnographic Atlas*, are "nomadic or fully migratory" or "semi-nomadic" (McGuirk and Nunn, 2025). Transhumant pastoralism is defined as the interaction between pastoralism and transhumance.

³⁵We lose significance only on one PPML specification on non-civil conflicts for the global sample. In the Africa sample, we lose significance for one OLS specification for the case of non-civil conflicts, and one OLS and one PPML specification for the case of localized conflicts. When we cluster the standard errors by country, the coefficients remain significant in the African sample, but fall below conventional significance levels in the global sample, for the specification that defines neighbors' transhumant pastoralism as the maximum value among neighbors.

Table 9: Coefficients on herding when controlling for different measures of transhumant pastoralism (THP)

	<i>Dependent variable: Number of conflict events</i>								
	All conflicts			Civil conflicts			Non-civil conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
No THP controls	0.949 (0.297)*** [0.391]**	0.912 (0.268)*** [0.337]***	2.883 (1.096)*** [1.036]***	0.773 (0.257)*** [0.329]**	0.740 (0.234)*** [0.284]***	2.875 (1.327)** [1.200]**	0.735 (0.226)*** [0.295]**	0.785 (0.206)*** [0.255]***	2.899 (1.185)** [1.236]**
Avg. neighbor THP (cont.)	0.951 (0.316)*** [0.440]**	0.835 (0.280)*** [0.360]**	3.094 (1.335)** [1.555]**	0.759 (0.272)*** [0.375]**	0.652 (0.241)*** [0.309]**	3.239 (1.612)** [1.824]*	0.722 (0.238)*** [0.319]**	0.727 (0.216)*** [0.267]***	2.650 (1.165)** [1.384]*
Max. neighbor THP (cont.)	0.627 (0.304)** [0.431]	0.540 (0.275)** [0.354]	2.641 (1.427)* [1.747]	0.481 (0.260)* [0.366]	0.404 (0.235)* [0.301]	2.784 (1.617)* [1.998]	0.494 (0.230)** [0.302]	0.521 (0.212)** [0.255]**	2.212 (1.382) [1.636]
Any neighbor THP > 0.15	0.835 (0.296)*** [0.425]*	0.751 (0.269)*** [0.345]**	3.189 (1.388)** [1.637]*	0.675 (0.252)*** [0.368]*	0.596 (0.232)** [0.295]**	3.136 (1.526)** [1.853]*	0.617 (0.230)*** [0.303]**	0.664 (0.210)*** [0.264]**	3.287 (1.768)* [1.924]*
Any neighbor THP > 0.35	0.644 (0.296)** [0.433]	0.616 (0.266)** [0.346]*	2.446 (1.149)** [1.337]*	0.494 (0.254)* [0.374]	0.462 (0.228)** [0.290]	2.386 (1.222)* [1.467]	0.508 (0.220)** [0.290]*	0.565 (0.204)*** [0.247]**	2.887 (1.596)* [1.796]
Any neighbor THP > 0.55	0.707 (0.300)** [0.393]*	0.668 (0.274)** [0.341]*	2.620 (1.160)** [1.206]**	0.551 (0.257)** [0.329]*	0.519 (0.237)** [0.289]*	2.463 (1.368)* [1.414]*	0.590 (0.230)** [0.294]**	0.632 (0.212)*** [0.253]**	2.787 (1.255)** [1.348]**
Any neighbor THP > 0.75	0.793 (0.293)*** [0.358]**	0.747 (0.277)*** [0.310]**	2.320 (1.061)** [1.000]**	0.634 (0.253)** [0.301]**	0.587 (0.240)** [0.262]**	2.223 (1.251)* [1.111]**	0.654 (0.222)*** [0.274]**	0.685 (0.212)*** [0.243]***	2.285 (1.049)** [1.169]*
Baseline Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	0.63	0.63	20.9	0.51	0.51	16.7	0.35	0.36	5.15
SD of dependent var	1.45	1.46	258.7	1.31	1.31	237.5	1.05	1.06	51.1
Number of Obs.	7,036	6,239	5,558	7,036	6,239	5,327	7,036	6,239	5,294
Number of Countries	211	211	106	211	211	100	211	211	84
Number of Ethnic Groups	1,104	985	934	1,104	985	879	1,104	985	906

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. The model controls for agricultural suitability within the polygon, the level of THP in neighboring polygons, the interaction between these two factors, and other baseline controls. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

A46 and A47 for Africa. We find that the relationship between a tradition of herding and conflict is found for both seasons and not just during the growing season. This pattern is consistent with a culture-of-honor mechanism, which should be active in both seasons. It is less consistent with the effects arising due to irregular migrations of transhumant pastoralists during the growing season.

5. The Culture of Honor Mechanism II: Data on Cultural Traits

Thus far, our results suggest that herding is linked to conflict primarily due to revenge-taking motives. We now provide additional evidence for this by examining whether descendants of traditional herders exhibit cultural traits and proclivities related to punishment and revenge-taking. We do this by first examining contemporary survey data and then analyzing historical folklore and ethnographic data.

5.1. Traditional Herding and a Psychology of Punishment: Global Survey Evidence

Data. To present additional direct evidence on the link between a tradition of herding and the desire to seek revenge, we leverage self-reports of the importance of punishment and revenge in survey data from the *Global Preferences Survey (GPS)*, a global dataset on the economic preferences of representative samples from 76 countries. The generally high quality of the *GPS* data has been confirmed by various studies that have linked responses to the *GPS* questions to various economic and social behaviors, both at the individual and at the country level (e.g., Falk et al., 2018; Enke, 2019; Becker, Enke and Falk, 2020; Sunde, Dohmen, Enke, Falk, Huffman and Meyerheim, 2022).

The survey measures attitudes toward punishment and revenge-taking using three questions:

1. How willing are you to punish someone who treats you unfairly, even if there may be costs for you? (0–10)
2. How willing are you to punish someone who treats others unfairly, even if there may be costs for you? (0–10)
3. How much do you agree with the following statement: “If I am treated very unjustly, I will take revenge at the first occasion, even if there is a cost to do so”? (0–10)

We view this set of questions as well-suited for our purposes because they directly capture the key psychological mechanism that underlies the culture of honor hypothesis: revenge-taking and

punishing behavior that is perceived as unfair. In our analysis, we use the summary measure constructed by Falk et al. (2018) as a weighted average of the three survey questions, normalized to have mean 0 and standard deviation 1. We also consider each survey question separately. At the individual level, the correlations among the three survey items range from 0.45 to 0.71.

Linkage to Historical Herding Data. Our analysis requires that we link individual-level responses in the GPS to historical ethnic groups to measure the extent to which an individual’s ancestors practiced herding. Because the GPS does not contain information on respondents’ ethnic or linguistic backgrounds, we use geographic subnational region identifiers in the GPS, which are usually states or provinces. We follow Giuliano and Nunn (2018) and create a population-weighted measure of the ancestral reliance on herding of the inhabitants of any country or district.³⁶ For nearly all of the 73,949 respondents from the GPS, living in 951 subnational regions and 75 countries, we are able to assign them a regional-level measure of the average ancestral herding index. For a subset of the observations (9,679 individuals from 12 small countries) we are only able to link respondents to the ancestral herding measure of their country.

Estimation Strategy and Covariates. The individual-level within-country estimates connect individuals to ancestral herding using the subnational region in which they live. Specifically, we estimate the following equation:

$$y_{j,r,c} = \alpha_c + \beta \text{Herding}_r + X_j \Gamma + X_r^H \Omega + \varepsilon_{j,r,c}, \quad (3)$$

where j indexes individuals in the GPS survey, r indexes their subnational region of residence, and c the country this region lies within. α_c denotes country fixed effects. $y_{j,r,c}$ is one of our measures of a psychology of punishment (either an aggregate summary measure or one of the underlying components) for individual j residing in subnational region r . Herding_r is subnational region r ’s average ancestral dependence on herding. X_j denotes a vector of individual-level covariates, including age, age squared, and the gender of the respondent.³⁷ X_r^H is our vector of historical covariates, measured at the regional level: average historical

³⁶See Appendix C for full details.

³⁷The inclusion of a larger set of controls, including education, income, religion, and cognitive skills, does not change the results, with the exception of the willingness to take revenge, for which the coefficient on historical herding remains of similar magnitude but loses statistical significance at conventional levels (Appendix Table A48). We also lose significance for historical herding, for the summary index of negative reciprocity, when standard errors are clustered at the country level.

settlement complexity, jurisdictional hierarchy, and (absolute) historical latitude. To account for the potential non-independence of the observations, we report standard errors clustered at the level of 951 subnational regions (districts), which is the level at which the herding index varies. To further document the robustness of the statistical significance of our findings and to account for within-country non-independence, we also report standard errors clustered at the country level.

Results. The estimates of equation (3) are reported in Table 10. We report estimates for the summary measure, as well as each component, with and without our baseline set of individual and regional historical covariates. All specifications include country fixed effects. We find a positive relationship between a tradition of herding and a psychology of revenge-taking, measured by our aggregate index (columns 1 and 2). The magnitudes of the estimates are modest but meaningful. To see this, take the standardized beta coefficient for column 2, which is reported in the bottom panel of the table: a one standard deviation increase in reliance on herding is associated with an increase in the summary measure by 7.3% of a standard deviation.

When we look at the components of the index, we find that the relationship with herding is slightly stronger for the questions that ask about a willingness to “punish” others when treated “unfairly” than for the question that asks about “revenge” when treated “unjustly” (columns 3–8). The beta coefficient for the revenge variable is smaller in magnitude and becomes insignificant at conventional levels with the country-level clustering (columns 7 and 8).³⁸

We also estimate the relationship between herding and the other measures of preferences contained in the *GPS* (altruism, trust, patience, risk aversion, and positive reciprocity). The results, reported in Appendix Table A49, show no systematic relationship between the culture of honor and most other economic preferences. The only exception is the belief that others have “only the best intentions” (referred to as ‘trust’ in the *GPS*), which is positively correlated with a tradition of herding.

Even though trust could be a reflection of a culture-of-honor effect, we examine the extent to which our results are robust to including trust as an additional covariate. We find that the

³⁸Figure A5 reports binned scatterplots of the summary measure, as well as each component as a function of historical dependence on herding, controlling for country fixed effects and the other covariates. As shown, the relationships appear quite general and not driven by a small number of influential observations or outliers.

Table 10: The historical origins of a psychology of punishment: Individual-level analysis (GPS)

	<i>Dependent variable:</i>							
	Negative reciprocity components, 0–10 index							
	Negative reciprocity		Punish if ... treated unfairly				Willingness to	
	Summary index		Self		Others		take revenge	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	0.453 (0.185)** [0.246]*	0.493 (0.216)** [0.271]*	1.337 (0.520)** [0.640]**	1.448 (0.628)** [0.732]*	1.366 (0.483)** [0.677]**	1.380 (0.539)** [0.689]**	0.813 (0.492)* [0.605]	0.982 (0.561)* [0.691]
<i>Historical ethnicity-level covariates:</i>								
Settlement complexity		0.014 (0.019) [0.019]		0.039 (0.057) [0.058]		0.018 (0.049) [0.045]		0.047 (0.050) [0.052]
Jurisdictional hierarchy		0.022 (0.024) [0.031]		0.064 (0.067) [0.084]		0.024 (0.067) [0.082]		0.076 (0.063) [0.077]
Historical latitude (abs.)		-0.004 (0.004) [0.006]		-0.009 (0.011) [0.014]		-0.018 (0.012) [0.023]		-0.005 (0.011) [0.012]
<i>Individual-level covariates:</i>								
Age		-0.004 (0.001)** [0.002]**		-0.006 (0.004) [0.006]		-0.003 (0.004) [0.005]		-0.022 (0.004)** [0.005]**
Age squared		-0.000 (0.000)** [0.000]**		-0.000 (0.000)** [0.000]**		-0.000 (0.000)** [0.000]**		0.000 (0.000) [0.000]
Female indicator		-0.159 (0.009)** [0.012]**		-0.425 (0.028)** [0.038]**		-0.376 (0.025)** [0.030]**		-0.415 (0.026)** [0.036]**
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.067	0.073	0.065	0.070	0.066	0.067	0.040	0.048
Oster δ		233.5		183.8		31.4		-36.5
Mean of dependent var	-0.0031	-0.0031	4.20	4.20	4.35	4.35	3.63	3.63
SD of dependent var	1.00	1.00	3.04	3.04	3.04	3.04	3.00	3.00
Adj. R-squared	0.071	0.095	0.050	0.070	0.061	0.078	0.080	0.096
Number of Obs.	74,182	73,949	74,264	74,030	74,252	74,018	75,024	74,781
Number of Countries	75	75	75	75	75	75	75	75
Number of Subnational Regions	951	951	951	951	951	951	951	951

Note. The unit of observation is an individual from the *Global Preferences Survey* (GPS). The dependent variable in columns 1 and 2 is the negative reciprocity index from the GPS, constructed as the weighted average of three self-assessments in GPS that measure people's propensity for different types of punishment and revenge taking. The dependent variables in columns 3–8 correspond to the individual survey questions. Coefficients are reported with standard errors in parentheses clustered at the subnational region level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

estimated effect of herding remains similar with this added control (Appendix Table A50).³⁹

5.2. Historical Evidence from Folklore and Ethnography

As our final step, we provide evidence that, in the past, herding societies were more likely to have developed a culture of honor. To this end, we rely on two data sources that quantify the cultural

³⁹While the magnitudes are very similar, the estimates are slightly less precise in general, and we lose significance in column 4 and also in column 1 when we cluster standard errors at the country level.

traits of historical societies: (i) folklore data and (ii) ethnographic information from the *Standard Cross-Cultural Sample*.

Historical folklore data. We follow Michalopoulos and Xue (2021) in quantifying ethnic groups' historical cultural beliefs and practices using textual data on folklore, which is the collection of traditional beliefs, customs, and stories of a community, often in the form of oral traditions such as tales, proverbs, and jokes, that get passed from one generation to the next by word of mouth. The anthropologist and folklorist Yuri Berezkin assembled a dataset that codes the presence of 2,564 motifs, each of which is given by a short text that summarizes the events of the folktale.⁴⁰ Given that folklorists are interested in collecting stories that are untouched by modernization, this catalog should be thought of as capturing pre-industrial societies' culture. Based on Berezkin's catalog of motifs, Michalopoulos and Xue (2021) use text analyses to construct a dataset that codes the presence of a large number of economic, psychological, and cultural concepts in a society's oral tradition. In these analyses, a concept is said to appear in a motif if either the seed word itself or one of the 50 most closely related terms, according to *ConceptNet*, is mentioned in the motif. The data contain many concepts that are related to the culture of honor hypothesis. Michalopoulos and Xue (2019) study the association between herding and 'anger' and 'retaliation.' Following this logic, we design a bag-of-words that proxies for the salience of a culture of honor in folklore. To discipline our construction of a bag-of-words, we select seed words that follow from Nisbett and Cohen's (1996) discussion of the culture of honor in the introduction of their book. These fall into two general categories. One is using violence, including to defend one's honor or against attacks by others. The other is the notion of punishing or seeking revenge after you or your loved ones have been wronged by others. The seed words that follow from these are:

1. Violence/deterrence concepts: violence, perpetrator, strength, toughness, predation, predator, aggressiveness, affront, deterrence, defend, mayhem, guard
2. Punishment/revenge concepts: punish, retaliation, revenge

Following the methodology proposed by Michalopoulos and Xue (2021), for each of these seed words, we retrieve the top-50 list of related terms from *ConceptNet*. We then select the concepts from the folklore catalogue developed by Michalopoulos and Xue that appear in the top-50 list of

⁴⁰Details on the dataset and procedure are provided in Appendix D.

our seed words.⁴¹ For each concept, we generate a binary indicator that equals one if the concept appears in an ethnic group’s folklore and zero otherwise. We then average across all concepts within a given domain (violence/deterrence and punishment/revenge) to arrive at a summary measure that captures the fraction of concepts in the domain present in a society’s folklore. We also compute an overall summary measure of a culture of honor by taking the average across all concepts. Thus, our variables capture the average probability that the culture-of-honor-related concepts appear in a society’s folklore. Since the probability that a given concept is mentioned in a society’s folklore will mechanically be higher in societies with a larger folklore corpus, following Michalopoulos and Xue (2021), we always control for the number of publications and the year of first publication (both expressed in log).

Results. The estimates are reported in Table 11. For each folklore variable, we show two specifications. In the first, we control for the log of the number of publications and the log of the year of first publication together with country fixed effects. In the second, we additionally control for historical ethnicity-level characteristics (settlement complexity, jurisdictional hierarchy, and historical latitude (abs.)). We report two types of standard errors: clustered at the country level (in parentheses) and clustered at the language phylum level (i.e., the broadest language family) as defined in the *Ethnographic Atlas* (in square brackets).

We find that a history of herding is associated with traditional folktales that are more likely to be about any of the culture-of-honor-related words (columns 1 and 2), violence- or-conflict-related words (columns 3 and 4) or punishment-or-revenge-related words (columns 5 and 6). The magnitudes of the estimated effects, which are similar across dependent variables, suggest that an increase in the dependence on herding from zero to one is associated with a 16–22 percentage-point increase in the average probability that a culture-of-honor concept appears in folklore. Standardized beta coefficients, reported at the bottom of the table, suggest that a one standard deviation increase in herding is associated with an increase in culture-of-honor folklore by about 12–17% of a standard deviation.

The quality of the folklore data likely differs across societies. For example, the maximum number of motifs in a society’s catalogue is 598, while the minimum is 2. To check whether this drives the results, Appendix Table A51 reports a robustness check that restricts the analysis to

⁴¹See Appendix B for the full list of words.

Table 11: Culture-of-honor related folklore in Ethnographic Atlas societies

	<i>Dependent variable:</i>					
	Share of words related to ... that appear in group's folklore					
	Violence/Deterrence/ Punishment/Revenge		Violence/Deterrence		Punishment/Revenge	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	0.167 (0.039)*** [0.052]***	0.194 (0.044)*** [0.054]***	0.160 (0.042)*** [0.054]***	0.190 (0.047)*** [0.055]***	0.218 (0.062)*** [0.062]***	0.223 (0.073)*** [0.076]***
ln(number of publications)	0.157 (0.011)*** [0.010]***	0.156 (0.011)*** [0.009]***	0.152 (0.011)*** [0.010]***	0.151 (0.011)*** [0.010]***	0.191 (0.019)*** [0.022]***	0.189 (0.020)*** [0.023]***
ln(year of first publication)	1.685 (0.703)** [0.552]***	1.742 (0.821)** [0.617]***	1.775 (0.724)** [0.581]***	1.866 (0.851)** [0.653]***	1.109 (1.038) [0.902]	0.940 (1.194) [1.005]
Settlement complexity		0.004 (0.005) [0.003]		0.005 (0.004) [0.003]		-0.003 (0.008) [0.007]
Jurisdictional hierarchy		-0.002 (0.006) [0.007]		-0.002 (0.006) [0.007]		-0.006 (0.011) [0.010]
Historical latitude (abs.)		0.002 (0.001)** [0.002]		0.002 (0.001)* [0.002]		0.004 (0.003) [0.002]**
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.15	0.17	0.14	0.17	0.12	0.12
Oster δ		4.85		10.2		3.54
Mean of dependent var	0.51	0.50	0.49	0.49	0.62	0.62
SD of dependent var	0.22	0.22	0.21	0.21	0.34	0.34
Adj. R-squared	0.61	0.62	0.61	0.61	0.43	0.44
Number of Obs.	1,135	1,107	1,135	1,107	1,135	1,107
Number of Countries	149	148	149	148	149	148
Number of Linguistic Phyla	115	114	115	114	115	114

Note. The unit of observation is a society from the *Ethnographic Atlas*. The dependent variable is the share of culture-of-honor-related terms tagged in a society's folklore by Michalopoulos and Xue (2021). The dependent variable in columns (1) and (2) is the share of all terms related to violence/deterrence or punishment/revenge. In columns (3) and (4), the dependent variable is the share of terms related to violence/deterrence and in columns (5) and (6), it is the share of terms related to punishment/revenge. Standard errors in parentheses are clustered at the country level. Standard errors in square brackets are clustered at the language phylum level (i.e., largest language family) as defined in the *Ethnographic Atlas*. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

societies whose folklore consists of at least 40 motifs (the 25th percentile of the distribution). The results are very similar.⁴²

A second concern is that the bag-of-words analysis ignores the context in which culture-of-

⁴²An alternative specification consists of weighting the specification by the number of motifs, expressed in log. The results, reported in Appendix Table A52, are very similar to our baseline results.

honor terms appear. For instance, “defend” or “punishment” may be expressions of a well-functioning judicial system rather than of revenge-taking. To check this, we use GPT to analyze the folklore motifs, prompting the model to determine whether any given motif contains content related to (i) violence/deterrence, and (ii) punishment/revenge.⁴³

We also test the robustness of the results using more traditional supervised machine learning methods. Specifically, we train logistic regression classifiers using two different text representation approaches: (i) TF-IDF (term frequency-inverse document frequency), which weights terms based on their importance within each motif relative to the entire corpus, and (ii) sentence embeddings, which represent each motif as a learned vector that captures semantic meaning and similarity to other motifs. The results, reported in Appendix Table A53, produce estimates in line with our main specification.⁴⁴

Evidence on Moral Views from the Standard Cross-Cultural Sample. While the analysis of folklore data shows an increased salience of punishment- and violence-related themes in the culture of herding societies, the results do not directly speak to the normative views of societies: whether people consider it morally right or wrong to engage in violent behavior, and how this depends on the social group the victim belongs to. To study this, we leverage information on the acceptability of violence in a small representative subset of independent ethnic groups from the *Ethnographic Atlas*, obtained from the *Standard Cross-Cultural Sample (SCCS)* (Murdock and White, 1969). While this dataset has the advantage of comprising a subset of groups from the *Ethnographic Atlas* chosen to be independent of each other and representative of the full sample, the sample size is relatively small (60 in total). Information is provided on the acceptability of violence towards three groups: members of the local community, members of the same society, and people of other societies. The original variables code each group as falling into one of the four categories: violence is (0) disapproved of, (1) tolerated, (2) accepted, (3) valued. We code the variables so that a higher value indicates greater acceptability of violence.

⁴³Specifically, the prompt is: “The following is a description of a motif from folklore. Title: {title}. Description: {description}. Please determine whether the motif references any of the following concepts: violence/deterrence or revenge/punishment. Note that not all motifs relate to these concepts, and some may relate to multiple concepts. Violence/Deterrence: (True/False) Revenge/Punishment: (True/False)” We set the temperature equal to zero.

⁴⁴To perform our analysis, we randomly selected 500 motifs from the full set of 2,564, and coded these motifs manually. Of these, 400 motifs were used for training, while 100 motifs served as a representative test set for validation. The models trained on the 400 motifs were used to predict all 2,564 motifs.

Table 12: The acceptability of violence in pre-industrial societies using the SCCS

	<i>Dependent variable:</i>									
	First		Acceptability of violence [0-3 index] against ...							
	principal component	Average effect size	Other society		Own society		Own local community			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dependence on herding	1.49 (0.69)**	1.46 (0.79)*	0.81 (0.35)**	0.79 (0.39)**	1.05 (0.53)*	0.90 (0.51)*	1.15 (0.57)**	1.15 (0.65)*	0.37 (0.38)	0.42 (0.41)
Settlement complexity		0.088 (0.094)		0.049 (0.046)		0.068 (0.062)		-0.00052 (0.083)		0.063 (0.047)
Jurisdictional hierarchy		-0.043 (0.19)		-0.018 (0.095)		0.34 (0.11)***		-0.22 (0.17)		-0.12 (0.11)
Historical latitude (abs.)		0.026 (0.014)*		0.014 (0.0072)**		0.0046 (0.010)		0.014 (0.012)		0.019 (0.0084)**
Continent FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.26	0.25	0.20	0.19	0.26	0.22	0.24	0.24	0.13	0.15
Mean of dependent var	0.0035	0.0035	0	0	2.33	2.33	1.37	1.37	0.43	0.43
SD of dependent var	1.38	1.38	1	1	0.97	0.97	1.16	1.16	0.70	0.70
Adj. R-squared	-0.00076	0.0034			-0.021	0.057	0.030	0.017	-0.048	0.032
Number of Obs.	60	60	60	60	60	60	60	60	60	60

Note. The unit of observation is a society from the *Standard Cross-Cultural Sample* (SCCS). The dependent variables are derived from three variables from the SCCS that measure the acceptability of violence, using the following scale: (0) disapproved, (1) tolerated, (2) accepted, and (3) valued. Columns (1) and (2) use the first principal component of the three variables. Columns (3) and (4) report average effect size estimates, following Kling et al. (2007), using the three measures. Columns (5)–(10) report estimates for each of the three variables individually. Coefficients are reported with standard errors in parentheses.

Table 12 reports OLS estimates showing the relationship between a greater dependence on herding and the acceptability of violence. Estimates without ethnicity-level covariates are reported in the odd-numbered columns, while those with the covariates are reported in the even-numbered columns. Both specifications include continent fixed effects. Estimates are shown for five dependent variables: the first principal component of the three violence measures (columns 1–2), their average effect size (columns 3–4), and the three measures separately (columns 5–10).⁴⁵

Despite the small sample size, we consistently find a positive relationship between traditional herding and the acceptability of violence. In terms of quantitative magnitude, the results suggest that a one standard deviation increase in dependence on herding increases the overall acceptability of violence by 13–26% of a standard deviation.

Interestingly, the results are largely driven by the acceptability of violence towards people who are not members of one's local community. This suggests that the 'culture of honor' logic does not exclusively (or even primarily) apply to very localized within-group violence but to broader cross-group violence. This is consistent with our results on conflicts, which show large effects on forms of out-group violence such as civil conflicts.

⁴⁵To calculate the average effect size in columns 3–4, we implement the procedure outlined in Kling, Liebman and Katz (2007) and used by Clingensmith, Khwaja and Kremer (2009).

6. Conclusions

Our study has examined the importance of norms of punishment and revenge-taking for explaining the prevalence of within-country conflicts across the world today. Given the endogeneity of revenge to conflict incidence, we focused on a determinant of revenge-taking that has been widely emphasized in the social psychology literature; namely, the importance of traditional herding activities for shaping a ‘culture of honor.’

Our analysis combined information from ethnographic sources with contemporary data on the incidence and intensity of conflicts, as well as contemporary survey data on individual values and preferences. Linking these data, we were able to test for associations between herding, revenge-taking, and conflict. We found that a tradition of herding is associated with a greater incidence and intensity of conflict and warfare, and that this is true for all types of conflicts, including civil conflicts where citizens are fighting against government agents. We found that this link between herding and conflict appears to largely reflect revenge-taking motives. Consistent with this, we also found that a history of herding is associated with participants’ willingness to take revenge and punish other people for unfair behavior in the *Global Preferences Survey*, and that similar revenge expressions are correlated with historical herding practices in data on cultural folklore. We also documented that, in line with Nisbett and Cohen’s (1996) original hypothesis, a culture of honor appears to be more prevalent in societies where formal institutions are weak.

Our results have implications for both the economics literature on conflict and the literature on culture. The fact that the culture of honor hypothesis is relevant for the emergence, duration, and severity of economically meaningful armed conflicts is particularly important since the occurrence of civil war has been viewed as a puzzle among scholars in the social sciences who take a rational perspective. Our results provide evidence that cultural values are important factors in explaining the incidence and severity of conflict. They also suggest that economic incentives shape people’s moral and cultural traits, and that these, in turn, feed back into economic outcomes such as conflict. We believe that this perspective—an economically functional psychology that is shaped by material incentives—is a promising path to advance the literature on morality and culture in moving beyond its traditional focus on documenting historical persistence per se.

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Online Appendix

A. Supplementary figures and tables

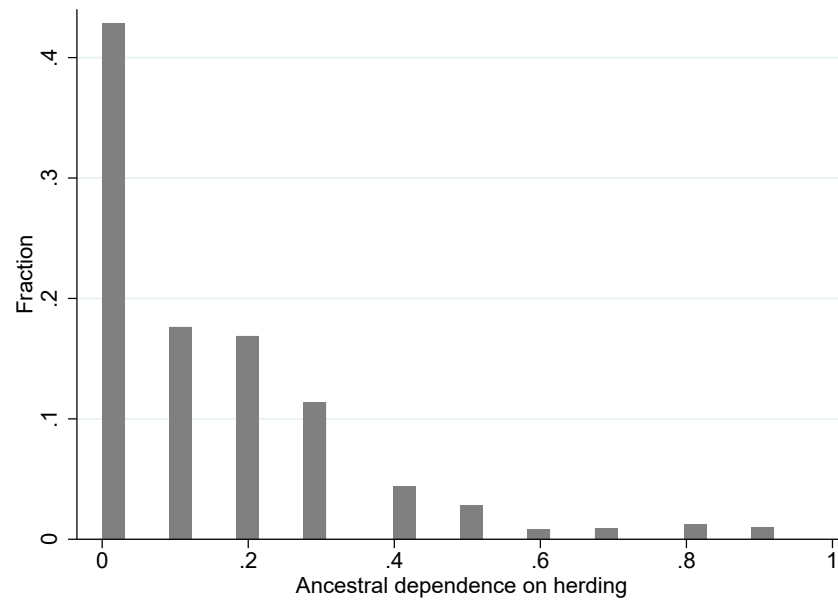


Figure A1: Distribution of dependence on herding in the *Ethnographic Atlas*.

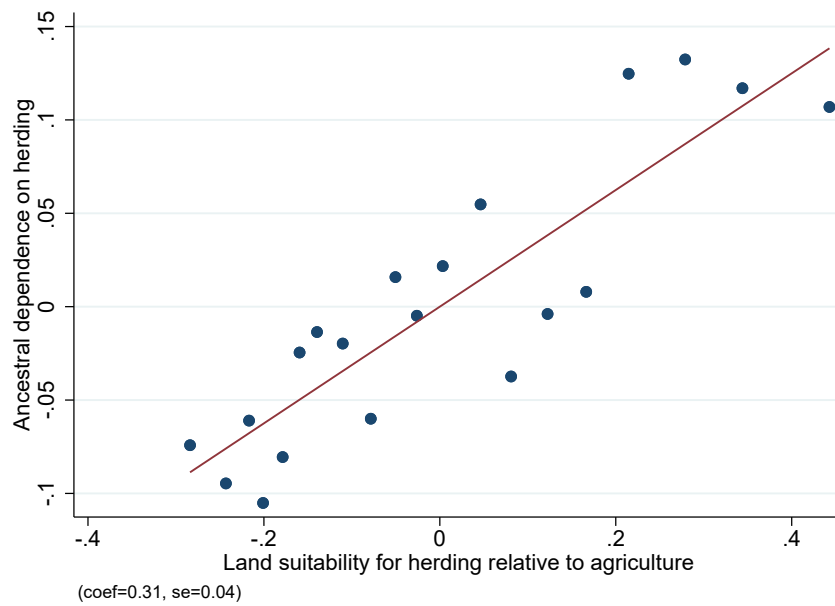


Figure A2: Binscatter plot: dependence on herding and land suitability for herding relative to agriculture for 637 societies in the *Ethnographic Atlas*. The plot controls for continent fixed effects.

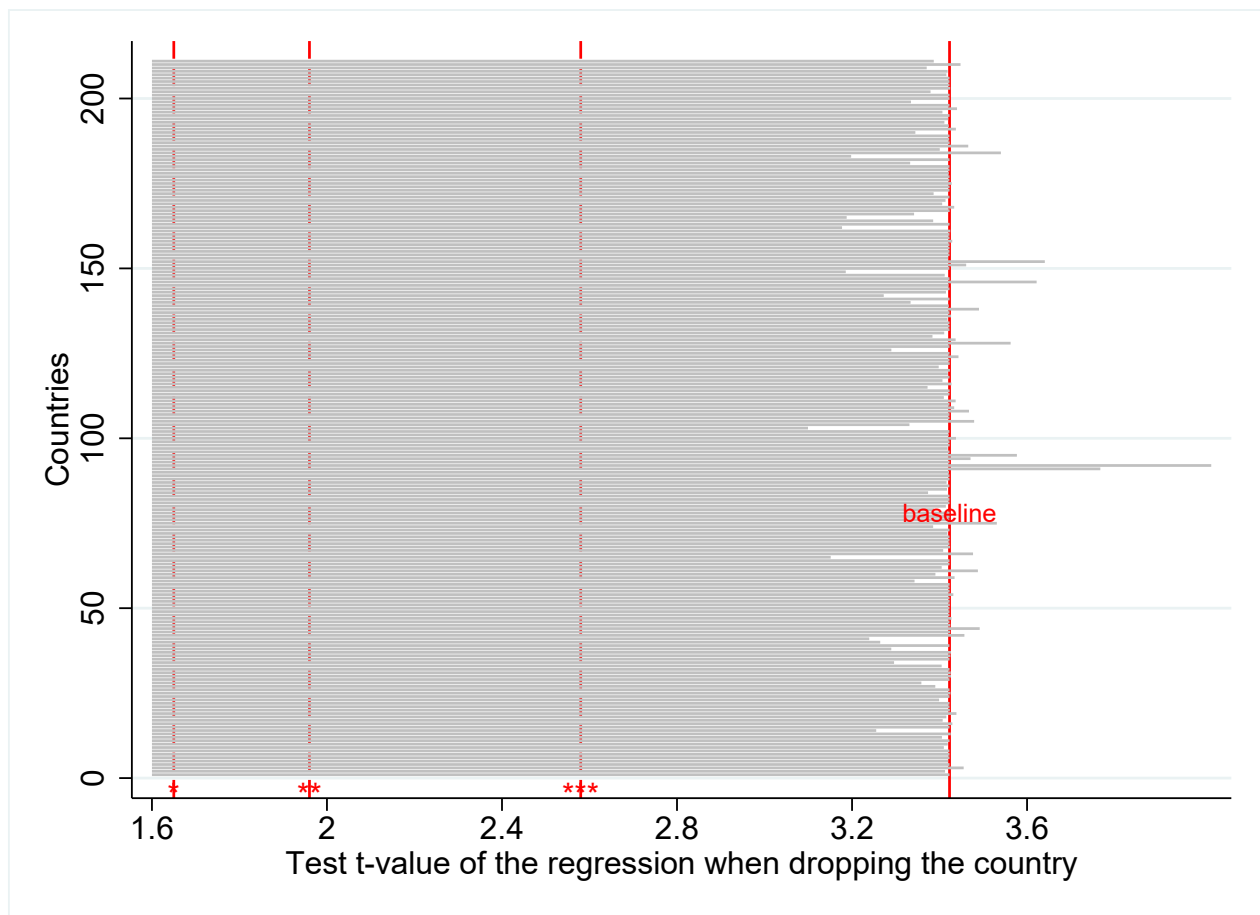


Figure A3: Dropping one country at a time, OLS

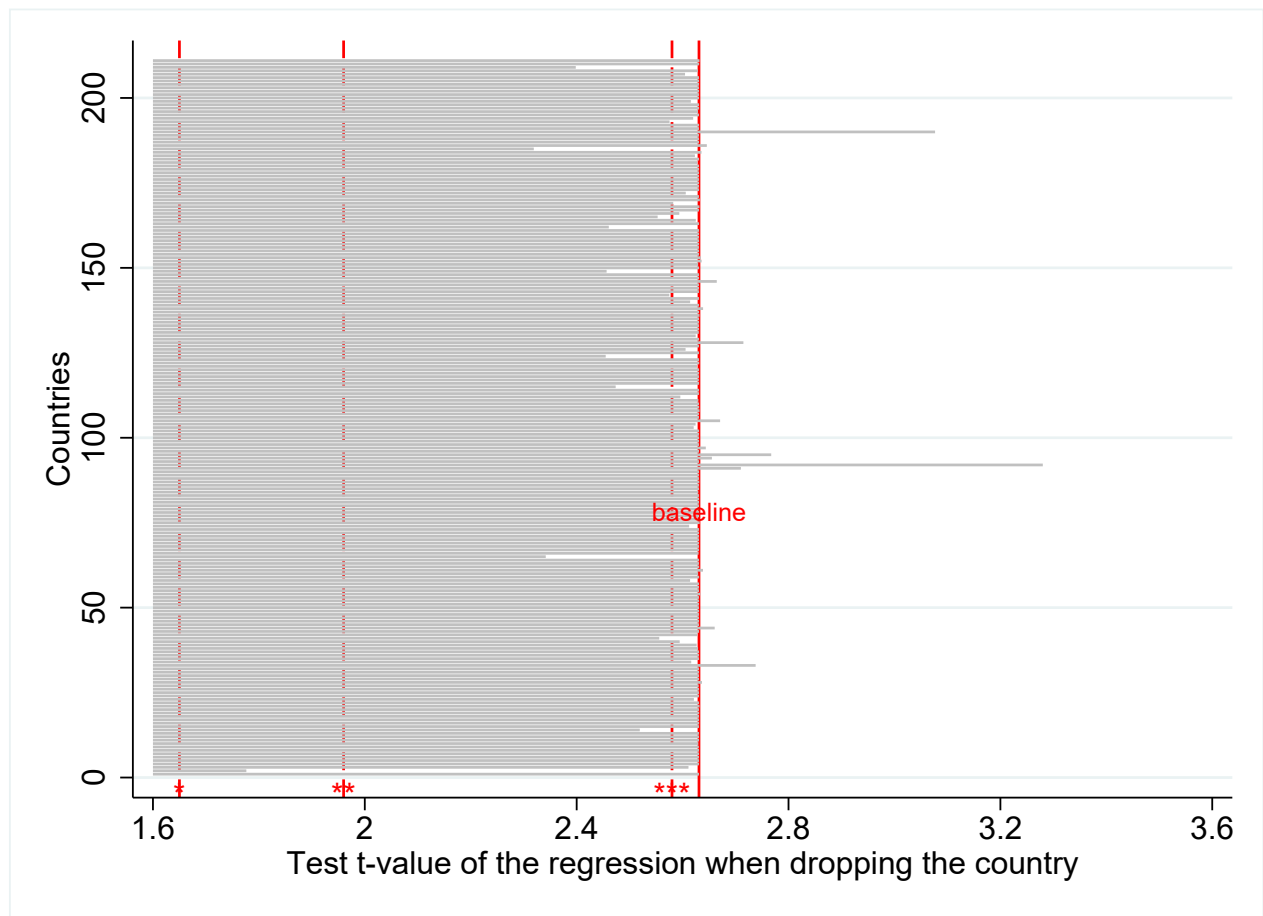
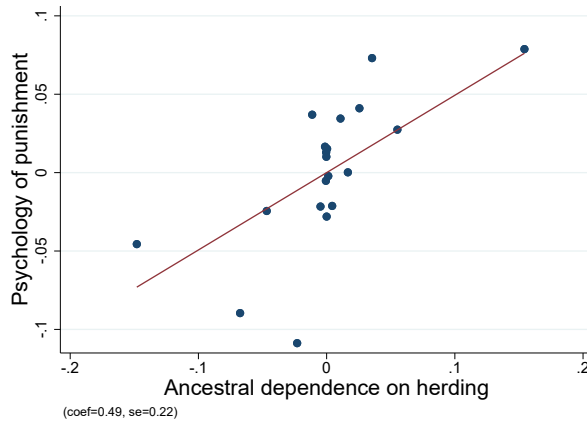
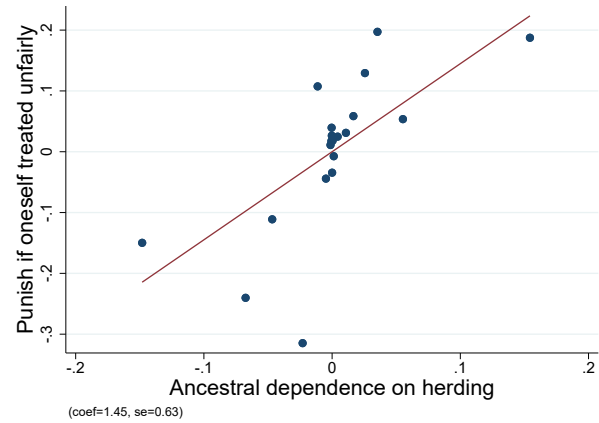


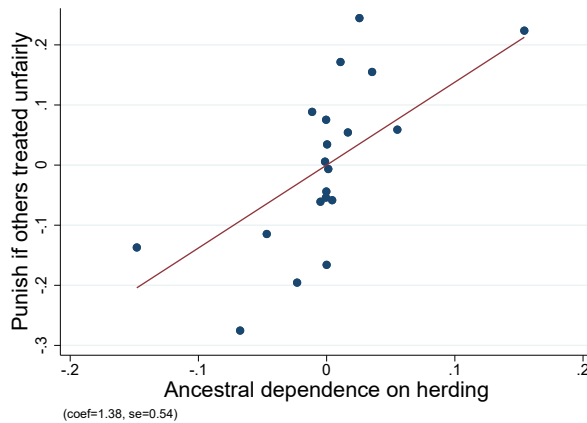
Figure A4: Dropping one country at a time, PPML



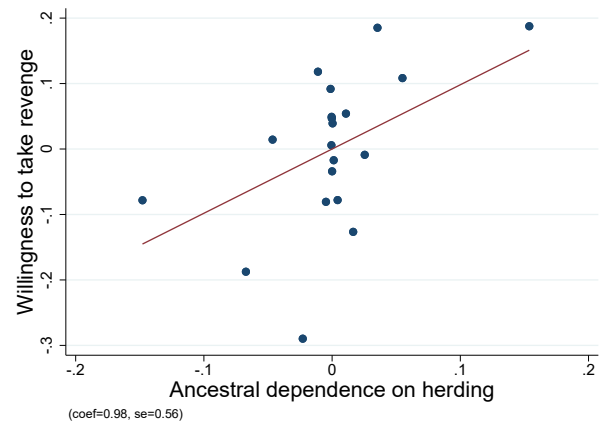
(a) Summary measure of psychology of punishment



(b) Punish if treated unfairly



(c) Punish if others treated unfairly



(d) Willingness to take revenge

Figure A5: Binscatter partial correlation plots for the relationship between a contemporary psychology of punishment in the GPS and a tradition of herding. In each plot, a unit of observation is a respondent in the GPS. Each dot shows the average of the dependent variable for a given range of values of dependence on herding. Each binscatter is constructed after first partialing out country fixed effects, settlement complexity, jurisdictional hierarchy, distance from the equator, age, age squared, and female indicator.

Table A1: Descriptive statistics

	Obs.	Mean	S.D.	Max.	Min.
Panel A: The global language group level sample based on <i>UCDP</i>					
Number of conflict events: All conflicts	7,038	18.4	233.8	14811	0
Number of conflict events: Civil conflicts	7,038	14.2	210.0	14150	0
Number of conflict events: Non-civil conflicts	7,038	4.26	45.0	1953	0
Dependence on herding	7,036	0.13	0.16	0.92	0
Settlement complexity	6,502	5.93	1.78	8	1
Jurisdictional hierarchy	6,319	2.01	1.23	5	1
Historical latitude (abs.)	7,038	14.4	12.9	72	0
Population density (ln)	6,952	-11.0	2.17	-3.53	-22.6
Ethnic fractionalization	7,037	0.79	0.23	0.98	0
Ethnic polarization	7,037	0.092	0.055	0.25	0
Nighttime lights	6,995	1.27	4.70	63	0
Share of land for herding	7,042	0.15	0.23	1	0
Ruggedness	6,995	0.15	0.18	1.49	0
Panel B: The African language group level sample based on <i>ACLED</i>					
Number of conflict events: All conflicts	2,285	56.6	427.0	14654	0
Number of conflict events: Civil conflicts	2,285	29.8	237.7	8441	0
Number of conflict events: Non-civil conflicts	2,285	26.8	206.3	6213	0
Number of conflict events: Localized conflicts	2,285	15.7	117.6	2598	0
Dependence on herding	2,285	0.18	0.15	0.92	0
Settlement complexity	2,199	6.10	1.45	8	1
Jurisdictional hierarchy	2,143	2.04	0.94	5	1
Historical latitude (abs.)	2,285	8.87	5.75	42	0
Population density (ln)	2,276	-10.3	1.55	-4.20	-22.6
Ethnic fractionalization	2,285	0.88	0.11	0.98	0
Ethnic polarization	2,285	0.079	0.042	0.24	0
Nighttime lights	2,279	0.60	3.50	59.9	0
Share of land for herding	2,285	0.19	0.24	1	0
Ruggedness	2,279	0.071	0.087	1.01	0
Panel C: The individual level sample from the <i>Global Preference Survey</i>					
Negative reciprocity index	74,182	-0.0031	1.00	2.33	-1.59
Punish if ... treated unfairly:					
Self	74,264	4.20	3.04	10	0
Others	74,252	4.35	3.04	10	0
Willingness to take revenge	75,024	3.63	3.00	10	0
Dependence on herding	75,176	0.28	0.15	0.92	0
Settlement complexity	75,176	6.34	1.73	8	0
Jurisdictional hierarchy	75,176	3.67	1.06	5	0
Historical latitude (abs.)	75,176	31.9	15.5	64.0	0.050
Age	74,931	41.5	17.4	99	15
Age squared	74,931	2026.8	1623.0	9801	225
Female indicator	75,176	0.54	0.50	1	0
Panel D: The ethnic group level sample from <i>Ethnographic Atlas</i>					
Share of words related to ... that appear in group's folklore:					
Violence/Deterrence/Punishment/Revenge	1,135	0.51	0.22	0.96	0
Violence/Deterrence	1,135	0.49	0.21	0.95	0
Punishment/Revenge	1,135	0.62	0.34	1	0
Dependence on herding	1,135	0.15	0.19	0.92	0
Settlement complexity	1,135	5.11	2.21	8	1
Jurisdictional hierarchy	1,107	1.90	1.04	5	1
Historical latitude (abs.)	1,135	20.8	17.2	78	0
Panel E: The ethnic group level sample from <i>Standard Cross Cultural Sample (SCCS)</i>					
First principal component of the acceptance of violence	60	0.0035	1.38	2.57	-2.35
Acceptability of Violence against:					
Other society	60	2.33	0.97	3	0
Same society	60	1.37	1.16	3	0
Same local comm.	60	0.43	0.70	2	0
Dependence on herding	60	0.15	0.24	0.92	0
Settlement complexity	60	4.05	2.42	7	1
Jurisdictional hierarchy	60	1.85	0.99	4	1
Historical latitude (abs.)	60	21.5	16.2	62	0.064

Table A2: Traditional herding and contemporary conflict: Negative binomial estimates

	<i>Dependent variable: Number of conflict events</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	6.165 (1.031)*** [1.355]***	3.474 (0.765)*** [0.926]***	5.398 (0.939)*** [1.119]***	3.344 (0.837)*** [1.036]***	5.749 (1.057)*** [1.482]***	3.663 (0.789)*** [0.995]***
<i>Historical ethnicity-level covariates:</i>						
Settlement complexity		-0.071 (0.056) [0.070]		-0.092 (0.066) [0.078]		-0.024 (0.064) [0.086]
Jurisdictional hierarchy		0.515 (0.096)*** [0.100]***		0.497 (0.110)*** [0.125]***		0.431 (0.093)*** [0.102]***
Historical latitude (abs.)		63.709 (20.603)*** [31.715]**		75.401 (24.794)*** [33.137]**		66.054 (20.160)*** [36.190]*
<i>Contemporary language-level covariates:</i>						
Population density (ln)		0.609 (0.055)*** [0.064]***		0.602 (0.062)*** [0.084]***		0.667 (0.060)*** [0.071]***
Ethnic fractionalization		-0.971 (0.749) [1.126]		-0.431 (0.745) [1.270]		0.311 (0.781) [1.177]
Ethnic polarization		4.473 (2.513)* [3.457]		9.586 (2.915)*** [4.790]**		-3.280 (2.261) [3.055]
Nighttime lights		-0.071 (0.014)*** [0.020]***		-0.088 (0.018)*** [0.024]***		-0.059 (0.014)*** [0.019]***
Share of land for herding		0.255 (0.492) [0.707]		-0.062 (0.517) [0.905]		0.307 (0.395) [0.558]
Ruggedness		-1.591 (0.611)*** [0.706]**		-1.522 (0.672)** [0.624]**		-2.286 (0.629)*** [0.786]***
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	18.4	19.3	14.2	14.8	4.27	4.54
SD of dependent var	233.8	245.4	210.0	220.2	45.0	47.6
Number of Obs.	7,036	6,239	7,036	6,239	7,036	6,239
Number of Countries	211	211	211	211	211	211
Number of Ethnic Groups	1,104	985	1,104	985	1,104	985

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A3: Traditional herding and contemporary conflict globally: Extensive margin

	<i>Dependent variable: Incidence of conflict events (indicator)</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	0.174 (0.066)*** [0.099]*	0.133 (0.068)* [0.086]	0.139 (0.060)** [0.095]	0.103 (0.062)* [0.077]	0.187 (0.063)*** [0.085]**	0.182 (0.061)*** [0.076]**
<i>Historical ethnicity-level covariates:</i>						
Settlement complexity		0.001 (0.004) [0.005]		0.001 (0.004) [0.004]		-0.002 (0.004) [0.005]
Jurisdictional hierarchy		0.003 (0.007) [0.013]		0.006 (0.006) [0.011]		0.005 (0.006) [0.011]
Historical latitude (abs.)		2.622 (1.373)* [2.014]		2.411 (1.217)** [1.545]		1.583 (1.212) [1.951]
<i>Contemporary language-level covariates:</i>						
Population density (ln)		0.033 (0.004)*** [0.007]***		0.029 (0.004)*** [0.008]***		0.025 (0.003)*** [0.006]***
Ethnic fractionalization		-0.094 (0.044)** [0.067]		-0.121 (0.041)*** [0.064]*		0.037 (0.043) [0.087]
Ethnic polarization		0.362 (0.167)** [0.237]		0.353 (0.152)** [0.225]		-0.103 (0.153) [0.261]
Nighttime lights		-0.003 (0.002) [0.003]		-0.004 (0.002)** [0.003]		-0.001 (0.002) [0.002]
Share of land for herding		0.056 (0.036) [0.048]		0.043 (0.033) [0.046]		0.035 (0.030) [0.039]
Ruggedness		-0.084 (0.049)* [0.050]*		-0.082 (0.045)* [0.054]		-0.117 (0.041)*** [0.056]**
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.065	0.049	0.056	0.041	0.083	0.080
Oster δ		3.48		2.93		6.04
Mean of dependent var	0.22	0.22	0.19	0.19	0.14	0.14
SD of dependent var	0.41	0.42	0.39	0.39	0.35	0.35
Adj. R-squared	0.30	0.32	0.30	0.33	0.23	0.25
Number of Obs.	7,036	6,239	7,036	6,239	7,036	6,239
Number of Countries	211	211	211	211	211	211
Number of Ethnic Groups	1,104	985	1,104	985	1,104	985

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are dummy variables that take the value of one if any conflict events occurred during the period 1989-2016 according to the *Uppsala Conflict Data Program* (UCDP). Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A4: Traditional herding and contemporary conflict globally: Intensive margin

	<i>Dependent variable: Number of conflict events (IHS)</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	1.775 (0.461)*** [0.431]***	1.855 (0.483)*** [0.446]***	1.748 (0.472)*** [0.421]***	1.801 (0.513)*** [0.485]***	1.486 (0.436)*** [0.451]***	1.574 (0.528)*** [0.433]***
<i>Historical ethnicity-level covariates:</i>						
Settlement complexity		-0.018 (0.044) [0.055]		-0.015 (0.045) [0.054]		-0.004 (0.047) [0.045]
Jurisdictional hierarchy		0.178 (0.059)*** [0.066]***		0.177 (0.060)*** [0.068]***		0.166 (0.068)** [0.072]**
Historical latitude (abs.)		8.928 (20.286) [17.131]		12.770 (21.191) [18.671]		24.072 (18.166) [14.067]*
<i>Contemporary language-level covariates:</i>						
Population density (ln)		0.260 (0.044)*** [0.053]***		0.279 (0.046)*** [0.053]***		0.185 (0.047)*** [0.046]***
Ethnic fractionalization		-0.429 (0.546) [0.714]		-0.394 (0.530) [0.662]		-1.134 (0.574)** [0.582]*
Ethnic polarization		-2.202 (2.179) [1.696]		-0.902 (2.003) [1.679]		-4.586 (2.027)** [1.683]***
Nighttime lights		-0.035 (0.012)*** [0.013]***		-0.054 (0.018)*** [0.018]***		-0.019 (0.015) [0.015]
Share of land for herding		0.132 (0.304) [0.354]		-0.001 (0.361) [0.371]		0.064 (0.295) [0.363]
Ruggedness		-1.039 (0.443)** [0.380]***		-1.031 (0.426)** [0.448]**		-1.335 (0.449)*** [0.385]***
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.17	0.17	0.17	0.17	0.17	0.17
Oster δ		3.14		2.82		2.56
Mean of dependent var	2.11	2.10	1.98	1.97	1.78	1.78
SD of dependent var	1.85	1.86	1.82	1.84	1.58	1.59
Adj. R-squared	0.17	0.22	0.19	0.25	0.13	0.19
Number of Obs.	1,552	1,387	1,311	1,163	985	886
Number of Countries	122	122	115	115	94	93
Number of Ethnic Groups	561	496	501	442	435	388

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. They are measured as the inverse hyperbolic sine (arsinh) of the value. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A5: Countries and territories in *Ethnologue* that provide within-country variation in herding

Country	# of groups in country	Herding indicator across groups:			Country	# of groups in country	Herding indicator across groups:		
		Mean	S.D.	C.V.			Mean	S.D.	C.V.
Australia	151	0.005	0.034	7.186	Guinea	28	0.180	0.108	0.598
New Caledonia	35	0.006	0.035	5.916	Saudi Arabia	3	0.643	0.384	0.597
Canada	72	0.011	0.043	4.105	Algeria	15	0.285	0.170	0.596
Venezuela	29	0.035	0.141	4.042	Iraq	9	0.338	0.200	0.591
Solomon Islands	67	0.015	0.054	3.548	Benin	46	0.160	0.095	0.590
Colombia	74	0.030	0.102	3.350	Niger	12	0.431	0.249	0.578
Suriname	11	0.028	0.092	3.317	Nigeria	466	0.156	0.090	0.576
Brazil	166	0.017	0.046	2.715	Sudan	120	0.263	0.149	0.566
Guyana	12	0.034	0.090	2.648	Gabon	38	0.080	0.045	0.565
Panama	11	0.037	0.094	2.528	Eritrea	8	0.530	0.296	0.559
United States	144	0.022	0.055	2.523	Oman	10	0.325	0.175	0.539
Paraguay	18	0.040	0.100	2.508	Libya	5	0.365	0.195	0.534
Cook Islands	5	0.061	0.136	2.236	Thailand	55	0.137	0.073	0.532
Costa Rica	8	0.051	0.109	2.125	Armenia	3	0.505	0.265	0.524
Mexico	286	0.042	0.082	1.946	Ethiopia	83	0.303	0.154	0.508
Brunei	8	0.051	0.095	1.852	Mauritania	6	0.472	0.234	0.496
El Salvador	3	0.102	0.176	1.732	Italy	18	0.222	0.110	0.495
Bolivia	32	0.055	0.094	1.711	Congo	57	0.081	0.039	0.477
Indonesia	696	0.058	0.093	1.617	Iran	38	0.421	0.197	0.467
Malaysia	110	0.066	0.094	1.427	Bosnia and Herzegovina	2	0.305	0.141	0.464
Honduras	9	0.091	0.129	1.417	Ireland	2	0.305	0.141	0.464
Spain	2	0.153	0.216	1.414	Tajikistan	11	0.405	0.185	0.458
Portugal	2	0.153	0.216	1.414	Latvia	2	0.155	0.071	0.456
French Guiana	10	0.052	0.073	1.405	Somalia	9	0.750	0.338	0.451
Taiwan	12	0.060	0.082	1.351	Myanmar	87	0.145	0.064	0.444
Mozambique	39	0.093	0.116	1.238	Syria	10	0.385	0.169	0.438
Philippines	160	0.061	0.075	1.233	South Africa	13	0.359	0.156	0.435
Argentina	20	0.109	0.128	1.175	Nepal	102	0.306	0.130	0.426
Peru	88	0.094	0.108	1.142	Hungary	3	0.272	0.115	0.425
Guatemala	52	0.056	0.063	1.121	Sierra Leone	14	0.155	0.065	0.420
Ghana	67	0.183	0.205	1.118	Finland	5	0.465	0.195	0.419
Kazakhstan	3	0.370	0.406	1.098	Turkey	15	0.378	0.158	0.418
Ecuador	22	0.107	0.112	1.042	Burkina Faso	58	0.198	0.079	0.400
Macedonia	3	0.203	0.203	0.996	Uganda	36	0.255	0.100	0.394
Togo	33	0.231	0.229	0.989	Cote d'Ivoire	67	0.120	0.047	0.393
Sri Lanka	5	0.143	0.137	0.956	Senegal	29	0.198	0.076	0.385
Namibia	18	0.259	0.247	0.955	Gambia	8	0.243	0.092	0.378
Romania	5	0.324	0.296	0.915	Switzerland	5	0.225	0.084	0.372
Nicaragua	5	0.124	0.111	0.895	Viet Nam	88	0.167	0.059	0.355
Belize	7	0.161	0.142	0.881	Afghanistan	34	0.355	0.124	0.350
Chile	7	0.204	0.175	0.860	Western Sahara	2	0.405	0.141	0.349
Kuwait	2	0.505	0.424	0.840	Egypt	6	0.355	0.122	0.345
Lithuania	2	0.255	0.212	0.832	United Kingdom	6	0.238	0.082	0.343
Uzbekistan	7	0.446	0.367	0.821	Botswana	4	0.380	0.126	0.331
Cameroon	257	0.141	0.116	0.818	Moldova	3	0.305	0.100	0.328
Equatorial Guinea	11	0.067	0.053	0.793	Madagascar	10	0.325	0.103	0.318
Central African Republic	59	0.116	0.091	0.786	Cambodia	19	0.221	0.069	0.312
Democratic Republic of the Congo	186	0.097	0.075	0.771	Azerbaijan	14	0.455	0.140	0.308
Mongolia	10	0.454	0.349	0.768	Pakistan	47	0.314	0.095	0.304
Chad	119	0.195	0.149	0.762	Austria	4	0.330	0.096	0.290
Kyrgyzstan	3	0.438	0.321	0.733	Georgia	7	0.419	0.121	0.290
Angola	36	0.175	0.126	0.720	Laos	72	0.183	0.051	0.279
Malawi	12	0.120	0.085	0.709	Bahrain	2	0.255	0.071	0.277
Kenya	55	0.331	0.230	0.692	United Arab Emirates	2	0.255	0.071	0.277
Guadeloupe	2	0.205	0.141	0.690	Israel	4	0.305	0.082	0.268
Albania	4	0.279	0.192	0.688	Slovakia	6	0.288	0.075	0.261
Guinea-Bissau	16	0.198	0.135	0.685	Norway	5	0.545	0.134	0.246
China	198	0.214	0.146	0.684	Liberia	26	0.113	0.027	0.241
Tanzania	107	0.251	0.168	0.671	Sweden	6	0.555	0.122	0.221
India	300	0.204	0.137	0.671	Cyprus	2	0.355	0.071	0.199
East Timor	17	0.215	0.143	0.665	Denmark	2	0.355	0.071	0.199
Bulgaria	6	0.355	0.235	0.661	Belgium	4	0.280	0.050	0.179
Russian Federation	87	0.331	0.216	0.652	Lesotho	4	0.355	0.058	0.163
Zambia	35	0.130	0.085	0.651	Morocco	5	0.325	0.045	0.138
Zimbabwe	13	0.200	0.128	0.640	Turkmenistan	3	0.438	0.058	0.132
Jordan	2	0.555	0.354	0.637	Yemen	7	0.291	0.038	0.130
Japan	12	0.145	0.092	0.634	Djibouti	2	0.863	0.081	0.094
Bangladesh	8	0.154	0.095	0.617	Bhutan	23	0.401	0.021	0.052
Mali	30	0.275	0.168	0.611					

Invariant countries (multiple polygons but no variation in herding): Trinidad and Tobago, Singapore, Liechtenstein, Dominica, Netherlands, Dominican Republic, Czech Republic, Sao Tome e Principe, Ukraine, Netherlands Antilles, Germany, Poland, Comoros, Serbia, Tunisia, Vanuatu, Papua New Guinea, Fiji

Singleton countries (countries with only one language group): Malta, Montserrat, Swaziland, Reunion, Rwanda, Uruguay, United States Virgin Islands, France, Cayman Islands, Slovenia, Turks and Caicos Islands, Belarus, Lebanon, Qatar, Croatia, Saint Lucia, Maldives, Cape Verde Islands, Puerto Rico, Martinique, Bermuda, Falkland Islands, Aruba, Andorra, Burundi, Antigua and Barbuda, Grenada, Cuba, Greece, Seychelles, Bahamas, Saint Pierre and Miquelon, Saint Kitts and Nevis, Mayotte, Korea, North, San Marino, Saint Vincent and the Grenadines, Jamaica, Korea, South, Luxembourg, Iceland, Haiti, New Zealand, British Virgin Islands, Norfolk Island, Barbados, Montenegro, Mauritius, Greenland, Estonia, Anguilla

The countries in bold are those that provide within-country variations in conflict.

The language groups are equally weighted in calculating the mean, standard deviation, and the coefficient of variation.

Table A6: Traditional herding and contemporary conflict globally: Number of deaths

	Dependent variable: Number of conflict deaths								
	All conflicts			Civil conflicts			Non-civil conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependence on herding	1.456 (0.424)*** [0.538]***	1.370 (0.409)*** [0.503]***	3.067 (0.879)*** [0.828]***	1.182 (0.367)*** [0.471]**	1.124 (0.354)*** [0.421]***	2.391 (1.046)** [0.918]***	1.303 (0.358)*** [0.436]***	1.383 (0.347)*** [0.442]***	4.345 (0.721)*** [0.635]***
<i>Historical ethnicity-level covariates:</i>									
Settlement complexity		-0.011 (0.025) [0.027]	0.208 (0.120)* [0.103]**		-0.008 (0.023) [0.025]	0.200 (0.138) [0.111]*		-0.012 (0.021) [0.025]	0.153 (0.082)* [0.073]**
Jurisdictional hierarchy		0.075 (0.037)** [0.046]	0.365 (0.151)** [0.177]**		0.065 (0.032)** [0.037]*	0.319 (0.178)* [0.218]		0.081 (0.031)*** [0.037]**	0.483 (0.121)*** [0.079]***
Historical latitude (abs.)		12.977 (9.308) [9.720]	56.288 (17.939)*** [24.026]**		10.624 (8.170) [8.296]	63.790 (23.120)*** [32.051]**		7.488 (6.915) [8.138]	47.133 (15.745)*** [21.002]**
<i>Contemporary language-level covariates:</i>									
Population density (ln)		0.174 (0.023)*** [0.047]***	0.595 (0.068)*** [0.067]***		0.145 (0.021)*** [0.046]***	0.586 (0.077)*** [0.065]***		0.111 (0.016)*** [0.031]***	0.665 (0.087)*** [0.079]***
Ethnic fractionalization		-0.357 (0.241) [0.404]	-2.119 (1.175)* [1.585]		-0.508 (0.222)** [0.362]	-2.331 (1.411)* [1.702]		0.112 (0.197) [0.353]	-1.200 (0.898) [1.426]
Ethnic polarization		0.986 (1.016) [1.461]	3.576 (3.445) [5.359]		1.596 (0.901)* [1.358]	8.761 (3.718)** [6.296]		-1.382 (0.766)* [1.163]	-6.411 (3.087)** [3.529]*
Nighttime lights		-0.022 (0.009)** [0.013]*	-0.163 (0.036)*** [0.037]***		-0.028 (0.008)*** [0.013]**	-0.200 (0.044)*** [0.047]***		-0.010 (0.008) [0.010]	-0.092 (0.026)*** [0.027]***
Share of land for herding		0.224 (0.188) [0.235]	1.243 (0.445)*** [0.435]***		0.075 (0.169) [0.205]	0.951 (0.488)* [0.507]*		0.217 (0.145) [0.178]	2.134 (0.497)*** [0.606]***
Ruggedness		-0.728 (0.274)*** [0.348]**	-2.119 (0.813)*** [0.890]**		-0.646 (0.251)** [0.328]*	-1.821 (0.853)** [0.942]*		-0.768 (0.212)*** [0.328]**	-2.659 (0.654)*** [1.083]**
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.10	0.095		0.094	0.088		0.12	0.12	
Oster δ		3.95			3.97			4.57	
Mean of dependent var	1.00	1.01	209.4	0.79	0.79	162.6	0.59	0.61	56.0
SD of dependent var	2.20	2.22	2122.1	1.96	1.97	1942.0	1.68	1.71	500.9
Adj. R-squared	0.30	0.32		0.30	0.32		0.23	0.25	
Pseudo R-squared			0.60			0.62			0.58
Number of Obs.	7,036	6,239	5,524	7,036	6,239	5,293	7,036	6,239	5,281
Number of Countries	211	211	104	211	211	98	211	211	82
Number of Ethnic Groups	1,104	985	927	1,104	985	871	1,104	985	905

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict deaths reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989–2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A7: Traditional herding and contemporary conflict globally: Winsorizing top 5%

	Dependent variable: Number of conflict events								
	All conflicts			Civil conflicts			Non-civil conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependence on herding	1.003 (0.387)*** [0.504]**	0.943 (0.334)*** [0.441]**	2.710 (1.606)* [1.693]	0.797 (0.324)** [0.418]*	0.735 (0.284)*** [0.362]**	2.039 (1.657) [1.674]	0.721 (0.295)** [0.373]*	0.796 (0.247)*** [0.335]**	4.035 (2.063)* [2.343]*
<i>Historical ethnicity-level covariates:</i>									
Settlement complexity		-0.018 (0.018) [0.022]	0.123 (0.138) [0.132]		-0.015 (0.016) [0.019]	0.118 (0.149) [0.140]		-0.018 (0.013) [0.017]	0.077 (0.085) [0.085]
Jurisdictional hierarchy		0.060 (0.025)** [0.035]*	0.245 (0.157) [0.178]		0.058 (0.022)*** [0.026]**	0.206 (0.186) [0.218]		0.056 (0.020)*** [0.025]**	0.397 (0.105)*** [0.095]**
Historical latitude (abs.)		8.197 (6.972) [7.624]	78.093 (30.789)** [32.267]**		6.450 (6.087) [6.463]	86.386 (36.493)** [38.487]**		6.546 (4.900) [5.629]	64.670 (34.104)* [37.157]*
<i>Contemporary language-level covariates:</i>									
Population density (ln)		0.122 (0.015)*** [0.033]***	0.570 (0.080)*** [0.069]***		0.105 (0.014)*** [0.031]***	0.582 (0.091)*** [0.072]***		0.069 (0.010)*** [0.020]***	0.595 (0.089)*** [0.081]***
Ethnic fractionalization		-0.210 (0.170) [0.252]	-2.118 (1.397) [1.642]		-0.291 (0.155)* [0.224]	-1.990 (1.555) [1.846]		0.042 (0.126) [0.200]	-2.014 (0.824)** [0.922]**
Ethnic polarization		0.381 (0.675) [0.856]	5.776 (3.822) [5.128]		0.734 (0.589) [0.761]	10.017 (4.129)** [5.728]*		-0.737 (0.490) [0.679]	-4.843 (3.077) [4.151]
Nighttime lights		-0.018 (0.007)*** [0.010]*	-0.145 (0.034)*** [0.031]***		-0.020 (0.006)*** [0.009]**	-0.186 (0.042)*** [0.037]***		-0.008 (0.005) [0.006]	-0.079 (0.021)*** [0.018]**
Share of land for herding		0.173 (0.123) [0.156]	1.683 (0.498)*** [0.467]***		0.111 (0.110) [0.138]	1.779 (0.563)*** [0.550]***		0.087 (0.090) [0.102]	1.519 (0.480)*** [0.632]**
Ruggedness		-0.625 (0.199)*** [0.263]**	-2.298 (0.790)*** [0.859]***		-0.565 (0.181)*** [0.245]**	-2.314 (0.824)*** [0.930]**		-0.541 (0.148)*** [0.210]**	-2.034 (0.632)*** [0.598]**
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.089	0.083		0.079	0.072		0.089	0.097	
Oster δ		3.62			3.46			4.38	
Mean of dependent var	0.63	0.63	20.9	0.51	0.50	16.7	0.35	0.36	5.15
SD of dependent var	1.45	1.45	258.7	1.30	1.30	237.5	1.04	1.05	51.1
Adj. R-squared	0.28	0.31		0.28	0.31		0.22	0.24	
Pseudo R-squared			0.59			0.61			0.55
Number of Obs.	6,979	6,174	5,558	6,979	6,174	5,327	6,979	6,174	5,294
Number of Countries	154	146	106	154	146	100	154	146	84
Number of Ethnic Groups	1,096	976	934	1,096	976	879	1,096	976	906

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A8: Correlation coefficients between herding and the covariates

Variable Name	Raw	Within country	Obs.
	(1)	(2)	(3)
Baseline controls:			
Settlement complexity	-0.127**	-0.190***	6,502
Jurisdictional hierarchy	0.453***	0.154**	6,319
Historical latitude (abs.)	0.334***	0.213***	7,036
Population density (ln)	0.234***	0.019	6,950
Ethnic fractionalization	-0.028	-0.052*	7,035
Ethnic polarization	0.267***	0.078***	7,035
Nighttime lights	0.069***	-0.019	6,993
Share of land for herding	0.249***	0.154***	7,036
Ruggedness	0.197***	0.157***	6,993
Extended controls:			
Distance to capital (ln)	-0.247***	-0.082***	6,973
Distance to country border (ln)	0.114***	0.005	7,036
Distance to coastline (ln)	0.335***	0.053**	7,036
On or cross a major river	0.101***	0.018	7,036
Tsetse suitability index	-0.157***	-0.140***	7,036
Temperature (Mean)	-0.319***	-0.169***	6,996
Temperature (SD)	0.151***	0.063**	6,996
Precipitation (Mean)	-0.453***	-0.159***	7,029
Precipitation (SD)	-0.381***	-0.125***	7,029
Nomadic indicator	0.113*	0.229***	6,502
Semi-sedentary indicator	0.033	0.004	6,502
Former slavery	0.111**	0.011	6,141
Excluded from state power	0.027	-0.028	6,009
Ethnic segregation	0.210***	0.051*	7,013
Share of cropland	0.187***	0.034	7,036
Share of urban areas	0.029***	-0.007	7,036
Share of agnostics	0.083**	0.024	6,898
Share of Buddhists	0.066**	0.001	6,898
Share of Christians	-0.377***	-0.135***	6,898
Share of Hindus	0.152***	0.024	6,898
Share of Jews	0.014***	-0.000	6,898
Share of Muslims	0.272***	0.136***	6,898
Religious fractionalization	0.113***	0.028	6,898
Religious polarization	0.112***	0.027	6,898

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. Column 1 reports the raw correlation coefficients between herding and the covariates. Column 2 reports the partial correlation conditional on country fixed effects. Significance levels are computed based on standard errors clustered at the ethnicity level. *, **, and *** indicate significance at the 10, 5, and 1% levels.

Table A9: Traditional herding and contemporary conflict globally: More controls

	Dependent variable: Number of conflict events (arsinh) — all conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	0.806 (0.266)*** [0.329]**	0.952 (0.287)*** [0.358]***	0.971 (0.280)*** [0.347]***	0.915 (0.269)*** [0.339]***	0.971 (0.270)*** [0.320]***	0.918 (0.266)*** [0.333]***	0.943 (0.300)*** [0.322]***	0.894 (0.278)*** [0.304]***
Distance to capital (ln)	-0.229 (0.031)*** [0.036]***						-0.227 (0.036)*** [0.037]***	-0.238 (0.034)*** [0.038]***
Distance to country border (ln)	0.044 (0.021)** [0.028]						0.063 (0.023)*** [0.030]**	0.039 (0.016)** [0.023]*
Distance to coastline (ln)	-0.003 (0.021) [0.029]						-0.027 (0.022) [0.032]	
On or cross a major river	0.568 (0.091)*** [0.146]***						0.562 (0.097)*** [0.154]***	0.511 (0.093)*** [0.147]***
Tsetse suitability index	-0.054 (0.036) [0.049]						-0.068 (0.039)* [0.050]	-0.036 (0.035) [0.047]
Temperature (Mean)	0.016 (0.007)** [0.008]**						0.014 (0.007)* [0.007]*	0.016 (0.007)** [0.008]**
Temperature (SD)	-0.032 (0.137) [0.168]						-0.199 (0.129) [0.169]	-0.231 (0.129)* [0.188]
Precipitation (Mean)	0.001 (0.000)* [0.000]						0.001 (0.000) [0.001]	
Precipitation (SD)	-0.004 (0.002)* [0.003]						-0.005 (0.003)* [0.004]	-0.004 (0.002)* [0.003]
Nomadic indicator		0.263 (0.243) [0.346]					0.139 (0.267) [0.370]	
Semi-sedentary indicator		0.109 (0.170) [0.225]					0.089 (0.186) [0.240]	
Former slavery		-0.014 (0.074) [0.071]					-0.045 (0.077) [0.076]	
Excluded from state power			0.164 (0.064)** [0.061]***				0.176 (0.063)*** [0.071]**	0.183 (0.060)*** [0.063]***
Ethnic segregation			0.768 (0.304)** [0.449]*				1.000 (0.310)*** [0.449]**	1.039 (0.301)*** [0.436]**
Share of cropland				-0.026 (0.115) [0.135]			0.080 (0.133) [0.158]	
Share of urban areas				-0.102 (0.915) [1.061]			0.573 (0.849) [0.931]	
Share of agnostics					-0.167 (0.111) [0.081]**		-0.006 (0.189) [0.116]	
Share of Buddhists					-0.659 (0.410) [0.508]		-0.668 (0.313)** [0.361]*	-0.749 (0.338)** [0.403]*
Share of Christians					0.323 (0.198) [0.206]		0.229 (0.181) [0.179]	
Share of Hindus					-1.067 (0.359)*** [0.231]***		-1.353 (0.363)*** [0.226]***	-1.655 (0.328)*** [0.296]***
Share of Jews					6.819 (0.747)*** [0.252]***		5.275 (0.861)*** [0.266]***	4.849 (0.870)*** [0.204]***
Share of Muslims					0.330 (0.191)* [0.229]		0.300 (0.169)* [0.175]*	
Religious fractionalization						0.087 (0.116) [0.121]	0.112 (0.132) [0.139]	0.009 (0.003)*** [0.004]**
Religious polarization						-0.148 (0.233) [0.241]	-0.206 (0.264) [0.277]	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.085	0.10	0.099	0.097	0.10	0.097	0.096	0.092
Oster δ	3.99	4.40	4.57	4.38	5.34	4.58	4.59	4.43
Mean of dependent var	0.64	0.62	0.72	0.63	0.64	0.64	0.70	0.72
SD of dependent var	1.46	1.44	1.55	1.46	1.46	1.46	1.53	1.55
Adj. R-squared	0.36	0.33	0.34	0.32	0.33	0.33	0.40	0.39
Number of Obs.	6,189	5,835	5,307	6,239	6,189	6,189	4,952	5,288
Number of Countries	202	209	162	211	209	209	160	161
Number of Ethnic Groups	980	903	939	985	982	982	861	939

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016, computed as the inverse hyperbolic sine (IHS) of the value. Baseline controls include historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A10: Traditional herding and contemporary conflict globally: More controls

	Dependent variable: Number of conflict events (arsinh) — civil conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	0.641 (0.236)*** [0.279]**	0.786 (0.252)*** [0.307]**	0.793 (0.248)*** [0.299]***	0.741 (0.235)*** [0.284]***	0.776 (0.235)*** [0.267]***	0.744 (0.233)*** [0.281]***	0.750 (0.271)*** [0.290]**	0.687 (0.252)*** [0.273]**
Distance to capital (ln)	-0.210 (0.029)*** [0.034]***						-0.205 (0.034)*** [0.035]***	-0.220 (0.032)*** [0.036]***
Distance to country border (ln)	0.038 (0.017)** [0.026]						0.054 (0.019)*** [0.025]**	0.031 (0.014)** [0.019]
Distance to coastline (ln)	-0.008 (0.018) [0.027]						-0.029 (0.020) [0.029]	
On or cross a major river	0.485 (0.083)*** [0.127]***						0.491 (0.088)*** [0.133]***	0.447 (0.085)*** [0.130]***
Tsetse suitability index	-0.061 (0.030)** [0.040]						-0.066 (0.034)* [0.042]	-0.041 (0.030) [0.037]
Temperature (Mean)	0.013 (0.006)** [0.007]**						0.011 (0.006)* [0.007]	0.012 (0.006)** [0.007]*
Temperature (SD)	-0.065 (0.116) [0.130]						-0.197 (0.117)* [0.182]	-0.217 (0.118)* [0.195]
Precipitation (Mean)	0.001 (0.000)** [0.000]**						0.001 (0.000)** [0.000]*	
Precipitation (SD)	-0.004 (0.002)** [0.003]*						-0.005 (0.002)** [0.003]	-0.003 (0.002) [0.003]
Nomadic indicator		0.262 (0.217) [0.301]					0.142 (0.240) [0.324]	
Semi-sedentary indicator		0.143 (0.153) [0.195]					0.118 (0.167) [0.210]	
Former slavery		0.026 (0.066) [0.062]					-0.000 (0.070) [0.066]	
Excluded from state power			0.171 (0.059)*** [0.061]***				0.185 (0.059)*** [0.071]***	0.200 (0.057)*** [0.064]***
Ethnic segregation			0.659 (0.284)** [0.387]*				0.851 (0.286)*** [0.405]**	0.856 (0.281)*** [0.394]**
Share of cropland				-0.006 (0.095) [0.102]			0.121 (0.110) [0.125]	
Share of urban areas				-0.201 (0.839) [0.966]			0.444 (0.766) [0.822]	
Share of agnostics					-0.171 (0.101)* [0.070]**		-0.030 (0.166) [0.092]	
Share of Buddhists					-0.609 (0.399) [0.469]		-0.649 (0.322)** [0.351]*	-0.663 (0.334)** [0.348]*
Share of Christians					0.252 (0.168) [0.159]		0.115 (0.158) [0.144]	
Share of Hindus					-0.857 (0.338)** [0.224]***		-1.134 (0.349)*** [0.235]***	-1.336 (0.327)*** [0.269]***
Share of Jews					6.217 (0.727)*** [0.201]***		4.785 (0.900)*** [0.232]***	4.534 (0.901)*** [0.230]***
Share of Muslims					0.299 (0.163)* [0.178]*		0.218 (0.149) [0.145]	0.112 (0.093) [0.155]
Religious fractionalization						0.066 (0.106) [0.097]	0.087 (0.117) [0.110]	0.008 (0.003)*** [0.003]**
Religious polarization						-0.110 (0.213) [0.193]	-0.157 (0.234) [0.220]	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.075	0.092	0.090	0.087	0.091	0.087	0.084	0.078
Oster δ	3.72	4.33	4.55	4.34	5.01	4.53	4.09	3.72
Mean of dependent var	0.51	0.50	0.58	0.51	0.51	0.51	0.57	0.59
SD of dependent var	1.31	1.30	1.40	1.31	1.32	1.32	1.39	1.40
Adj. R-squared	0.36	0.33	0.34	0.32	0.33	0.33	0.40	0.39
Number of Obs.	6,189	5,835	5,307	6,239	6,189	6,189	4,952	5,288
Number of Countries	202	209	162	211	209	209	160	161
Number of Ethnic Groups	980	903	939	985	982	982	861	939

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016, computed as the inverse hyperbolic sine (IHS) of the value. Baseline controls include historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A11: Traditional herding and contemporary conflict globally: More controls

	Dependent variable: Number of conflict events (arsinh) — non-civil conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	0.708 (0.201)*** [0.251]***	0.832 (0.220)*** [0.263]***	0.830 (0.219)*** [0.258]***	0.787 (0.207)*** [0.256]***	0.828 (0.210)*** [0.257]***	0.791 (0.206)*** [0.254]***	0.827 (0.231)*** [0.245]***	0.833 (0.228)*** [0.240]***
Distance to capital (ln)	-0.142 (0.024)*** [0.028]***						-0.151 (0.028)*** [0.029]***	-0.152 (0.027)*** [0.030]***
Distance to country border (ln)	0.024 (0.015) [0.019]						0.034 (0.017)* [0.021]	0.031 (0.012)*** [0.015]**
Distance to coastline (ln)	0.006 (0.015) [0.019]						-0.004 (0.016) [0.022]	
On or cross a major river	0.384 (0.071)*** [0.108]***						0.380 (0.077)*** [0.114]***	0.375 (0.077)*** [0.113]***
Tsetse suitability index	-0.039 (0.029) [0.037]						-0.055 (0.033)* [0.038]	-0.052 (0.032) [0.037]
Temperature (Mean)	0.013 (0.005)** [0.006]**						0.010 (0.005)* [0.005]**	0.010 (0.005)* [0.005]**
Temperature (SD)	-0.043 (0.101) [0.127]						-0.148 (0.095) [0.117]	-0.154 (0.096) [0.117]
Precipitation (Mean)	0.000 (0.000) [0.000]						0.000 (0.000) [0.000]	
Precipitation (SD)	-0.002 (0.002) [0.003]						-0.003 (0.002) [0.003]	-0.003 (0.002) [0.003]
Nomadic indicator		0.217 (0.177) [0.243]					0.156 (0.198) [0.263]	0.099 (0.120) [0.134]
Semi-sedentary indicator		0.062 (0.126) [0.155]					0.059 (0.139) [0.161]	
Former slavery		-0.047 (0.052) [0.049]					-0.064 (0.055) [0.055]	-0.066 (0.055) [0.054]
Excluded from state power			0.071 (0.043) [0.043]				0.067 (0.043) [0.048]	0.071 (0.042)* [0.043]
Ethnic segregation			0.475 (0.205)** [0.309]				0.593 (0.210)*** [0.295]**	0.599 (0.211)*** [0.306]*
Share of cropland				-0.016 (0.089) [0.116]			-0.011 (0.103) [0.129]	
Share of urban areas				-0.249 (0.664) [0.643]			-0.041 (0.616) [0.579]	
Share of agnostics					-0.075 (0.076) [0.062]		0.054 (0.126) [0.089]	
Share of Buddhists					-0.456 (0.259)* [0.399]		-0.391 (0.204)* [0.271]	-0.445 (0.211)** [0.295]
Share of Christians					0.165 (0.157) [0.178]		0.145 (0.146) [0.157]	
Share of Hindus					-0.741 (0.298)** [0.166]***		-0.880 (0.292)*** [0.165]***	-1.011 (0.262)*** [0.192]***
Share of Jews					6.839 (0.180)*** [0.212]***		6.182 (0.230)*** [0.238]***	5.972 (0.175)*** [0.161]***
Share of Muslims					0.166 (0.154) [0.197]		0.170 (0.140) [0.156]	
Religious fractionalization						0.060 (0.076) [0.092]	0.083 (0.092) [0.107]	0.006 (0.003)** [0.003]**
Religious polarization						-0.104 (0.152) [0.184]	-0.155 (0.185) [0.216]	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.10	0.12	0.12	0.11	0.12	0.12	0.12	0.12
Oster δ	4.84	4.79	4.86	4.80	6.18	5.13	5.33	5.60
Mean of dependent var	0.36	0.35	0.41	0.36	0.36	0.36	0.40	0.40
SD of dependent var	1.06	1.04	1.12	1.06	1.06	1.06	1.11	1.11
Adj. R-squared	0.29	0.27	0.27	0.26	0.27	0.26	0.33	0.33
Number of Obs.	6,189	5,835	5,307	6,239	6,189	6,189	4,952	4,952
Number of Countries	202	209	162	211	209	209	160	160
Number of Ethnic Groups	980	903	939	985	982	982	861	861

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016, computed as the inverse hyperbolic sine (IHS) of the value. Baseline controls include historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A12: Traditional herding and contemporary conflict globally: More controls, PPML estimates

	Dependent variable: Number of conflict events — all conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	1.784 (1.154) [1.225]	3.483 (0.887)*** [0.863]***	2.858 (1.177)** [1.159]**	2.852 (1.157)** [1.150]**	3.248 (1.006)*** [0.922]***	2.967 (1.140)*** [1.079]***	3.028 (1.125)*** [1.287]***	2.188 (1.123)* [1.170]*
Distance to capital (ln)	-0.337 (0.074)*** [0.107]***						-0.404 (0.072)*** [0.095]***	-0.367 (0.075)*** [0.096]***
Distance to country border (ln)	0.362 (0.122)*** [0.147]**						0.367 (0.160)** [0.178]**	0.393 (0.136)*** [0.147]***
Distance to coastline (ln)	0.010 (0.135) [0.171]						0.023 (0.161) [0.174]	
On or cross a major river	0.338 (0.316) [0.455]						0.583 (0.249)** [0.375]	0.488 (0.238)** [0.359]
Tsetse suitability index	-0.048 (0.160) [0.161]						-0.168 (0.181) [0.154]	-0.093 (0.185) [0.171]
Temperature (Mean)	-0.020 (0.031) [0.039]						0.003 (0.020) [0.023]	0.003 (0.024) [0.026]
Temperature (SD)	-0.110 (0.763) [0.762]						-0.577 (1.049) [0.891]	-1.151 (1.088) [0.852]
Precipitation (Mean)	0.004 (0.003) [0.003]						0.006 (0.003)** [0.002]**	
Precipitation (SD)	-0.037 (0.034) [0.034]						-0.012 (0.028) [0.031]	-0.010 (0.031) [0.033]
Nomadic indicator		0.466 (0.840) [0.682]					-0.094 (0.957) [0.837]	
Semi-sedentary indicator		0.347 (0.845) [0.822]					0.488 (0.691) [0.623]	
Former slavery		0.472 (0.289) [0.309]					0.632 (0.274)** [0.313]**	
Excluded from state power			0.178 (0.522) [0.597]				0.512 (0.272)* [0.314]	0.706 (0.274)*** [0.326]**
Ethnic segregation			4.576 (1.641)*** [1.953]**				4.344 (2.017)** [2.249]*	4.582 (1.910)** [2.186]**
Share of cropland				-1.387 (0.426)*** [0.497]***			-0.681 (0.485) [0.475]	
Share of urban areas				-0.635 (3.680) [3.957]			-4.776 (4.780) [5.321]	
Share of agnostics					0.669 (1.032) [0.678]		-1.864 (1.855) [2.052]	
Share of Buddhists					-2.866 (0.994)*** [0.648]***		-3.911 (1.274)*** [1.487]***	-3.637 (1.224)*** [1.463]**
Share of Christians					-0.175 (0.705) [0.702]		-1.068 (0.593)* [0.557]*	
Share of Hindus					-1.690 (0.670)** [0.339]***		-2.413 (0.857)*** [0.613]***	-3.378 (0.923)*** [0.452]***
Share of Jews					8.502 (1.810)*** [1.021]***		6.878 (1.840)*** [0.952]***	6.613 (1.860)*** [0.928]***
Share of Muslims					1.962 (0.717)*** [0.663]***		0.347 (0.580) [0.451]	
Religious fractionalization						0.032 (0.084) [0.058]	0.252 (0.117)** [0.115]**	0.022 (0.008)*** [0.009]**
Religious polarization						-0.022 (0.178) [0.132]	-0.486 (0.246)** [0.240]**	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	21.0	20.4	23.9	20.9	21.1	21.1	23.2	23.9
SD of dependent var	259.2	258.4	276.5	258.7	259.5	259.5	275.9	276.8
Pseudo R-squared	0.68	0.63	0.60	0.60	0.63	0.60	0.75	0.73
Number of Obs.	5,535	5,186	4,861	5,558	5,521	5,521	4,545	4,850
Number of Countries	106	105	103	106	106	106	103	103
Number of Ethnic Groups	932	856	903	934	932	932	830	903

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Baseline controls include historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A13: Traditional herding and contemporary conflict globally: More controls, PPML estimates

	Dependent variable: Number of conflict events — civil conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	1.753 (1.254) [1.250]	3.632 (0.971)*** [0.905]***	2.828 (1.446)* [1.362]**	2.833 (1.401)** [1.336]**	3.433 (1.239)*** [1.106]***	2.989 (1.401)** [1.269]**	3.876 (1.057)*** [1.218]***	2.403 (1.092)** [1.106]**
Distance to capital (ln)	-0.368 (0.075)*** [0.110]***						-0.434 (0.078)*** [0.099]***	-0.412 (0.074)*** [0.098]***
Distance to country border (ln)	0.439 (0.134)*** [0.155]***						0.473 (0.163)*** [0.180]***	0.515 (0.143)*** [0.164]***
Distance to coastline (ln)	0.047 (0.167) [0.204]						0.052 (0.178) [0.175]	
On or cross a major river	0.280 (0.346) [0.484]						0.640 (0.243)*** [0.363]*	0.455 (0.254)* [0.362]
Tsetse suitability index	-0.022 (0.197) [0.244]						-0.057 (0.190) [0.212]	-0.077 (0.200) [0.251]
Temperature (Mean)	-0.019 (0.033) [0.041]						0.006 (0.019) [0.022]	0.029 (0.024) [0.027]
Temperature (SD)	-0.008 (0.658) [0.687]						-0.644 (0.952) [0.818]	-0.424 (0.971) [0.848]
Precipitation (Mean)	0.004 (0.004) [0.003]						0.005 (0.003)* [0.002]***	
Precipitation (SD)	-0.044 (0.032) [0.032]						-0.009 (0.024) [0.025]	-0.016 (0.029) [0.030]
Nomadic indicator		0.627 (0.795) [0.860]					-0.029 (0.755) [0.912]	
Semi-sedentary indicator		0.528 (0.827) [0.965]					0.581 (0.597) [0.673]	
Former slavery		0.492 (0.335) [0.354]					0.615 (0.271)** [0.308]**	
Excluded from state power			0.076 (0.556) [0.629]				0.584 (0.295)** [0.330]*	0.724 (0.264)*** [0.347]**
Ethnic segregation			4.872 (1.811)*** [2.149]**				3.575 (2.179) [2.206]	3.181 (1.833)* [2.008]
Share of cropland				-1.541 (0.413)*** [0.493]***			-0.638 (0.575) [0.587]	
Share of urban areas				-1.118 (4.678) [4.953]			-5.852 (5.396) [5.279]	
Share of agnostics					0.956 (1.098) [0.642]		-2.249 (1.743) [1.820]	
Share of Buddhists					-3.295 (1.072)*** [0.723]***		-4.467 (1.380)*** [1.592]***	-3.214 (1.148)*** [1.373]**
Share of Christians					-0.618 (0.831) [0.714]		-1.727 (0.703)** [0.605]***	
Share of Hindus					-1.780 (0.792)** [0.407]***		-3.136 (0.973)*** [0.643]***	-2.559 (0.836)*** [0.757]***
Share of Jews					8.983 (1.965)*** [1.162]***		6.335 (1.952)*** [0.932]***	8.666 (1.926)*** [1.114]***
Share of Muslims					2.256 (0.843)*** [0.733]***		-0.022 (0.693) [0.481]	2.004 (0.518)*** [0.780]**
Religious fractionalization						0.059 (0.090) [0.057]	0.321 (0.111)*** [0.103]***	0.026 (0.008)*** [0.010]***
Religious polarization						-0.073 (0.191) [0.129]	-0.615 (0.235)*** [0.214]***	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	16.8	16.3	19.1	16.7	16.9	16.9	18.6	19.1
SD of dependent var	238.0	238.9	253.8	237.5	238.3	238.3	255.1	254.1
Pseudo R-squared	0.72	0.66	0.63	0.63	0.66	0.62	0.80	0.78
Number of Obs.	5,304	4,970	4,661	5,327	5,290	5,290	4,356	4,650
Number of Countries	100	100	98	100	100	100	98	98
Number of Ethnic Groups	877	806	850	879	877	877	781	850

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Baseline controls include historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A14: Traditional herding and contemporary conflict globally: More controls, PPML estimates

	Dependent variable: Number of conflict events — non-civil conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	2.313 (1.389)* [1.471]	3.609 (1.258)*** [1.238]***	2.949 (1.249)** [1.355]**	2.928 (1.253)** [1.356]**	3.037 (1.174)*** [1.183]**	2.937 (1.191)** [1.243]**	3.097 (1.411)** [1.442]**	3.130 (1.452)** [1.477]**
Distance to capital (ln)	-0.200 (0.060)*** [0.067]***						-0.277 (0.074)*** [0.091]***	-0.289 (0.076)*** [0.089]***
Distance to country border (ln)	0.153 (0.097) [0.095]						0.126 (0.126) [0.115]	0.123 (0.113) [0.098]
Distance to coastline (ln)	0.067 (0.139) [0.139]						0.107 (0.155) [0.188]	
On or cross a major river	0.634 (0.351)* [0.364]*						0.624 (0.326)* [0.426]	0.658 (0.342)* [0.449]
Tsetse suitability index	-0.106 (0.177) [0.133]						-0.239 (0.212) [0.142]*	-0.219 (0.227) [0.148]
Temperature (Mean)	-0.011 (0.029) [0.037]						0.013 (0.021) [0.024]	0.009 (0.021) [0.024]
Temperature (SD)	-0.781 (1.166) [1.018]						-0.916 (1.286) [1.070]	-0.806 (1.296) [1.098]
Precipitation (Mean)	0.004 (0.004) [0.003]						0.007 (0.003)** [0.003]**	
Precipitation (SD)	-0.017 (0.053) [0.057]						-0.018 (0.045) [0.052]	0.009 (0.039) [0.044]
Nomadic indicator		0.013 (1.251) [0.958]					-0.338 (1.084) [0.678]	-0.492 (0.715) [0.484]
Semi-sedentary indicator		0.050 (1.001) [0.862]					0.176 (0.805) [0.692]	
Former slavery		0.372 (0.394) [0.423]					0.403 (0.320) [0.351]	0.408 (0.350) [0.378]
Excluded from state power			0.602 (0.343)* [0.376]				0.605 (0.259)** [0.343]*	0.615 (0.260)** [0.323]*
Ethnic segregation			3.256 (1.996) [2.263]				4.366 (2.014)** [2.628]*	4.406 (2.334)* [2.898]
Share of cropland				-0.991 (0.611) [0.740]			-0.768 (0.515) [0.534]	
Share of urban areas				-1.181 (1.933) [1.859]			-1.779 (2.648) [3.452]	
Share of agnostics					-65.638 (240.472) [185.101]		970.014 (589.730) [674.517]	
Share of Buddhists					-0.446 (1.301) [0.962]		-2.966 (1.405)** [1.406]**	-1.486 (1.610) [1.183]
Share of Christians					0.651 (0.453) [0.455]		-0.017 (0.608) [0.565]	
Share of Hindus					-1.388 (0.506)*** [0.302]***		-1.692 (0.864)* [0.606]***	-2.048 (0.770)*** [0.349]***
Share of Jews					283.246 (905.181) [529.764]		2284.126 (1152.745)** [1290.215]*	2118.053 (1122.253)* [1235.064]*
Share of Muslims					1.551 (0.518)*** [0.500]***		1.061 (0.541)** [0.417]**	
Religious fractionalization						-0.210 (0.133) [0.185]	-0.316 (0.161)** [0.158]**	0.011 (0.008) [0.008]
Religious polarization						0.464 (0.275)* [0.386]	0.664 (0.338)** [0.333]**	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	5.17	5.00	5.79	5.15	5.30	5.19	5.76	5.60
SD of dependent var	51.2	51.0	54.5	51.1	51.9	51.3	55.1	54.3
Pseudo R-squared	0.59	0.57	0.56	0.55	0.57	0.55	0.65	0.64
Number of Obs.	5,272	4,936	4,657	5,294	5,127	5,261	4,220	4,344
Number of Countries	84	82	82	84	83	84	80	80
Number of Ethnic Groups	904	827	883	906	896	904	801	806

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Baseline controls include historical settlement complexity, historical jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A15: Other pre-colonial ethnic features and contemporary conflict globally

	Dependent variable: Number of conflict events (IHS) — all conflicts				
	Specification A		Specification B		
	Additional variable	Obs.	Additional variable	Herding	Obs.
	(1)	(2)	(3)	(4)	(5)
Herding	0.101 (0.032)***	7036			
Jurisd. hierarchy beyond local, 1-5	0.094 (0.024)***	6319	0.080 (0.023)***	0.104 (0.031)***	6319
Political centralization, 0/1	0.085 (0.022)***	6319	0.073 (0.021)***	0.107 (0.031)***	6319
Jurisd. hierarchy local, 1-3	-0.010 (0.021)	6385	-0.008 (0.021)	0.117 (0.033)***	6385
Headman elected, 0/1	0.050 (0.022)**	5283	0.040 (0.024)	0.099 (0.039)**	5283
Property rights in land, 0/1	0.024 (0.024)	5003	0.025 (0.024)	0.123 (0.034)***	5003
Single inheritor for land, 0/1	-0.034 (0.022)	4828	-0.037 (0.023)	0.128 (0.034)***	4828
Institutional characteristics (PCA)	0.012 (0.017)	7038	-0.001 (0.017)	0.102 (0.031)***	7036
Patrilineality, 0/1	-0.011 (0.017)	6958	-0.013 (0.017)	0.108 (0.033)***	6956
Matrilineality, 0/1	-0.021 (0.016)	6958	-0.021 (0.017)	0.108 (0.033)***	6956
Patrilocality, 0/1	-0.009 (0.018)	6935	-0.017 (0.018)	0.106 (0.032)***	6933
Matrilocality, 0/1	0.001 (0.011)	6935	0.001 (0.012)	0.105 (0.032)***	6933
Polygyny, 0/1	-0.024 (0.020)	6815	-0.030 (0.021)	0.115 (0.031)***	6813
Clan communities, 0/1	-0.013 (0.018)	6290	-0.011 (0.017)	0.132 (0.034)***	6288
Kinship tightness, 0-1	-0.003 (0.015)	6890	0.002 (0.015)	0.114 (0.032)***	6888
Cousin marriage, 0/1	-0.010 (0.020)	5951	-0.010 (0.020)	0.108 (0.037)***	5949
Bride price, 0/1	0.045 (0.016)***	6901	0.036 (0.018)**	0.096 (0.032)***	6899
Female particip. in agriculture, 1-6	-0.073 (0.021)***	5115	-0.064 (0.021)***	0.089 (0.037)**	5115
Settlement complexity, 1-8	0.005 (0.022)	6502	0.027 (0.021)	0.139 (0.032)***	6502
Historical latitude (abs.)	0.090 (0.067)	7038	0.052 (0.064)	0.095 (0.030)***	7036
Nomadic, 0/1	0.029 (0.023)	6502	0.002 (0.022)	0.132 (0.032)***	6502
Semi-sedentary, 0/1	-0.015 (0.018)	6502	-0.015 (0.019)	0.133 (0.032)***	6502
Former slavery, 0/1	0.003 (0.022)	6143	0.002 (0.023)	0.131 (0.033)***	6141
High Gods Moral, 0/1	0.120 (0.037)***	4376	0.081 (0.036)**	0.124 (0.042)***	4376

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. They are measured as the inverse hyperbolic sine (arsinh) of the value. In specification A (columns 1–2) we regress conflict on various ethnic traits from Murdock (1967). In specification B (columns 3–5) we regress conflict on each of Murdock's additional variables and ancestral dependence on herding. Both specifications include country fixed effects. Standardized beta coefficients are reported with standard errors in parentheses clustered at the ethnicity level.

Table A16: Other pre-colonial ethnic features and contemporary conflict globally: PPML estimates

	Dependent variable: Number of conflict events — all conflicts				
	Specification A		Specification B		
	Additional variable	Obs.	Additional variable	Herding	Obs.
	(1)	(2)	(3)	(4)	(5)
Herding	1.867 (0.539)***	6311			
Jurisd. hierarchy beyond local, 1-5	0.412 (0.073)***	5626	0.378 (0.072)***	1.926 (0.480)***	5626
Political centralization, 0/1	1.015 (0.198)***	5626	0.899 (0.188)***	1.820 (0.500)***	5626
Jurisd. hierarchy local, 1-3	0.139 (0.136)	5690	0.104 (0.144)	2.202 (0.527)***	5690
Headman elected, 0/1	0.694 (0.224)***	4700	0.584 (0.230)**	2.136 (0.601)***	4700
Property rights in land, 0/1	0.677 (0.379)*	4474	0.747 (0.379)**	2.736 (0.634)***	4474
Single inheritor for land, 0/1	-0.319 (0.167)*	4304	-0.342 (0.166)**	2.500 (0.550)***	4304
Institutional characteristics (PCA)	0.065 (0.053)	6313	0.043 (0.054)	1.806 (0.517)***	6311
Patrilineality, 0/1	-0.393 (0.161)**	6235	-0.430 (0.163)***	1.911 (0.528)***	6233
Matrilineality, 0/1	-0.463 (0.292)	6235	-0.445 (0.290)	1.871 (0.558)***	6233
Patrilocality, 0/1	-0.096 (0.177)	6218	-0.227 (0.190)	2.002 (0.545)***	6216
Matrilocality, 0/1	-0.424 (0.574)	6218	-0.221 (0.610)	1.920 (0.551)***	6216
Polygyny, 0/1	-0.220 (0.166)	6127	-0.342 (0.169)**	2.251 (0.495)***	6125
Clan communities, 0/1	-0.225 (0.183)	5587	-0.236 (0.184)	2.445 (0.513)***	5585
Kinship tightness, 0-1	-0.426 (0.397)	6171	-0.247 (0.403)	2.128 (0.495)***	6169
Cousin marriage, 0/1	0.071 (0.224)	5281	0.062 (0.230)	1.993 (0.610)***	5279
Bride price, 0/1	0.163 (0.168)	6206	0.134 (0.185)	1.865 (0.532)***	6204
Female particip. in agriculture, 1-6	-0.341 (0.053)***	4669	-0.335 (0.053)***	1.836 (0.778)**	4669
Settlement complexity, 1-8	0.010 (0.045)	5807	0.119 (0.053)**	2.941 (0.616)***	5807
Historical latitude (abs.)	33.980 (31.053)	6313	23.993 (27.304)	1.734 (0.475)***	6311
Nomadic, 0/1	0.364 (0.231)	5807	-0.212 (0.279)	2.468 (0.601)***	5807
Semi-sedentary, 0/1	-0.344 (0.314)	5807	-0.424 (0.325)	2.319 (0.473)***	5807
Former slavery, 0/1	0.048 (0.172)	5483	0.031 (0.182)	2.557 (0.497)***	5481
High Gods Moral, 0/1	0.885 (0.290)***	3862	0.686 (0.297)**	1.851 (0.529)***	3862

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. In specification A (columns 1–2) we regress conflict on various ethnic traits from Murdock (1967). In specification B (columns 3–5) we regress conflict on each of Murdock's additional variables and ancestral dependence on herding. Both specifications include country fixed effects. Standardized beta coefficients are reported with standard errors in parentheses clustered at the ethnicity level.

Table A17: Correlation coefficients between herding and the covariates in Africa

Variable Name	Raw	Within country	Obs.
	(1)	(2)	(3)
Baseline controls:			
Settlement complexity	-0.522***	-0.434***	2,199
Jurisdictional hierarchy	0.175***	0.069	2,143
Historical latitude (abs.)	0.272***	0.256***	2,285
Population density (ln)	-0.019	-0.087***	2,276
Ethnic fractionalization	-0.223***	-0.098***	2,285
Ethnic polarization	0.185***	0.020	2,285
Nighttime lights	-0.004	-0.054***	2,279
Share of land for herding	0.262***	0.117***	2,285
Ruggedness	0.146***	0.071**	2,279
Extended controls:			
Distance to capital (ln)	-0.111***	-0.012	2,277
Distance to country border (ln)	0.002	-0.027	2,285
Distance to coastline (ln)	0.069*	0.107***	2,285
On or cross a major river	0.019	0.045	2,285
Tsetse suitability index	-0.432***	-0.291***	2,285
Temperature (Mean)	0.034	0.044	2,284
Temperature (SD)	0.147***	0.198***	2,284
Precipitation (Mean)	-0.399***	-0.248***	2,284
Precipitation (SD)	-0.223***	-0.199***	2,284
Nomadic indicator	0.553***	0.478***	2,199
Semi-sedentary indicator	0.058	0.080*	2,199
Former slavery	-0.081	-0.086*	2,016
Excluded from state power	0.063*	0.015	1,903
Ethnic segregation	0.179***	0.037	2,283
Share of cropland	0.083**	0.065*	2,285
Share of urban areas	0.003	-0.025**	2,285
Share of Christians	-0.168***	-0.089*	2,275
Share of Muslims	0.183***	0.085**	2,275
Religious fractionalization	0.055*	0.035	2,275
Religious polarization	0.056*	0.035	2,275

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. Column 1 reports the raw correlation coefficients between herding and the covariates. Column 2 reports the partial correlation conditional on country fixed effects. Significance levels are computed based on standard errors clustered at the ethnicity level. *, **, and *** indicate significance at the 10, 5, and 1% levels.

Table A18: Traditional herding and contemporary conflict in Africa: More controls

	Dependent variable: Number of conflict events (IHS) — all conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	1.768 (0.559)*** [0.464]***	1.697 (0.660)** [0.637]**	2.010 (0.553)*** [0.504]***	1.974 (0.593)*** [0.493]***	1.923 (0.602)*** [0.500]***	1.939 (0.576)*** [0.480]***	1.564 (0.559)*** [0.585]**	1.638 (0.534)*** [0.467]***
Distance to capital (ln)	-0.423 (0.045)*** [0.065]***						-0.408 (0.049)*** [0.070]***	-0.393 (0.045)*** [0.064]***
Distance to country border (ln)	0.131 (0.040)*** [0.064]**						0.142 (0.045)*** [0.062]**	0.103 (0.042)** [0.064]
Distance to coastline (ln)	-0.033 (0.074) [0.086]						-0.020 (0.076) [0.086]	
On or cross a major river	1.020 (0.171)*** [0.184]***						1.006 (0.185)*** [0.194]***	0.977 (0.173)*** [0.189]***
Tsetse suitability index	-0.038 (0.092) [0.142]						-0.133 (0.100) [0.118]	
Temperature (Mean)	0.051 (0.023)** [0.035]						0.040 (0.028) [0.048]	
Temperature (SD)	-0.430 (0.594) [0.941]						0.069 (0.675) [0.840]	
Precipitation (Mean)	0.004 (0.002) [0.002]						0.003 (0.003) [0.003]	
Precipitation (SD)	-0.050 (0.012)*** [0.010]***						-0.046 (0.014)*** [0.011]**	-0.039 (0.011)*** [0.008]***
Nomadic indicator		1.228 (0.525)** [0.560]**					0.924 (0.506)* [0.512]*	0.592 (0.430) [0.339]*
Semi-sedentary indicator		0.654 (0.509) [0.465]					0.633 (0.475) [0.460]	
Former slavery		0.017 (0.124) [0.105]					-0.008 (0.127) [0.110]	
Excluded from state power			0.368 (0.129)*** [0.144]**				0.340 (0.134)** [0.155]**	0.353 (0.122)*** [0.135]**
Ethnic segregation			5.063 (1.117)*** [1.129]***				4.338 (1.221)*** [1.230]***	4.347 (1.157)*** [1.249]***
Share of cropland				-0.395 (0.209)* [0.253]			-0.407 (0.281) [0.289]	-0.522 (0.220)** [0.235]**
Share of urban areas				1.122 (4.613) [5.823]			-1.489 (3.708) [4.401]	
Share of Christians					0.163 (0.282) [0.250]		-0.132 (0.265) [0.259]	
Share of Muslims					-0.009 (0.247) [0.226]		-0.030 (0.247) [0.238]	
Religious fractionalization						0.152 (0.101) [0.116]	0.123 (0.045)*** [0.059]**	0.015 (0.004)*** [0.004]***
Religious polarization						-0.264 (0.204) [0.233]	-0.218 (0.093)** [0.119]*	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.13	0.12	0.14	0.14	0.14	0.14	0.11	0.12
Oster δ	6.79	4.79	12.7	10.1	8.84	9.63	4.43	5.13
Mean of dependent var	1.66	1.69	1.86	1.65	1.65	1.65	1.90	1.86
SD of dependent var	2.08	2.09	2.15	2.08	2.08	2.08	2.16	2.15
Adj. R-squared	0.42	0.32	0.36	0.33	0.34	0.34	0.44	0.45
Number of Obs.	2,130	1,893	1,780	2,133	2,132	2,132	1,592	1,778
Number of Countries	54	55	51	56	56	56	50	51
Number of Ethnic Groups	449	396	439	449	449	449	388	439

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A19: Traditional herding and contemporary conflict in Africa: More controls

	Dependent variable: Number of conflict events (IHS) — civil conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	1.503 (0.492)*** [0.428]***	1.520 (0.585)*** [0.569]**	1.795 (0.510)*** [0.510]***	1.760 (0.523)*** [0.461]***	1.727 (0.521)*** [0.465]***	1.749 (0.509)*** [0.459]***	1.293 (0.531)** [0.527]**	1.373 (0.510)*** [0.457]***
Distance to capital (ln)	-0.383 (0.044)*** [0.064]***						-0.380 (0.047)*** [0.067]***	-0.353 (0.044)*** [0.062]***
Distance to country border (ln)	0.078 (0.035)** [0.058]						0.083 (0.039)** [0.052]	0.058 (0.037) [0.054]
Distance to coastline (ln)	0.004 (0.065) [0.082]						0.019 (0.070) [0.080]	
On or cross a major river	0.826 (0.157)*** [0.162]***						0.817 (0.173)*** [0.178]***	0.811 (0.161)*** [0.157]***
Tsetse suitability index	-0.047 (0.084) [0.129]						-0.103 (0.094) [0.110]	
Temperature (Mean)	0.052 (0.021)** [0.033]						0.041 (0.026) [0.047]	
Temperature (SD)	0.074 (0.487) [0.713]						0.396 (0.577) [0.653]	
Precipitation (Mean)	0.004 (0.002)* [0.002]						0.003 (0.002) [0.003]	
Precipitation (SD)	-0.049 (0.011)*** [0.010]***						-0.047 (0.013)*** [0.011]***	-0.037 (0.010)*** [0.007]***
Nomadic indicator		1.078 (0.479)** [0.503]**					0.838 (0.469)* [0.487]*	0.506 (0.429) [0.345]
Semi-sedentary indicator		0.731 (0.459) [0.447]					0.716 (0.458) [0.446]	
Former slavery		-0.075 (0.109) [0.103]					-0.109 (0.115) [0.106]	
Excluded from state power			0.327 (0.123)*** [0.152]**				0.362 (0.126)*** [0.159]**	0.357 (0.119)*** [0.142]**
Ethnic segregation			3.514 (1.114)*** [1.103]***				2.944 (1.224)** [1.156]**	3.262 (1.164)*** [1.165]***
Share of cropland				-0.249 (0.183) [0.250]			-0.358 (0.242) [0.290]	
Share of urban areas				1.591 (4.097) [4.872]			-0.967 (3.404) [3.905]	
Share of Christians					0.048 (0.233) [0.210]		-0.292 (0.224) [0.210]	
Share of Muslims					0.050 (0.205) [0.198]		-0.090 (0.211) [0.199]	
Religious fractionalization						0.150 (0.094) [0.107]	0.120 (0.045)*** [0.058]**	0.013 (0.004)*** [0.004]***
Religious polarization						-0.266 (0.190) [0.215]	-0.215 (0.092)** [0.116]*	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.12	0.12	0.14	0.14	0.14	0.14	0.10	0.11
Oster δ	4.25	3.32	6.17	5.26	4.91	5.48	2.93	3.33
Mean of dependent var	1.24	1.27	1.41	1.24	1.24	1.24	1.43	1.41
SD of dependent var	1.84	1.84	1.92	1.84	1.84	1.84	1.92	1.92
Adj. R-squared	0.41	0.32	0.35	0.33	0.33	0.34	0.43	0.43
Number of Obs.	2,130	1,893	1,780	2,133	2,132	2,132	1,592	1,778
Number of Countries	54	55	51	56	56	56	50	51
Number of Ethnic Groups	449	396	439	449	449	449	388	439

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A20: Traditional herding and contemporary conflict in Africa: More controls

	Dependent variable: Number of conflict events (IHS) — non-civil conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	1.478 (0.452)*** [0.357]***	1.428 (0.562)** [0.522]***	1.682 (0.464)*** [0.384]***	1.640 (0.496)*** [0.386]***	1.596 (0.510)*** [0.397]***	1.601 (0.482)*** [0.376]***	1.346 (0.451)*** [0.479]***	1.351 (0.423)*** [0.360]***
Distance to capital (ln)	-0.433 (0.044)*** [0.061]***						-0.423 (0.046)*** [0.066]***	-0.409 (0.043)*** [0.061]***
Distance to country border (ln)	0.128 (0.034)*** [0.054]**						0.141 (0.040)*** [0.058]**	0.118 (0.037)*** [0.059]**
Distance to coastline (ln)	-0.035 (0.065) [0.072]						-0.013 (0.070) [0.078]	
On or cross a major river	0.874 (0.151)*** [0.180]***						0.917 (0.158)*** [0.199]***	0.867 (0.149)*** [0.187]***
Tsetse suitability index	-0.058 (0.079) [0.119]						-0.153 (0.088)* [0.100]	
Temperature (Mean)	0.031 (0.019) [0.032]						0.025 (0.024) [0.043]	
Temperature (SD)	-0.438 (0.520) [0.850]						-0.012 (0.619) [0.832]	
Precipitation (Mean)	0.003 (0.002) [0.002]						0.003 (0.002) [0.003]	
Precipitation (SD)	-0.039 (0.010)*** [0.011]***						-0.037 (0.011)*** [0.012]***	-0.030 (0.009)*** [0.008]***
Nomadic indicator		0.966 (0.462)** [0.505]*					0.734 (0.434)* [0.458]	0.570 (0.363) [0.313]*
Semi-sedentary indicator		0.214 (0.471) [0.413]					0.217 (0.427) [0.393]	
Former slavery		0.047 (0.106) [0.088]					0.014 (0.108) [0.094]	
Excluded from state power			0.272 (0.107)** [0.107]**				0.187 (0.114) [0.117]	0.245 (0.102)** [0.117]**
Ethnic segregation			4.739 (0.914)*** [0.855]***				4.066 (0.970)*** [1.065]***	4.112 (0.902)*** [1.020]***
Share of cropland				-0.356 (0.174)** [0.187]*			-0.344 (0.246) [0.239]	-0.421 (0.186)** [0.184]**
Share of urban areas				1.298 (4.457) [5.558]			-1.588 (3.519) [4.059]	
Share of Christians					0.250 (0.246) [0.214]		-0.038 (0.232) [0.210]	
Share of Muslims					-0.055 (0.211) [0.179]		-0.067 (0.209) [0.183]	
Religious fractionalization						0.122 (0.093) [0.101]	0.084 (0.038)** [0.044]*	0.010 (0.004)** [0.003]***
Religious polarization						-0.211 (0.188) [0.203]	-0.149 (0.079)* [0.090]	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.12	0.12	0.14	0.14	0.13	0.13	0.11	0.11
Oster δ	7.46	4.97	13.7	11.4	10.0	10.4	4.58	5.18
Mean of dependent var	1.21	1.25	1.37	1.21	1.21	1.21	1.41	1.38
SD of dependent var	1.78	1.80	1.86	1.78	1.78	1.78	1.88	1.86
Adj. R-squared	0.41	0.30	0.34	0.31	0.31	0.31	0.44	0.45
Number of Obs.	2,130	1,893	1,780	2,133	2,132	2,132	1,592	1,778
Number of Countries	54	55	51	56	56	56	50	51
Number of Ethnic Groups	449	396	439	449	449	449	388	439

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A21: Traditional herding and contemporary conflict in Africa: More controls

	Dependent variable: Number of conflict events (IHS) — localized conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	1.223 (0.381)*** [0.302]***	1.225 (0.502)** [0.482]**	1.423 (0.415)*** [0.344]**	1.343 (0.437)*** [0.351]**	1.292 (0.452)*** [0.361]**	1.297 (0.424)*** [0.344]**	1.224 (0.390)*** [0.432]**	1.176 (0.365)*** [0.324]**
Distance to capital (ln)	-0.458 (0.040)*** [0.053]**						-0.450 (0.042)*** [0.057]**	-0.435 (0.040)*** [0.052]**
Distance to country border (ln)	0.114 (0.028)*** [0.046]**						0.130 (0.034)*** [0.050]**	0.118 (0.031)*** [0.050]**
Distance to coastline (ln)	-0.021 (0.057) [0.064]						-0.001 (0.061) [0.071]	
On or cross a major river	0.874 (0.137)*** [0.176]**						0.942 (0.145)*** [0.200]**	0.897 (0.137)*** [0.176]**
Tsetse suitability index	-0.034 (0.065) [0.100]						-0.113 (0.075) [0.088]	
Temperature (Mean)	0.023 (0.016) [0.024]						0.023 (0.019) [0.031]	
Temperature (SD)	-0.588 (0.405) [0.466]						-0.299 (0.500) [0.541]	-0.557 (0.410) [0.532]
Precipitation (Mean)	0.003 (0.002) [0.002]						0.003 (0.002) [0.003]	
Precipitation (SD)	-0.030 (0.009)*** [0.010]**						-0.030 (0.010)*** [0.011]**	-0.024 (0.008)*** [0.009]**
Nomadic indicator		0.787 (0.407)* [0.464]*					0.627 (0.369)* [0.412]	0.503 (0.319) [0.300]*
Semi-sedentary indicator		0.042 (0.440) [0.403]					0.119 (0.393) [0.387]	
Former slavery		0.077 (0.089) [0.090]					0.037 (0.087) [0.090]	
Excluded from state power			0.205 (0.102)** [0.098]**				0.121 (0.100) [0.095]	0.162 (0.092)* [0.096]*
Ethnic segregation			3.807 (0.782)*** [0.733]**				3.269 (0.797)*** [0.864]**	3.308 (0.718)*** [0.776]**
Share of cropland				-0.383 (0.143)*** [0.162]**			-0.321 (0.205) [0.215]	-0.343 (0.157)** [0.169]**
Share of urban areas				2.130 (4.216) [5.114]			-0.715 (3.267) [3.636]	
Share of Christians					0.332 (0.213) [0.174]*		0.027 (0.184) [0.151]	
Share of Muslims					0.034 (0.184) [0.124]		-0.000 (0.176) [0.148]	
Religious fractionalization						0.141 (0.094) [0.102]	0.096 (0.035)*** [0.040]**	0.008 (0.004)** [0.003]**
Religious polarization						-0.252 (0.189) [0.205]	-0.176 (0.072)** [0.082]**	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.11	0.11	0.13	0.12	0.12	0.12	0.11	0.11
Oster δ	7.33	4.84	11.4	10.3	8.81	9.17	4.80	5.06
Mean of dependent var	0.93	0.96	1.06	0.93	0.93	0.93	1.09	1.06
SD of dependent var	1.58	1.60	1.67	1.58	1.58	1.58	1.68	1.67
Adj. R-squared	0.45	0.30	0.34	0.31	0.31	0.32	0.47	0.48
Number of Obs.	2,130	1,893	1,780	2,133	2,132	2,132	1,592	1,778
Number of Countries	54	55	51	56	56	56	50	51
Number of Ethnic Groups	449	396	439	449	449	449	388	439

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A22: Traditional herding and contemporary conflict in Africa: More controls, PPML estimates

Dependent variable: Number of conflict events — all conflicts								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	2.346 (0.628)*** [0.770]***	1.478 (0.896)* [1.002]	1.900 (0.721)*** [0.776]**	2.121 (0.727)*** [0.749]***	2.179 (0.693)*** [0.734]***	2.068 (0.740)*** [0.765]***	1.934 (0.759)** [1.026]*	1.280 (0.674)* [0.704]*
Distance to capital (ln)	-0.366 (0.039)*** [0.048]***						-0.378 (0.045)*** [0.057]***	-0.369 (0.045)*** [0.056]***
Distance to country border (ln)	0.240 (0.087)*** [0.112]**						0.197 (0.092)** [0.114]*	0.192 (0.078)** [0.093]**
Distance to coastline (ln)	0.036 (0.148) [0.193]						0.073 (0.157) [0.207]	
On or cross a major river	0.766 (0.241)*** [0.204]***						0.813 (0.234)*** [0.223]***	0.774 (0.246)*** [0.183]***
Tsetse suitability index	0.085 (0.136) [0.169]						0.105 (0.132) [0.130]	
Temperature (Mean)	0.025 (0.041) [0.054]						-0.016 (0.040) [0.055]	
Temperature (SD)	-2.605 (1.004)*** [1.232]**						-2.699 (1.179)** [1.191]**	
Precipitation (Mean)	0.003 (0.004) [0.006]						0.002 (0.004) [0.004]	
Precipitation (SD)	-0.085 (0.025)*** [0.029]***						-0.065 (0.027)** [0.030]**	-0.091 (0.024)*** [0.021]***
Nomadic indicator		1.290 (0.871) [0.941]					0.729 (0.628) [0.451]	0.533 (0.561) [0.408]
Semi-sedentary indicator		0.214 (0.942) [0.941]					0.610 (0.679) [0.619]	
Former slavery		0.153 (0.241) [0.301]					-0.081 (0.220) [0.235]	
Excluded from state power			0.344 (0.306) [0.282]				0.406 (0.165)** [0.183]**	0.628 (0.186)*** [0.210]***
Ethnic segregation			-0.488 (1.381) [1.443]				-0.050 (2.475) [2.657]	0.206 (1.443) [1.554]
Share of cropland				-0.586 (0.467) [0.488]			-0.105 (0.475) [0.447]	-0.604 (0.562) [0.564]
Share of urban areas				-1.254 (2.804) [2.457]			3.708 (7.207) [7.221]	
Share of Christians					0.262 (0.436) [0.437]		-0.805 (0.524) [0.384]**	
Share of Muslims					-0.270 (0.435) [0.466]		-0.598 (0.515) [0.328]*	
Religious fractionalization						0.016 (0.065) [0.065]	0.010 (0.057) [0.051]	0.011 (0.007) [0.006]*
Religious polarization						0.020 (0.133) [0.129]	-0.001 (0.120) [0.105]	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	55.9	55.7	64.3	55.9	55.9	55.9	65.2	64.4
SD of dependent var	419.4	424.2	452.3	419.3	419.4	419.4	461.1	452.6
Pseudo R-squared	0.76	0.62	0.62	0.62	0.62	0.62	0.78	0.76
Number of Obs.	2,118	1,879	1,771	2,119	2,118	2,118	1,584	1,769
Number of Countries	45	44	44	45	45	45	44	44
Number of Ethnic Groups	448	395	437	448	448	448	387	437

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A23: Traditional herding and contemporary conflict in Africa: More controls, PPML estimates

	Dependent variable: Number of conflict events — civil conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	2.547 (0.870)*** [0.851]***	1.850 (1.013)* [1.244]	2.235 (0.800)*** [0.943]**	2.480 (0.780)*** [0.863]***	2.487 (0.777)*** [0.839]***	2.394 (0.796)*** [0.889]***	2.239 (0.966)** [1.142]**	1.695 (0.845)** [0.896]*
Distance to capital (ln)	-0.327 (0.046)*** [0.052]***						-0.347 (0.055)*** [0.060]***	-0.334 (0.051)*** [0.062]***
Distance to country border (ln)	0.174 (0.085)** [0.094]*						0.094 (0.077) [0.081]	0.123 (0.070)* [0.069]*
Distance to coastline (ln)	0.089 (0.177) [0.210]						0.145 (0.153) [0.202]	
On or cross a major river	0.791 (0.301)*** [0.269]***						0.781 (0.292)*** [0.289]***	0.783 (0.307)** [0.247]***
Tsetse suitability index	-0.090 (0.188) [0.216]						0.006 (0.162) [0.151]	
Temperature (Mean)	0.026 (0.046) [0.067]						-0.029 (0.041) [0.059]	
Temperature (SD)	-1.486 (1.082) [1.144]						-1.553 (1.139) [0.935]*	
Precipitation (Mean)	0.001 (0.005) [0.006]						0.000 (0.004) [0.004]	
Precipitation (SD)	-0.083 (0.027)*** [0.030]***						-0.063 (0.027)** [0.029]**	-0.096 (0.024)*** [0.022]***
Nomadic indicator		1.526 (1.004) [0.959]					0.662 (0.822) [0.502]	0.574 (0.687) [0.557]
Semi-sedentary indicator		0.469 (0.895) [0.903]					0.418 (0.787) [0.619]	
Former slavery		0.215 (0.241) [0.302]					0.027 (0.228) [0.230]	
Excluded from state power			0.445 (0.283) [0.279]				0.409 (0.191)** [0.222]*	0.684 (0.260)*** [0.301]**
Ethnic segregation			-0.576 (1.525) [1.753]				-0.114 (2.772) [3.033]	0.177 (1.788) [2.024]
Share of cropland				-0.614 (0.505) [0.645]			0.022 (0.422) [0.400]	
Share of urban areas				-0.846 (3.212) [2.742]			5.913 (8.820) [8.905]	
Share of Christians					0.053 (0.484) [0.587]		-0.897 (0.572) [0.499]*	
Share of Muslims					-0.239 (0.428) [0.507]		-0.485 (0.560) [0.422]	
Religious fractionalization						0.032 (0.063) [0.063]	0.041 (0.059) [0.055]	0.008 (0.008) [0.007]
Religious polarization						-0.018 (0.130) [0.123]	-0.071 (0.128) [0.118]	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	28.9	28.6	33.6	28.9	28.9	28.9	33.5	33.6
SD of dependent var	227.5	231.5	247.5	227.4	227.5	227.5	251.7	247.6
Pseudo R-squared	0.72	0.63	0.61	0.61	0.61	0.61	0.75	0.73
Number of Obs.	2,118	1,879	1,771	2,119	2,118	2,118	1,584	1,769
Number of Countries	45	44	44	45	45	45	44	44
Number of Ethnic Groups	448	395	437	448	448	448	387	437

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A24: Traditional herding and contemporary conflict in Africa: More controls, PPML estimates

	Dependent variable: Number of conflict events — non-civil conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	2.223 (0.627)*** [0.903]**	1.246 (0.883) [0.902]	1.693 (0.747)** [0.711]**	1.835 (0.786)** [0.744]**	1.907 (0.743)** [0.743]**	1.805 (0.802)** [0.760]**	1.714 (0.687)** [1.021]*	0.909 (0.709) [0.678]
Distance to capital (ln)	-0.419 (0.035)*** [0.043]***						-0.424 (0.035)*** [0.043]***	-0.422 (0.044)*** [0.053]***
Distance to country border (ln)	0.343 (0.094)*** [0.131]***						0.349 (0.101)*** [0.156]**	0.277 (0.099)*** [0.141]*
Distance to coastline (ln)	-0.058 (0.130) [0.156]						-0.056 (0.143) [0.170]	
On or cross a major river	0.749 (0.220)*** [0.202]***						0.792 (0.211)*** [0.195]***	0.815 (0.239)*** [0.211]***
Tsetse suitability index	0.150 (0.114) [0.136]						0.075 (0.125) [0.097]	
Temperature (Mean)	0.021 (0.041) [0.038]						0.013 (0.045) [0.051]	
Temperature (SD)	-4.164 (1.049)*** [1.486]***						-4.157 (1.249)*** [1.620]**	
Precipitation (Mean)	0.003 (0.004) [0.006]						0.004 (0.004) [0.004]	
Precipitation (SD)	-0.080 (0.023)*** [0.023]***						-0.072 (0.026)*** [0.027]***	-0.076 (0.026)*** [0.027]***
Nomadic indicator		0.870 (0.839) [1.063]					0.971 (0.548)* [0.455]**	0.429 (0.539) [0.469]
Semi-sedentary indicator		-0.027 (1.076) [1.130]					0.959 (0.627) [0.607]	
Former slavery		0.113 (0.285) [0.341]					-0.201 (0.242) [0.231]	
Excluded from state power			0.230 (0.396) [0.383]				0.350 (0.162)** [0.169]**	0.596 (0.169)*** [0.188]***
Ethnic segregation			-0.307 (1.517) [1.420]				0.283 (1.500) [1.616]	0.815 (1.240) [1.237]
Share of cropland				-0.445 (0.511) [0.488]			-0.130 (0.522) [0.603]	-0.686 (0.517) [0.508]
Share of urban areas				-2.339 (2.702) [2.505]			2.288 (3.277) [3.308]	
Share of Christians					0.401 (0.406) [0.343]		-0.815 (0.507) [0.326]**	
Share of Muslims					-0.269 (0.477) [0.392]		-0.657 (0.519) [0.299]**	
Religious fractionalization						0.003 (0.069) [0.070]	0.002 (0.060) [0.055]	0.015 (0.007)** [0.006]**
Religious polarization						0.060 (0.142) [0.140]	0.021 (0.126) [0.109]	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	27.0	27.1	30.7	27.0	27.0	27.0	31.7	30.7
SD of dependent var	208.7	208.6	222.3	208.7	208.7	208.7	226.8	222.4
Pseudo R-squared	0.80	0.63	0.63	0.64	0.64	0.65	0.81	0.80
Number of Obs.	2,118	1,879	1,771	2,119	2,118	2,118	1,584	1,769
Number of Countries	45	44	44	45	45	45	44	44
Number of Ethnic Groups	448	395	437	448	448	448	387	437

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A25: Traditional herding and contemporary conflict in Africa: More controls, PPML estimates

	Dependent variable: Number of conflict events — localized conflicts							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	1.481 (0.665)** [0.801]*	1.785 (0.880)** [0.869]**	2.197 (0.772)** [0.838]**	2.320 (0.804)** [0.796]**	2.454 (0.738)** [0.811]**	2.333 (0.817)** [0.829]**	1.046 (0.672) [0.920]	1.394 (0.665)** [0.997]
Distance to capital (ln)	-0.451 (0.036)** [0.042]**						-0.444 (0.038)** [0.038]**	-0.408 (0.035)** [0.038]**
Distance to country border (ln)	0.359 (0.087)** [0.107]**						0.361 (0.091)** [0.125]**	0.253 (0.077)** [0.107]**
Distance to coastline (ln)	-0.153 (0.108) [0.133]						-0.096 (0.118) [0.139]	
On or cross a major river	1.069 (0.234)** [0.196]**						0.973 (0.226)** [0.206]**	1.010 (0.223)** [0.184]**
Tsetse suitability index	0.271 (0.104)** [0.091]**						0.109 (0.133) [0.094]	
Temperature (Mean)	0.041 (0.040) [0.028]						0.061 (0.048) [0.046]	
Temperature (SD)	-4.146 (1.035)** [1.031]**						-3.506 (1.458)** [1.600]**	-3.348 (1.140)** [1.417]**
Precipitation (Mean)	0.005 (0.004) [0.004]						0.006 (0.004) [0.004]	
Precipitation (SD)	-0.090 (0.029)** [0.026]**						-0.081 (0.030)** [0.028]**	-0.058 (0.025)** [0.026]**
Nomadic indicator		0.966 (0.809) [1.007]					0.676 (0.541) [0.508]	0.120 (0.639) [0.479]
Semi-sedentary indicator		0.414 (1.049) [1.026]					1.019 (0.588)* [0.581]*	
Former slavery		0.050 (0.234) [0.260]					-0.253 (0.235) [0.202]	
Excluded from state power			0.540 (0.321)* [0.312]*				0.378 (0.194)* [0.200]*	0.562 (0.184)** [0.193]**
Ethnic segregation			-1.548 (1.225) [0.983]				-0.500 (1.337) [1.383]	0.581 (1.169) [1.233]
Share of cropland				-0.745 (0.519) [0.502]			-0.414 (0.546) [0.641]	-0.813 (0.523) [0.607]
Share of urban areas				-3.211 (2.644) [2.203]			1.966 (2.112) [2.269]	
Share of Christians					0.773 (0.414)* [0.354]**		-0.449 (0.568) [0.388]	
Share of Muslims					-0.189 (0.484) [0.432]		-0.773 (0.597) [0.379]**	
Religious fractionalization						0.023 (0.068) [0.070]	0.009 (0.053) [0.047]	0.016 (0.007)** [0.005]**
Religious polarization						0.030 (0.140) [0.138]	0.010 (0.112) [0.093]	
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	16.2	16.0	18.1	16.2	16.2	16.2	18.8	18.1
SD of dependent var	121.6	117.5	124.3	121.6	121.6	121.6	127.8	124.3
Pseudo R-squared	0.83	0.65	0.66	0.67	0.67	0.68	0.84	0.82
Number of Obs.	2,118	1,879	1,771	2,119	2,118	2,118	1,584	1,769
Number of Countries	45	44	44	45	45	45	44	44
Number of Ethnic Groups	448	395	437	448	448	448	387	437

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997–2016. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A26: Other pre-colonial ethnic features and contemporary conflict in Africa

	Dependent variable: Number of conflict events (IHS) — all conflicts				
	Specification A		Specification B		
	Additional variable	Obs.	Additional variable	Herding	Obs.
	(1)	(2)	(3)	(4)	(5)
Herding	0.124 (0.037)***	2285			
Jurisd. hierarchy beyond local, 1-5	0.104 (0.029)***	2143	0.096 (0.028)***	0.118 (0.040)***	2143
Political centralization, 0/1	0.105 (0.029)***	2143	0.098 (0.029)***	0.118 (0.039)***	2143
Jurisd. hierarchy local, 1-3	-0.004 (0.030)	2149	0.015 (0.029)	0.129 (0.041)***	2149
Headman elected, 0/1	0.090 (0.031)***	1701	0.092 (0.028)***	0.116 (0.038)***	1701
Property rights in land, 0/1	-0.007 (0.026)	1821	0.007 (0.024)	0.108 (0.041)***	1821
Single inheritor for land, 0/1	0.004 (0.031)	1767	0.006 (0.032)	0.113 (0.043)***	1767
Institutional characteristics (PCA)	-0.013 (0.028)	2285	-0.007 (0.027)	0.124 (0.037)***	2285
Patrilineality, 0/1	0.025 (0.030)	2233	0.017 (0.031)	0.149 (0.033)***	2233
Matrilineality, 0/1	-0.035 (0.033)	2233	-0.026 (0.035)	0.148 (0.033)***	2233
Patrilocality, 0/1	0.006 (0.032)	2265	-0.008 (0.033)	0.135 (0.039)***	2265
Matrilocality, 0/1	0.015 (0.019)	2265	0.023 (0.019)	0.136 (0.039)***	2265
Polygyny, 0/1	0.023 (0.029)	2246	0.027 (0.030)	0.128 (0.037)***	2246
Clan communities, 0/1	-0.029 (0.029)	1917	-0.021 (0.028)	0.144 (0.043)***	1917
Kinship tightness, 0-1	-0.022 (0.025)	2258	-0.011 (0.026)	0.136 (0.039)***	2258
Cousin marriage, 0/1	-0.020 (0.037)	1710	-0.019 (0.036)	0.176 (0.035)***	1710
Bride price, 0/1	0.045 (0.024)*	2284	0.027 (0.024)	0.119 (0.038)***	2284
Female particip. in agriculture, 1-6	0.011 (0.040)	1477	0.020 (0.040)	0.089 (0.038)**	1477
Settlement complexity, 1-8	-0.023 (0.035)	2199	0.040 (0.033)	0.152 (0.038)***	2199
Historical latitude (abs.)	0.044 (0.040)	2285	-0.004 (0.040)	0.125 (0.038)***	2285
Nomadic, 0/1	0.076 (0.039)**	2199	0.022 (0.039)	0.122 (0.039)***	2199
Semi-sedentary, 0/1	0.017 (0.034)	2199	0.007 (0.031)	0.133 (0.039)***	2199
Former slavery, 0/1	0.006 (0.031)	2016	0.017 (0.030)	0.136 (0.040)***	2016
High Gods Moral, 0/1	0.112 (0.050)**	1197	0.062 (0.050)	0.133 (0.064)**	1197

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. In specification A (columns 1–2) we regress conflict on various ethnic traits from Murdock (1967). In specification B (columns 3–5) we regress conflict on each of Murdock's additional variables and ancestral dependence on herding. Both specifications include country fixed effects. Standardized beta coefficients are reported with standard errors in parentheses clustered at the ethnicity level.

Table A27: Other pre-colonial ethnic features and contemporary conflict in Africa: PPML estimates

	Dependent variable: Number of conflict events — all conflicts				
	Specification A		Specification B		
	Additional variable	Obs.	Additional variable	Herding	Obs.
	(1)	(2)	(3)	(4)	(5)
Herding	2.468 (0.430)***	2272			
Jurisd. hierarchy beyond local, 1-5	0.516 (0.115)***	2130	0.470 (0.113)***	2.295 (0.452)***	2130
Political centralization, 0/1	1.005 (0.199)***	2130	0.854 (0.195)***	2.152 (0.465)***	2130
Jurisd. hierarchy local, 1-3	0.068 (0.140)	2136	0.120 (0.132)	2.630 (0.448)***	2136
Headman elected, 0/1	0.878 (0.261)***	1687	0.759 (0.212)***	2.358 (0.519)***	1687
Property rights in land, 0/1	1.007 (0.294)***	1807	1.343 (0.320)***	2.964 (0.507)***	1807
Single inheritor for land, 0/1	0.057 (0.203)	1753	0.072 (0.196)	2.541 (0.523)***	1753
Institutional characteristics (PCA)	-0.139 (0.064)**	2272	-0.147 (0.058)**	2.595 (0.442)***	2272
Patrilineality, 0/1	-0.137 (0.223)	2220	-0.141 (0.218)	2.538 (0.442)***	2220
Matrilineality, 0/1	-0.410 (0.315)	2220	-0.363 (0.306)	2.538 (0.437)***	2220
Patrilocality, 0/1	-0.108 (0.265)	2252	-0.155 (0.255)	2.638 (0.437)***	2252
Matrilocality, 0/1	-0.135 (0.626)	2252	0.529 (0.458)	2.657 (0.438)***	2252
Polygyny, 0/1	0.074 (0.284)	2233	0.085 (0.267)	2.540 (0.433)***	2233
Clan communities, 0/1	-0.181 (0.204)	1903	-0.151 (0.197)	2.578 (0.446)***	1903
Kinship tightness, 0-1	-0.283 (0.624)	2245	-0.093 (0.645)	2.638 (0.441)***	2245
Cousin marriage, 0/1	-0.150 (0.241)	1697	-0.044 (0.218)	2.692 (0.434)***	1697
Bride price, 0/1	0.390 (0.227)*	2271	0.296 (0.204)	2.428 (0.434)***	2271
Female particip. in agriculture, 1-6	-0.151 (0.103)	1463	-0.115 (0.103)	3.050 (0.898)***	1463
Settlement complexity, 1-8	-0.103 (0.057)*	2186	0.051 (0.062)	2.884 (0.498)***	2186
Historical latitude (abs.)	25.861 (18.608)	2272	12.780 (17.108)	2.432 (0.437)***	2272
Nomadic, 0/1	0.811 (0.277)***	2186	0.051 (0.300)	2.511 (0.475)***	2186
Semi-sedentary, 0/1	-0.406 (0.589)	2186	-0.510 (0.556)	2.583 (0.429)***	2186
Former slavery, 0/1	0.111 (0.205)	2003	0.192 (0.205)	2.628 (0.431)***	2003
High Gods Moral, 0/1	0.932 (0.292)***	1177	0.535 (0.305)*	2.469 (0.575)***	1177

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. In specification A (columns 1–2) we regress conflict on various ethnic traits from Murdock (1967). In specification B (columns 3–5) we regress conflict on each of Murdock's additional variables and ancestral dependence on herding. Both specifications include country fixed effects. Standardized beta coefficients are reported with standard errors in parentheses clustered at the ethnicity level.

Table A28: Traditional herding, ethnic fractionalization and contemporary conflict globally: Interaction with previous conflicts

	<i>Dependent variable: Number of conflict events (IHS)</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	-0.010 (0.029) [0.036]	-0.011 (0.025) [0.036]	-0.015 (0.025) [0.031]	-0.012 (0.022) [0.031]	0.017 (0.016) [0.018]	0.011 (0.015) [0.020]
Ethnic fractionalization	0.006 (0.016) [0.029]	0.001 (0.019) [0.025]	0.004 (0.016) [0.028]	-0.006 (0.018) [0.023]	-0.006 (0.007) [0.011]	0.001 (0.008) [0.011]
Herding × Previous Conflict (1y)	1.481 (0.220)*** [0.231]***	1.575 (0.234)*** [0.276]***	1.632 (0.227)*** [0.236]***	1.698 (0.245)*** [0.292]***	0.678 (0.189)*** [0.279]**	0.812 (0.189)*** [0.284]***
Fractionalization × Previous Conflict (1y)	-0.883 (0.353)** [0.239]***	-0.942 (0.348)*** [0.242]***	-0.763 (0.292)*** [0.296]**	-0.799 (0.286)*** [0.296]***	-0.743 (0.345)** [0.165]***	-0.787 (0.343)** [0.189]***
Previous conflict indicator (1y)	1.755 (0.294)*** [0.204]***	1.742 (0.293)*** [0.206]***	1.568 (0.245)*** [0.245]***	1.545 (0.246)*** [0.244]***	1.544 (0.295)*** [0.151]***	1.537 (0.296)*** [0.171]***
<i>Historical ethnicity-level covariates:</i>						
Settlement complexity		0.002 (0.003) [0.003]		0.002 (0.003) [0.003]		0.001 (0.001) [0.001]
Jurisdictional hierarchy		0.017 (0.004)*** [0.005]***		0.013 (0.003)*** [0.004]***		0.008 (0.002)*** [0.002]***
Historical latitude (abs.)		1.455 (1.208) [1.145]		0.980 (1.083) [1.056]		1.143 (0.579)** [0.620]*
<i>Contemporary language-level covariates:</i>						
Population density (ln)		0.009 (0.002)*** [0.003]***		0.008 (0.002)*** [0.003]***		0.004 (0.001)*** [0.001]***
Ethnic polarization		-0.081 (0.112) [0.115]		-0.003 (0.082) [0.104]		-0.073 (0.056) [0.054]
Nighttime lights		-0.003 (0.001)*** [0.001]**		-0.003 (0.001)*** [0.001]***		-0.001 (0.000)* [0.000]
Share of land for herding		0.001 (0.014) [0.016]		-0.000 (0.012) [0.014]		0.001 (0.007) [0.009]
Ruggedness		-0.093 (0.030)*** [0.035]***		-0.077 (0.027)*** [0.033]**		-0.042 (0.015)*** [0.015]***
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	0.092	0.092	0.072	0.072	0.039	0.040
SD of dependent var	0.50	0.51	0.45	0.45	0.31	0.31
Adj. R-squared	0.42	0.42	0.41	0.41	0.34	0.35
Number of Obs.	196,980	174,692	196,980	174,692	196,980	174,692
Number of Countries	211	211	211	211	211	211
Number of Ethnic Groups	1,104	985	1,104	985	1,104	985

Note. The unit of observation is a within-country language group from the *Ethnologue* and a year. The dependent variables are based on information from the *Uppsala Conflict Data Program* (UCDP) about conflict events around the globe for each year during the period 1989-2016. They are measured as the inverse hyperbolic sine (IHS) of the value. the Previous conflict indicator is based on the specification's dependent variable measured in the previous year. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A29: Traditional herding, ethnic polarization and contemporary conflict globally: Interaction with previous conflicts

	<i>Dependent variable: Number of conflict events (IHS)</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	0.007 (0.035) [0.036]	-0.003 (0.025) [0.035]	-0.002 (0.029) [0.032]	-0.005 (0.022) [0.030]	0.024 (0.019) [0.019]	0.013 (0.014) [0.019]
Ethnic polarization	-0.110 (0.079) [0.085]	-0.191 (0.103)* [0.102]*	-0.075 (0.064) [0.077]	-0.131 (0.082) [0.087]	-0.044 (0.043) [0.040]	-0.103 (0.057)* [0.053]*
Herding × Previous Conflict (1y)	1.478 (0.229)*** [0.226]***	1.566 (0.249)*** [0.270]***	1.617 (0.236)*** [0.231]***	1.661 (0.259)*** [0.286]***	0.738 (0.198)*** [0.294]**	0.879 (0.208)*** [0.297]***
Polarization × Previous Conflict (1y)	2.798 (1.106)** [1.387]**	2.856 (1.190)** [1.375]**	3.920 (1.121)*** [1.558]**	4.082 (1.199)*** [1.554]***	0.921 (1.079) [1.164]	1.035 (1.154) [1.196]
Previous conflict indicator (1y)	0.750 (0.130)*** [0.135]***	0.688 (0.131)*** [0.137]***	0.517 (0.132)*** [0.158]***	0.455 (0.133)*** [0.159]***	0.850 (0.124)*** [0.130]***	0.795 (0.125)*** [0.129]***
<i>Historical ethnicity-level covariates:</i>						
Settlement complexity		0.002 (0.003) [0.003]		0.002 (0.003) [0.003]		0.001 (0.001) [0.001]
Jurisdictional hierarchy		0.018 (0.004)*** [0.005]***		0.014 (0.003)*** [0.004]***		0.009 (0.002)*** [0.002]***
Historical latitude (abs.)		1.883 (1.547) [1.213]		1.317 (1.335) [1.117]		1.386 (0.784)* [0.694]**
<i>Contemporary language-level covariates:</i>						
Population density (ln)		0.010 (0.002)*** [0.003]***		0.008 (0.002)*** [0.003]***		0.004 (0.001)*** [0.001]***
Ethnic fractionalization		-0.038 (0.032) [0.033]		-0.030 (0.026) [0.030]		-0.018 (0.015) [0.015]
Nighttime lights		-0.003 (0.001)*** [0.001]**		-0.003 (0.001)*** [0.001]***		-0.001 (0.000)** [0.000]*
Share of land for herding		0.001 (0.013) [0.017]		-0.002 (0.012) [0.014]		-0.000 (0.007) [0.009]
Ruggedness		-0.093 (0.030)*** [0.034]***		-0.076 (0.027)*** [0.031]**		-0.044 (0.016)*** [0.015]***
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	0.092	0.092	0.072	0.072	0.039	0.040
SD of dependent var	0.50	0.51	0.45	0.45	0.31	0.31
Adj. R-squared	0.42	0.42	0.41	0.42	0.34	0.34
Number of Obs.	196,980	174,692	196,980	174,692	196,980	174,692
Number of Countries	211	211	211	211	211	211
Number of Ethnic Groups	1,104	985	1,104	985	1,104	985

Note. The unit of observation is a within-country language group from the *Ethnologue* and a year. The dependent variables are based on information from the *Uppsala Conflict Data Program* (UCDP) about conflict events around the globe for each year during the period 1989-2016. They are measured as the inverse hyperbolic sine (IHS) of the value. The Previous conflict indicator is based on the specification's dependent variable measured in the previous year. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A30: Traditional herding and contemporary conflict in Africa: Interaction with previous conflicts

	<i>Dependent variable: Number of conflict events (IHS)</i>							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	0.129 (0.064)** [0.054]**	0.129 (0.085) [0.069]*	0.128 (0.049)*** [0.045]***	0.120 (0.061)** [0.057]**	0.101 (0.048)** [0.039]**	0.109 (0.066) [0.048]**	0.056 (0.037) [0.032]*	0.061 (0.054) [0.039]
Herding × Previous conflict (1y)	1.388 (0.298)*** [0.377]**	1.495 (0.304)*** [0.394]**	1.427 (0.345)*** [0.423]**	1.501 (0.354)*** [0.430]**	0.968 (0.237)*** [0.360]**	1.060 (0.244)*** [0.397]**	1.096 (0.243)*** [0.360]**	1.143 (0.247)*** [0.396]**
Previous conflict indicator (1y)	1.140 (0.084)*** [0.085]**	1.063 (0.086)*** [0.080]**	0.972 (0.090)*** [0.094]**	0.910 (0.095)*** [0.084]**	1.012 (0.077)*** [0.078]**	0.951 (0.080)*** [0.088]**	0.882 (0.078)*** [0.070]**	0.838 (0.081)*** [0.087]**
<i>Historical ethnicity-level covariates:</i>								
Settlement complexity		-0.006 (0.011) [0.011]		-0.007 (0.009) [0.009]		-0.002 (0.008) [0.007]		-0.002 (0.006) [0.006]
Jurisdictional hierarchy		0.048 (0.014)*** [0.014]**		0.035 (0.011)*** [0.010]**		0.037 (0.010)*** [0.012]**		0.027 (0.008)*** [0.009]**
Historical latitude (abs.)		1.196 (3.857) [3.479]		0.490 (2.696) [2.715]		0.884 (3.402) [2.827]		1.211 (3.097) [2.568]
<i>Contemporary language-level covariates:</i>								
Population density (ln)		0.066 (0.010)*** [0.016]**		0.047 (0.008)*** [0.013]**		0.046 (0.007)*** [0.010]**		0.036 (0.005)*** [0.008]**
Ethnic fractionalization		-0.058 (0.310) [0.165]		-0.104 (0.261) [0.140]		-0.052 (0.253) [0.128]		-0.040 (0.204) [0.115]
Ethnic polarization		0.410 (0.687) [0.620]		0.164 (0.596) [0.473]		0.371 (0.536) [0.493]		0.237 (0.416) [0.440]
Nighttime lights		-0.001 (0.006) [0.007]		-0.004 (0.004) [0.005]		0.001 (0.004) [0.005]		0.000 (0.004) [0.004]
Share of land for herding		0.096 (0.055)* [0.078]		0.061 (0.042) [0.058]		0.060 (0.041) [0.056]		0.054 (0.030)* [0.040]
Ruggedness		0.087 (0.128) [0.204]		0.124 (0.107) [0.173]		0.016 (0.089) [0.134]		-0.044 (0.062) [0.083]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	0.35	0.35	0.23	0.23	0.22	0.23	0.16	0.16
SD of dependent var	0.96	0.96	0.77	0.77	0.74	0.75	0.61	0.62
Adj. R-squared	0.44	0.44	0.40	0.41	0.41	0.42	0.40	0.41
Number of Obs.	45,700	42,660	45,700	42,660	45,700	42,660	45,700	42,660
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	497	449	497	449	497	449	497	449

Note. The unit of observation is a within-country language group from the *Ethnologue* and a year. The dependent variables are based on information from the *Armed Conflict Location and Event Data Project* (ACLED) about the number of conflict events in Africa for each year during the period 1997-2016. They are measured as the inverse hyperbolic sine (IHS) of the value. the Previous conflict indicator is based on the specification's dependent variable measured in the previous year. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A31: Traditional herding, ethnic fractionalization and contemporary conflict in Africa: Interaction with previous conflicts

	Dependent variable: Number of conflict events (IHS)							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	0.168 (0.064)*** [0.059]***	0.186 (0.086)** [0.067]***	0.151 (0.048)*** [0.049]***	0.158 (0.061)** [0.057]***	0.126 (0.049)** [0.041]***	0.146 (0.067)** [0.047]***	0.070 (0.037)* [0.033]**	0.087 (0.053)* [0.037]**
Ethnic fractionalization	0.174 (0.133) [0.148]	0.466 (0.223)** [0.179]**	0.112 (0.105) [0.133]	0.304 (0.187) [0.137]**	0.067 (0.106) [0.107]	0.293 (0.173)* [0.148]*	0.059 (0.088) [0.071]	0.242 (0.146)* [0.145]
Herding × Previous Conflict (1y)	0.972 (0.273)*** [0.319]***	1.062 (0.288)*** [0.342]***	1.011 (0.339)*** [0.397]**	1.081 (0.357)*** [0.414]**	0.515 (0.200)** [0.283]*	0.578 (0.223)*** [0.321]*	0.601 (0.196)*** [0.255]**	0.621 (0.208)*** [0.288]**
Fractionalization × Previous Conflict (1y)	-1.774 (0.410)*** [0.399]***	-1.803 (0.414)*** [0.418]***	-1.564 (0.398)*** [0.411]***	-1.567 (0.398)*** [0.416]***	-1.841 (0.421)*** [0.420]***	-1.869 (0.448)*** [0.464]***	-1.761 (0.347)*** [0.387]***	-1.818 (0.374)*** [0.436]***
Previous conflict indicator (1y)	2.760 (0.379)*** [0.358]***	2.714 (0.385)*** [0.385]***	2.399 (0.370)*** [0.374]***	2.343 (0.373)*** [0.388]***	2.703 (0.390)*** [0.393]***	2.676 (0.420)*** [0.436]***	2.509 (0.321)*** [0.360]***	2.526 (0.347)*** [0.407]***
<i>Historical ethnicity-level covariates:</i>								
Settlement complexity		-0.005 (0.010) [0.010]		-0.006 (0.008) [0.008]		-0.001 (0.007) [0.007]		-0.001 (0.006) [0.006]
Jurisdictional hierarchy		0.046 (0.014)*** [0.013]***		0.033 (0.011)*** [0.009]***		0.034 (0.010)*** [0.011]***		0.023 (0.008)*** [0.008]***
Historical latitude (abs.)		1.115 (3.582) [3.545]		0.569 (2.527) [2.781]		0.904 (3.098) [2.919]		1.125 (2.813) [2.591]
<i>Contemporary language-level covariates:</i>								
Population density (ln)		0.065 (0.010)*** [0.015]***		0.045 (0.008)*** [0.013]***		0.044 (0.007)*** [0.010]***		0.033 (0.005)*** [0.007]***
Ethnic polarization		0.944 (0.551)* [0.659]		0.626 (0.480) [0.496]		0.761 (0.407)* [0.503]		0.625 (0.316)** [0.461]
Nighttime lights		-0.000 (0.005) [0.006]		-0.003 (0.004) [0.004]		0.002 (0.004) [0.004]		0.001 (0.003) [0.004]
Share of land for herding		0.111 (0.054)** [0.074]		0.073 (0.041)* [0.056]		0.068 (0.040)* [0.052]		0.063 (0.029)** [0.037]*
Ruggedness		0.105 (0.123) [0.195]		0.137 (0.102) [0.165]		0.030 (0.086) [0.127]		-0.025 (0.058) [0.077]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	0.35	0.35	0.23	0.23	0.22	0.23	0.16	0.16
SD of dependent var	0.96	0.96	0.77	0.77	0.74	0.75	0.61	0.62
Adj. R-squared	0.44	0.45	0.41	0.42	0.42	0.43	0.41	0.42
Number of Obs.	45,700	42,660	45,700	42,660	45,700	42,660	45,700	42,660
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	497	449	497	449	497	449	497	449

Note. The unit of observation is a within-country language group from the *Ethnologue* and a year. The dependent variables are based on information from the *Armed Conflict Location and Event Data Project* (ACLED) about the number of conflict events in Africa for each year during the period 1997-2016. They are measured as the inverse hyperbolic sine (IHS) of the value. The Previous conflict indicator is based on the specification's dependent variable measured in the previous year. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A32: Traditional herding, ethnic polarization and contemporary conflict in Africa: Interaction with previous conflicts

<i>Dependent variable: Number of conflict events (IHS)</i>								
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	0.163 (0.065)** [0.060]***	0.163 (0.085)* [0.070]**	0.150 (0.049)*** [0.049]***	0.142 (0.061)** [0.058]**	0.127 (0.050)** [0.041]***	0.133 (0.066)** [0.048]***	0.072 (0.037)* [0.032]**	0.076 (0.052) [0.037]**
Ethnic polarization	-0.072 (0.268) [0.449]	-0.448 (0.643) [0.523]	-0.065 (0.222) [0.356]	-0.437 (0.550) [0.414]	0.088 (0.203) [0.331]	-0.226 (0.508) [0.391]	0.052 (0.156) [0.254]	-0.230 (0.409) [0.334]
Herding × Previous Conflict (1y)	1.149 (0.291)*** [0.346]***	1.217 (0.298)*** [0.367]***	1.200 (0.334)*** [0.408]***	1.241 (0.341)*** [0.424]***	0.715 (0.257)*** [0.330]**	0.753 (0.273)*** [0.361]**	0.823 (0.261)*** [0.312]**	0.821 (0.265)*** [0.338]**
Polarization × Previous Conflict (1y)	4.399 (1.139)*** [1.130]***	4.352 (1.181)*** [1.216]***	3.810 (1.251)*** [1.247]***	3.787 (1.296)*** [1.331]***	4.558 (1.254)*** [1.455]***	4.697 (1.343)*** [1.561]***	4.528 (1.185)*** [1.462]***	4.751 (1.242)*** [1.532]***
Previous conflict indicator (1y)	0.809 (0.115)*** [0.137]***	0.750 (0.118)*** [0.127]***	0.679 (0.130)*** [0.147]***	0.631 (0.136)*** [0.131]***	0.672 (0.112)*** [0.135]***	0.617 (0.115)*** [0.142]***	0.548 (0.112)*** [0.133]***	0.502 (0.115)*** [0.142]***
<i>Historical ethnicity-level covariates:</i>								
Settlement complexity		-0.008 (0.011) [0.010]		-0.009 (0.009) [0.008]		-0.004 (0.008) [0.007]		-0.004 (0.006) [0.006]
Jurisdictional hierarchy		0.049 (0.014)*** [0.014]***		0.036 (0.011)*** [0.010]***		0.036 (0.010)*** [0.011]***		0.026 (0.008)*** [0.008]***
Historical latitude (abs.)		0.494 (3.650) [3.508]		0.025 (2.566) [2.764]		0.260 (3.163) [2.781]		0.520 (2.852) [2.373]
<i>Contemporary language-level covariates:</i>								
Population density (ln)		0.065 (0.010)*** [0.015]***		0.046 (0.008)*** [0.013]***		0.045 (0.007)*** [0.010]***		0.034 (0.005)*** [0.007]***
Ethnic fractionalization		-0.126 (0.315) [0.158]		-0.154 (0.267) [0.132]		-0.116 (0.261) [0.122]		-0.111 (0.213) [0.105]
Nighttime lights		-0.001 (0.005) [0.006]		-0.004 (0.004) [0.005]		0.002 (0.004) [0.005]		0.001 (0.003) [0.004]
Share of land for herding		0.096 (0.053)* [0.074]		0.062 (0.041) [0.056]		0.058 (0.039) [0.053]		0.055 (0.029)* [0.038]
Ruggedness		0.071 (0.126) [0.203]		0.111 (0.105) [0.171]		-0.002 (0.087) [0.135]		-0.058 (0.060) [0.084]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	0.35	0.35	0.23	0.23	0.22	0.23	0.16	0.16
SD of dependent var	0.96	0.96	0.77	0.77	0.74	0.75	0.61	0.62
Adj. R-squared	0.44	0.45	0.41	0.41	0.42	0.43	0.40	0.41
Number of Obs.	45,700	42,660	45,700	42,660	45,700	42,660	45,700	42,660
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	497	449	497	449	497	449	497	449

Note. The unit of observation is a within-country language group from the *Ethnologue* and a year. The dependent variables are based on information from the *Armed Conflict Location and Event Data Project* (ACLED) about the number of conflict events in Africa for each year during the period 1997-2016. They are measured as the inverse hyperbolic sine (IHS) of the value. The Previous conflict indicator is based on the specification's dependent variable measured in the previous year. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A33: Traditional herding and contemporary conflict globally: Number of months

	Dependent variable: Number of months experiencing a conflict event								
	All conflicts			Civil conflicts			Non-civil conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Dependence on herding	0.803 (0.250)*** [0.349]**	0.755 (0.229)*** [0.288]**	2.044 (0.481)*** [0.550]**	0.672 (0.219)*** [0.296]**	0.627 (0.202)*** [0.243]**	2.048 (0.547)*** [0.633]**	0.629 (0.201)*** [0.271]**	0.676 (0.183)*** [0.226]**	2.274 (0.655)*** [0.679]**
<i>Historical ethnicity-level covariates:</i>									
Settlement complexity		-0.008 (0.015) [0.018]	0.021 (0.047) [0.048]		-0.006 (0.014) [0.016]	0.032 (0.053) [0.055]		-0.007 (0.011) [0.014]	0.056 (0.062) [0.057]
Jurisdictional hierarchy		0.044 (0.021)** [0.034]	0.262 (0.058)*** [0.072]**		0.045 (0.019)** [0.025]*	0.271 (0.065)*** [0.075]**		0.044 (0.017)** [0.024]*	0.288 (0.068)*** [0.073]**
Historical latitude (abs.)		6.390 (5.695) [6.458]	24.520 (15.269) [18.683]		5.015 (5.040) [5.562]	20.889 (18.927) [22.659]		5.188 (4.262) [4.811]	47.647 (14.604)*** [17.095]**
<i>Contemporary language-level covariates:</i>									
Population density (ln)		0.104 (0.013)*** [0.029]**	0.414 (0.044)*** [0.053]**		0.091 (0.012)*** [0.028]**	0.426 (0.048)*** [0.059]**		0.060 (0.009)*** [0.018]**	0.463 (0.058)*** [0.055]**
Ethnic fractionalization		-0.172 (0.147) [0.222]	-1.079 (0.409)*** [0.477]**		-0.249 (0.136)* [0.200]	-1.128 (0.452)** [0.508]**		0.028 (0.111) [0.180]	-1.529 (0.509)*** [0.635]**
Ethnic polarization		0.322 (0.559) [0.745]	-0.630 (1.860) [2.249]		0.575 (0.500) [0.662]	0.633 (1.953) [2.470]		-0.568 (0.414) [0.599]	-2.345 (2.097) [2.516]
Nighttime lights		-0.015 (0.006)*** [0.009]*	-0.073 (0.017)*** [0.019]**		-0.016 (0.005)*** [0.008]**	-0.087 (0.018)*** [0.020]**		-0.007 (0.004) [0.006]	-0.066 (0.018)*** [0.019]**
Share of land for herding		0.145 (0.105) [0.135]	0.737 (0.307)** [0.318]**		0.083 (0.094) [0.123]	0.629 (0.336)* [0.336]*		0.067 (0.080) [0.091]	1.141 (0.339)*** [0.383]**
Ruggedness		-0.514 (0.170)*** [0.230]**	-1.306 (0.376)*** [0.393]**		-0.469 (0.156)*** [0.216]**	-1.172 (0.405)*** [0.399]**		-0.495 (0.132)*** [0.193]**	-1.866 (0.436)*** [0.444]**
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.10	0.095		0.093	0.087		0.11	0.11	
Oster δ		4.42			4.41			5.03	
Mean of dependent var	0.54	0.55	4.60	0.44	0.44	3.79	0.31	0.31	1.94
SD of dependent var	1.23	1.23	20.8	1.12	1.12	19.0	0.90	0.91	10.6
Adj. R-squared	0.29	0.33		0.29	0.33		0.23	0.25	
Pseudo R-squared			0.50			0.51			0.47
Number of Obs.	7,036	6,239	5,558	7,036	6,239	5,327	7,036	6,239	5,294
Number of Countries	211	211	106	211	211	100	211	211	84
Number of Ethnic Groups	1,104	985	934	1,104	985	879	1,104	985	906

Note. The unit of observation is a within-country language group from the *Ethnologue*. The dependent variables are the number of months experiencing a conflict reported in the *Uppsala Conflict Data Program* (UCDP) about conflict events globally for the period 1989-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A34: Traditional herding and contemporary conflict in Africa: Number of months

	Dependent variable: Number of months experiencing a conflict event											
	All conflicts			Civil conflicts			Non-civil conflicts			Localized conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Dependence on herding	1.332 (0.419)*** [0.342]***	1.528 (0.487)*** [0.409]***	1.953 (0.442)*** [0.387]***	1.361 (0.375)*** [0.315]***	1.406 (0.429)*** [0.387]***	2.188 (0.527)*** [0.455]***	1.145 (0.376)*** [0.309]***	1.328 (0.427)*** [0.327]***	1.922 (0.445)*** [0.408]***	0.929 (0.330)*** [0.303]***	1.060 (0.377)*** [0.298]***	1.925 (0.509)*** [0.493]***
<i>Historical ethnicity-level covariates:</i>												
Settlement complexity		0.001 (0.040) [0.035]	-0.030 (0.050) [0.045]		-0.001 (0.036) [0.036]	-0.029 (0.058) [0.054]		-0.008 (0.036) [0.035]	-0.030 (0.056) [0.048]		-0.026 (0.032) [0.032]	-0.030 (0.061) [0.057]
Jurisdictional hierarchy		0.110 (0.049)** [0.043]**	0.301 (0.070)*** [0.072]***		0.123 (0.044)*** [0.045]***	0.351 (0.082)*** [0.071]***		0.113 (0.045)** [0.041]***	0.358 (0.081)*** [0.087]***		0.095 (0.039)** [0.039]**	0.374 (0.086)*** [0.088]***
Historical latitude (abs.)		5.845 (12.326) [12.551]	-7.008 (13.941) [14.447]		10.419 (11.734) [12.789]	-7.088 (15.498) [15.359]		-0.783 (10.976) [12.668]	-5.241 (15.025) [16.186]		-3.103 (9.867) [12.670]	4.986 (16.358) [17.556]
<i>Contemporary language-level covariates:</i>												
Population density (ln)		0.324 (0.037)*** [0.060]***	0.454 (0.060)*** [0.054]***		0.239 (0.034)*** [0.052]***	0.437 (0.065)*** [0.054]***		0.291 (0.033)*** [0.053]***	0.578 (0.070)*** [0.067]***		0.264 (0.028)*** [0.043]***	0.652 (0.078)*** [0.084]***
Ethnic fractionalization		0.399 (0.815) [0.719]	-0.001 (0.645) [0.368]		0.167 (0.741) [0.742]	-0.203 (0.626) [0.393]		0.348 (0.820) [0.601]	-0.168 (0.801) [0.380]		0.231 (0.758) [0.560]	-0.440 (0.844) [0.389]
Ethnic polarization		1.443 (2.095) [3.095]	2.774 (1.991) [2.698]		1.577 (1.862) [2.574]	2.776 (2.107) [2.740]		0.826 (1.992) [2.912]	3.946 (2.299)* [2.875]		-0.164 (1.764) [2.428]	3.276 (2.381) [3.429]
Nighttime lights		0.013 (0.017) [0.032]	-0.029 (0.014)** [0.017]*		0.004 (0.015) [0.026]	-0.038 (0.015)*** [0.015]**		0.012 (0.016) [0.030]	-0.035 (0.015)** [0.019]*		0.005 (0.014) [0.025]	-0.047 (0.013)*** [0.017]***
Share of land for herding		0.256 (0.212) [0.251]	0.676 (0.326)** [0.472]		0.182 (0.187) [0.231]	0.577 (0.403) [0.523]		0.318 (0.181)* [0.220]	1.012 (0.362)*** [0.550]*		0.372 (0.159)** [0.193]*	1.309 (0.380)*** [0.573]**
Ruggedness		-0.807 (0.459)* [0.538]	-0.667 (0.605) [0.945]		-0.198 (0.429) [0.488]	-0.511 (0.646) [0.962]		-0.821 (0.388)** [0.626]	-0.706 (0.750) [1.236]		-0.867 (0.325)*** [0.517]*	-1.463 (0.841)* [1.318]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.12	0.13		0.14	0.14		0.11	0.13		0.10	0.12	
Oster δ		12.0			5.71			12.1			9.33	
Mean of dependent var	1.39	1.41	11.1	1.05	1.06	7.05	1.05	1.06	7.06	0.81	0.82	4.81
SD of dependent var	1.71	1.71	28.6	1.52	1.52	21.3	1.51	1.51	21.5	1.35	1.35	16.5
Adj. R-squared	0.27	0.32		0.28	0.32		0.24	0.30		0.24	0.30	
Pseudo R-squared			0.38			0.41			0.41			0.44
Number of Obs.	2,285	2,133	2,119	2,285	2,133	2,119	2,285	2,133	2,119	2,285	2,133	2,119
Number of Countries	56	56	45	56	56	45	56	56	45	56	56	45
Number of Ethnic Groups	497	449	448	497	449	448	497	449	448	497	449	448

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of months experiencing a conflict reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A35: Traditional herding and contemporary conflict in Africa: Prolonged events

	<i>Dependent variable: Incidence of conflict events (multinomial)</i>							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cat. 1 (omitted): no incidence								
Cat. 2: no prolonged								
Dependence on herding	-0.102 (0.579) [0.535]	0.527 (0.747) [0.742]	0.197 (0.499) [0.555]	0.573 (0.649) [0.734]	0.078 (0.541) [0.525]	0.749 (0.717) [0.598]	-0.064 (0.511) [0.496]	0.313 (0.685) [0.623]
Cat. 3: prolonged								
Dependence on herding	1.932 (0.706)*** [0.696]***	2.284 (0.911)** [0.795]***	2.734 (0.663)*** [0.715]***	2.730 (0.889)*** [0.783]***	1.892 (0.690)*** [0.624]***	2.249 (0.861)*** [0.714]***	1.941 (0.728)*** [0.680]***	2.127 (0.898)** [0.727]***
Baseline controls	No	Yes	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Elasticity of herding on:								
Cat. 1	-0.10	-0.16	-0.13	-0.15	-0.089	-0.14	-0.070	-0.095
Cat. 2	-0.12	-0.058	-0.091	-0.041	-0.075	0.0022	-0.082	-0.035
Cat. 3	0.25	0.27	0.38	0.37	0.26	0.29	0.29	0.31
Share of Cat.2	0.277	0.280	0.258	0.262	0.242	0.244	0.206	0.208
Share of Cat.3	0.241	0.244	0.175	0.175	0.193	0.195	0.149	0.150
Equality of Coefficients (p-value)	0.001	0.015	0.000	0.004	0.002	0.037	0.001	0.023
Pseudo R-squared	0.14	0.19	0.16	0.20	0.13	0.18	0.13	0.19
Number of Obs.	2,285	2,133	2,285	2,133	2,285	2,133	2,285	2,133
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	497	449	497	449	497	449	497	449

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables take the value of 0 if no conflicts occurred according to the *Armed Conflict Location and Event Data Project* (ACLED) during 1997 to 2016, 1 if at least one conflict occurred but those between the same actors never spread over 2 months, and 2 if conflicts between the same actors spread over more than 2 months. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A36: Traditional herding and contemporary conflict in Africa: Multinomial logit analysis with GPT labeling

	<i>Dependent variable: Incidence of conflict events (multinomial)</i>							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cat. 1 (omitted): no incidence								
Cat. 2: no revenge-taking								
Dependence on herding	0.029 (0.595) [0.560]	0.755 (0.754) [0.723]	0.676 (0.504) [0.571]	1.035 (0.664) [0.716]	0.116 (0.539) [0.594]	0.831 (0.697) [0.616]	0.067 (0.518) [0.569]	0.596 (0.678) [0.568]
Cat. 3: revenge-taking								
Dependence on herding	2.105 (0.723)*** [0.605]***	2.187 (0.910)** [0.787]***	2.525 (0.707)*** [0.641]***	2.225 (0.898)** [0.818]***	2.408 (0.721)*** [0.558]***	2.506 (0.926)*** [0.752]***	2.091 (0.698)*** [0.505]***	1.765 (0.876)** [0.694]**
Baseline controls	No	Yes	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Elasticity of herding on:								
Cat. 1	-0.091	-0.14	-0.11	-0.13	-0.090	-0.14	-0.068	-0.083
Cat. 2	-0.086	-0.00088	0.013	0.065	-0.068	0.021	-0.056	0.029
Cat. 3	0.30	0.27	0.35	0.29	0.35	0.33	0.31	0.24
Share of Cat.2	0.331	0.338	0.317	0.321	0.287	0.292	0.225	0.229
Share of Cat.3	0.186	0.186	0.116	0.116	0.147	0.147	0.129	0.128
Equality of Coefficients (p-value)	0.000	0.032	0.001	0.093	0.000	0.022	0.001	0.139
Pseudo R-squared	0.14	0.19	0.15	0.19	0.13	0.18	0.13	0.19
Number of Obs.	2,285	2,133	2,285	2,133	2,285	2,133	2,285	2,133
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	497	449	497	449	497	449	497	449

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables take the value of 0 if no conflicts occurred according to the *Armed Conflict Location and Event Data Project* (ACLED) during 1997 to 2016, 1 if at least one conflict occurred but none of them were described as revenge-motivated, and 2 if at least one event that occurred during this time was described as revenge-motivated. Baseline controls include settlement complexity, jurisdictional hierarchy, historical latitude (abs.), population density (ln), ethnic fractionalization and polarization, nighttime lights, the share of land used for herding, and terrain ruggedness. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A37: Traditional herding and contemporary conflict in Africa: Multinomial logit analysis, logistic regression classifier with TF-IDF features

	<i>Dependent variable: Incidence of conflict events (multinomial)</i>							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cat. 1 (omitted): no incidence								
Cat. 2: no revenge-taking								
Dependence on herding	-1.162 (0.779) [0.751]	-0.451 (1.004) [0.978]	-0.148 (0.624) [0.713]	-0.189 (0.793) [0.887]	-0.350 (0.737) [0.752]	0.751 (0.926) [0.825]	-0.767 (0.727) [0.730]	0.131 (0.944) [0.756]
Cat. 3: revenge-taking								
Dependence on herding	1.252 (0.595)** [0.574]**	1.688 (0.792)** [0.780]**	1.619 (0.529)*** [0.525]**	1.915 (0.698)*** [0.616]**	1.307 (0.585)** [0.583]**	1.587 (0.755)** [0.752]**	1.407 (0.575)** [0.546]**	1.382 (0.721)* [0.659]**
Baseline controls	No	Yes	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Elasticity of herding on:								
Cat. 1	-0.089	-0.14	-0.11	-0.14	-0.083	-0.13	-0.069	-0.085
Cat. 2	-0.30	-0.23	-0.14	-0.17	-0.15	0.014	-0.21	-0.061
Cat. 3	0.14	0.17	0.19	0.22	0.16	0.17	0.19	0.18
Share of Cat.2	0.105	0.108	0.108	0.110	0.115	0.119	0.097	0.100
Share of Cat.3	0.413	0.416	0.325	0.328	0.319	0.320	0.257	0.258
Equality of Coefficients (p-value)	0.001	0.034	0.004	0.004	0.028	0.378	0.005	0.199
Pseudo R-squared	0.14	0.19	0.15	0.19	0.13	0.17	0.13	0.19
Number of Obs.	2,285	2,133	2,285	2,133	2,285	2,133	2,285	2,133
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	497	449	497	449	497	449	497	449

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables take the value of 0 if no conflicts occurred according to the *Armed Conflict Location and Event Data Project* (ACLED) during 1997 to 2016, 1 if at least one conflict occurred but none of them were described as revenge-motivated, and 2 if at least one event that occurred during this time was described as revenge-motivated. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A38: Traditional herding and contemporary conflict in Africa: Multinomial logit analysis, logistic regression classifier with embedding features

<i>Dependent variable: Incidence of conflict events (multinomial)</i>								
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cat. 1 (omitted): no incidence								
Cat. 2: no revenge-taking								
Dependence on herding	-0.777 (0.770) [0.658]	-0.145 (0.996) [0.970]	0.200 (0.675) [0.707]	0.215 (0.881) [1.000]	-0.429 (0.677) [0.616]	0.629 (0.862) [0.660]	-0.495 (0.631) [0.602]	0.176 (0.781) [0.658]
Cat. 3: revenge-taking								
Dependence on herding	1.224 (0.605)** [0.602]**	1.654 (0.791)** [0.750]**	1.462 (0.534)*** [0.576]**	1.711 (0.689)** [0.658]***	1.537 (0.609)** [0.582]***	1.760 (0.779)** [0.695]**	1.545 (0.607)** [0.560]***	1.648 (0.776)** [0.650]**
Baseline controls	No	Yes	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Elasticity of herding on:								
Cat. 1	-0.088	-0.14	-0.11	-0.13	-0.087	-0.13	-0.070	-0.094
Cat. 2	-0.23	-0.17	-0.071	-0.090	-0.17	-0.013	-0.16	-0.060
Cat. 3	0.14	0.17	0.16	0.19	0.20	0.20	0.22	0.22
Share of Cat.2	0.122	0.124	0.109	0.111	0.144	0.146	0.129	0.130
Share of Cat.3	0.396	0.400	0.324	0.327	0.290	0.293	0.226	0.228
Equality of Coefficients (p-value)	0.005	0.051	0.058	0.062	0.004	0.185	0.002	0.073
Pseudo R-squared	0.14	0.19	0.14	0.18	0.13	0.18	0.14	0.19
Number of Obs.	2,285	2,133	2,285	2,133	2,285	2,133	2,285	2,133
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	497	449	497	449	497	449	497	449

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables take the value of 0 if no conflicts occurred according to the *Armed Conflict Location and Event Data Project* (ACLED) during 1997 to 2016, 1 if at least one conflict occurred but none of them were described as revenge-motivated, and 2 if at least one event that occurred during this time was described as revenge-motivated. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A39: Herding and revenge-taking in the descriptions of ACLED events

	<i>Dependent variable: Indicator of containing revenge-related terms</i>							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Herding-related terms	0.030 (0.007)*** [0.007]***	0.024 (0.006)*** [0.006]***	0.007 (0.005) [0.005]	0.004 (0.005) [0.005]	0.038 (0.008)*** [0.009]***	0.028 (0.008)*** [0.008]***	0.046 (0.011)*** [0.012]***	0.029 (0.009)*** [0.008]***
Length of description (ln)		0.015 (0.002)*** [0.003]***		0.011 (0.001)*** [0.002]***		0.019 (0.004)*** [0.004]***		0.022 (0.005)*** [0.005]***
Country FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Source FE	No	Yes	No	Yes	No	Yes	No	Yes
Mean of dependent var	0.014	0.015	0.0093	0.010	0.020	0.021	0.026	0.026
SD of dependent var	0.12	0.12	0.096	0.10	0.14	0.14	0.16	0.16
Adj. R-squared	0.0014	0.017	0.000046	0.011	0.0024	0.036	0.0030	0.065
Number of Obs.	129,257	120,688	68,044	62,415	61,213	58,273	35,837	34,965

Note. The unit of observation is an event from the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. The dependent variables are based on the text description of the event. Coefficients are reported with standard errors clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A40: Herding and revenge-taking in the descriptions of ACLED events: Alternative text analysis methods

<i>Dependent variable: Indicator of being revenge-related</i>								
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Logistic regression classifier with TF-IDF								
Herding-related terms	0.232 (0.012)*** [0.016]***	0.169 (0.010)*** [0.010]***	0.247 (0.016)*** [0.023]***	0.179 (0.014)*** [0.016]***	0.215 (0.014)*** [0.020]***	0.155 (0.010)*** [0.014]***	0.215 (0.016)*** [0.022]***	0.162 (0.014)*** [0.019]***
Length of description (ln)		0.101 (0.005)*** [0.007]***		0.101 (0.007)*** [0.008]***		0.099 (0.008)*** [0.009]***		0.090 (0.007)*** [0.008]***
Country FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Source FE	No	Yes	No	Yes	No	Yes	No	Yes
Mean of dependent var	0.31	0.34	0.32	0.35	0.31	0.32	0.28	0.28
SD of dependent var	0.46	0.47	0.47	0.48	0.46	0.47	0.45	0.45
Adj. R-squared	0.015	0.056	0.017	0.068	0.013	0.054	0.014	0.052
Number of Obs.	129,257	120,688	68,044	62,415	61,213	58,273	35,837	34,965
Panel B: Logistic regression classifier with sentence embeddings								
Herding-related terms	0.180 (0.036)*** [0.037]***	0.136 (0.034)*** [0.036]***	0.111 (0.042)*** [0.042]**	0.070 (0.037)* [0.038]*	0.249 (0.032)*** [0.033]***	0.195 (0.032)*** [0.037]***	0.274 (0.031)*** [0.033]***	0.209 (0.039)*** [0.045]***
Length of description (ln)		-0.008 (0.008) [0.010]		-0.021 (0.007)*** [0.012]*		-0.002 (0.012) [0.014]		0.016 (0.006)*** [0.007]**
Country FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Source FE	No	Yes	No	Yes	No	Yes	No	Yes
Mean of dependent var	0.28	0.30	0.31	0.33	0.25	0.26	0.19	0.20
SD of dependent var	0.45	0.46	0.46	0.47	0.43	0.44	0.39	0.40
Adj. R-squared	0.0075	0.040	0.0024	0.035	0.017	0.070	0.028	0.062
Number of Obs.	129,257	120,688	68,044	62,415	61,213	58,273	35,837	34,965
Panel C: Classification by GPT								
Herding-related terms	0.082 (0.019)*** [0.018]***	0.069 (0.018)*** [0.016]***	0.050 (0.025)** [0.021]**	0.044 (0.025)* [0.022]*	0.080 (0.024)*** [0.026]***	0.065 (0.021)*** [0.021]***	0.084 (0.023)*** [0.025]***	0.056 (0.017)*** [0.016]***
Length of description (ln)		0.024 (0.002)*** [0.003]***		0.018 (0.002)*** [0.002]***		0.033 (0.004)*** [0.005]***		0.034 (0.006)*** [0.006]***
Country FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Source FE	No	Yes	No	Yes	No	Yes	No	Yes
Mean of dependent var	0.027	0.027	0.017	0.017	0.038	0.038	0.048	0.048
SD of dependent var	0.16	0.16	0.13	0.13	0.19	0.19	0.21	0.21
Adj. R-squared	0.0023	0.022	0.00046	0.015	0.0027	0.032	0.0029	0.048
Number of Obs.	120,688	120,688	62,415	62,415	58,273	58,273	34,965	34,965

Note. The unit of observation is an event from the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. The dependent variables are based on the text description of the event. Coefficients are reported with standard errors clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A41: Traditional herding and contemporary conflict by rule of law

	<i>Dependent variable: Number of all conflict events</i>							
	Low		Lower-middle		Upper-middle		High	
	Count (IHS)	Count	Count (IHS)	Count	Count (IHS)	Count	Count (IHS)	Count
	OLS	PPML	OLS	PPML	OLS	PPML	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	2.021 (0.640)*** [0.523]***	4.211 (1.479)*** [0.940]***	1.593 (0.500)*** [0.418]***	4.800 (1.229)*** [1.130]***	0.324 (0.318) [0.377]	-0.298 (3.497) [3.762]	-0.633 (0.435) [0.545]	-0.380 (2.128) [2.143]
<i>Historical ethnicity-level covariates:</i>								
Settlement complexity	-0.030 (0.047) [0.053]	0.339 (0.175)* [0.123]***	-0.026 (0.041) [0.031]	0.241 (0.124)* [0.124]*	0.025 (0.018) [0.020]	0.221 (0.304) [0.314]	0.023 (0.029) [0.044]	-0.134 (0.162) [0.186]
Jurisdictional hierarchy	0.063 (0.057) [0.051]	0.004 (0.325) [0.508]	0.048 (0.047) [0.041]	0.474 (0.150)*** [0.170]***	0.068 (0.037)* [0.027]**	-0.650 (0.361)* [0.460]	0.041 (0.047) [0.098]	0.467 (0.137)*** [0.151]***
Historical latitude (abs.)	-19.730 (13.748) [18.561]	-87.423 (57.505) [48.401]*	21.329 (19.714) [15.026]	68.735 (19.053)*** [13.129]***	16.552 (9.521)* [9.590]	284.422 (70.758)*** [46.789]***	6.475 (6.018) [12.925]	63.103 (95.059) [102.755]
<i>Contemporary language-level covariates:</i>								
Population density (ln)	0.279 (0.041)*** [0.066]***	0.599 (0.144)*** [0.139]***	0.209 (0.037)*** [0.048]***	0.734 (0.115)*** [0.158]***	0.017 (0.011) [0.012]	0.830 (0.430)* [0.474]*	0.133 (0.024)*** [0.055]**	0.841 (0.368)** [0.242]***
Ethnic fractionalization	0.519 (0.824) [0.316]	-5.045 (2.403)** [3.231]	-0.286 (0.309) [0.497]	-3.890 (2.554) [1.414]***	-0.519 (0.186)*** [0.218]**	-4.113 (2.550) [2.507]	0.370 (0.369) [0.320]	0.887 (1.124) [0.532]*
Ethnic polarization	2.857 (2.328) [2.959]	8.755 (4.021)** [2.384]***	-0.028 (1.891) [1.203]	3.873 (11.153) [8.343]	1.406 (0.974) [0.922]	8.921 (12.447) [12.027]	-0.707 (1.104) [1.129]	-3.216 (3.793) [3.897]
Nighttime lights	-0.013 (0.013) [0.020]	-0.163 (0.039)*** [0.043]***	-0.012 (0.022) [0.026]	-0.135 (0.046)** [0.054]**	0.010 (0.019) [0.006]	-0.104 (0.042)** [0.047]**	-0.037 (0.009)*** [0.013]***	-0.201 (0.100)** [0.082]**
Share of land for herding	0.185 (0.228) [0.276]	1.942 (0.693)*** [0.853]**	0.510 (0.242)** [0.260]*	2.416 (0.606)*** [0.771]***	-0.446 (0.225)** [0.465]	0.081 (1.370) [0.928]	0.155 (0.214) [0.200]	1.060 (1.551) [1.697]
Ruggedness	-0.568 (0.495) [0.587]	-4.633 (1.516)*** [1.110]***	0.324 (0.314) [0.284]	-1.332 (1.060) [1.080]	-0.812 (0.299)*** [0.729]	-3.763 (0.844)*** [1.063]***	-0.598 (0.280)** [0.138]***	0.569 (1.043) [0.930]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.16		0.19		0.049		-0.066	
Oster δ	13.1		1.89		1.82		7.20	
Mean of dependent var	1.07	32.8	0.69	18.7	0.21	5.70	0.66	33.3
SD of dependent var	1.75	409.2	1.45	170.2	0.87	96.6	1.56	245.7
Adj. R-squared	0.28		0.32		0.35		0.28	
Pseudo R-squared		0.68		0.77		0.72		0.45
Number of Obs.	1,565	1,551	1,112	986	1,936	1,926	1,500	1,076
Number of Countries	30	27	29	26	20	19	62	32
Number of Ethnic Groups	300	298	307	300	301	298	388	324

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A42: Traditional herding and contemporary conflict in Africa: Transhumant and non-transhumant herding

	Dependent variable: Number of conflict events											
	All conflicts			Civil conflicts			Non-civil conflicts			Localized conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Transhumant herding	1.978 (0.686)*** [0.567]***	2.351 (0.907)*** [0.695]***	2.789 (0.788)*** [0.739]***	1.931 (0.613)*** [0.525]***	2.020 (0.783)** [0.613]***	3.273 (0.899)*** [0.920]***	1.645 (0.599)*** [0.509]***	1.996 (0.781)** [0.592]***	2.219 (0.875)** [0.742]***	1.428 (0.534)*** [0.481]***	1.759 (0.691)** [0.524]***	2.552 (0.930)*** [0.765]***
Non-transhumant herding	1.587 (0.612)*** [0.655]**	1.450 (0.632)** [0.567]**	1.324 (1.019) [1.050]	1.640 (0.544)*** [0.534]***	1.407 (0.570)** [0.535]**	1.177 (1.209) [1.256]	1.229 (0.509)** [0.619]*	1.152 (0.514)** [0.433]**	1.471 (0.922) [0.939]	0.845 (0.459)* [0.567]	0.785 (0.475)* [0.426]*	2.097 (0.906)** [1.057]**
<i>Historical ethnicity-level covariates:</i>												
Settlement complexity		0.027 (0.054) [0.039]	0.058 (0.106) [0.102]		0.012 (0.048) [0.037]	0.122 (0.127) [0.144]		0.018 (0.047) [0.033]	-0.020 (0.109) [0.092]		0.003 (0.040) [0.030]	0.018 (0.111) [0.091]
Jurisdictional hierarchy		0.153 (0.062)** [0.051]***	0.639 (0.137)*** [0.141]***		0.156 (0.054)*** [0.053]***	0.616 (0.141)*** [0.136]***		0.146 (0.055)*** [0.048]***	0.671 (0.151)*** [0.167]***		0.121 (0.046)*** [0.044]***	0.649 (0.124)*** [0.136]***
Historical latitude (abs.)		9.695 (15.218) [15.127]	-21.135 (24.700) [27.196]		13.676 (14.161) [15.328]	-43.398 (33.253) [33.423]		1.364 (13.084) [14.281]	-9.593 (19.623) [21.692]		-3.567 (11.858) [13.772]	11.571 (21.625) [26.390]
<i>Contemporary language-level covariates:</i>												
Population density (ln)		0.393 (0.045)*** [0.072]**	0.657 (0.086)*** [0.062]***		0.289 (0.040)*** [0.063]***	0.518 (0.089)*** [0.077]**		0.340 (0.038)*** [0.062]**	0.832 (0.096)*** [0.078]**		0.308 (0.033)*** [0.052]**	0.884 (0.098)*** [0.086]**
Ethnic fractionalization		0.472 (1.017) [0.799]	-1.896 (1.017)* [1.171]		0.244 (0.892) [0.804]	-1.615 (1.082) [1.160]		0.351 (1.008) [0.680]	-2.455 (1.121)** [1.221]**		0.205 (0.929) [0.651]	-2.220 (1.119)** [1.006]**
Ethnic polarization		1.747 (2.561) [3.729]	8.483 (3.422)** [3.515]**		1.903 (2.251) [3.162]	7.495 (3.636)** [3.711]**		0.710 (2.398) [3.388]	10.146 (3.547)*** [3.663]***		-0.316 (2.119) [2.916]	8.900 (3.018)*** [3.826]**
Nighttime lights		0.004 (0.021) [0.039]	-0.052 (0.014)*** [0.010]***		-0.004 (0.018) [0.032]	-0.053 (0.017)*** [0.011]***		0.006 (0.019) [0.035]	-0.058 (0.014)*** [0.012]***		-0.000 (0.017) [0.030]	-0.067 (0.014)*** [0.013]***
Share of land for herding		0.349 (0.259) [0.297]	1.452 (0.817)* [0.835]*		0.238 (0.226) [0.269]	1.634 (0.890)* [0.890]*		0.415 (0.214)* [0.248]*	1.384 (0.749)* [0.828]*		0.489 (0.188)*** [0.214]**	1.742 (0.615)*** [0.729]**
Ruggedness		-0.813 (0.561) [0.655]	0.554 (1.144) [1.543]		-0.181 (0.519) [0.577]	0.835 (1.119) [1.554]		-0.809 (0.453)* [0.740]	0.243 (1.294) [1.641]		-0.953 (0.374)** [0.597]	-1.231 (1.238) [1.382]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Diff(Transhumant vs. non-transhumant) p-value	0.607	0.334	0.151	0.674	0.465	0.107	0.531	0.288	0.368	0.334	0.174	0.596
Mean of dependent var	1.63	1.65	55.9	1.22	1.24	28.9	1.20	1.21	27.0	0.92	0.93	16.2
SD of dependent var	2.07	2.08	419.3	1.83	1.84	227.4	1.77	1.78	208.7	1.57	1.58	121.6
Adj. R-squared	0.28	0.33		0.29	0.33		0.25	0.31		0.26	0.31	
Pseudo R-squared			0.62			0.61			0.64			0.67
Number of Obs.	2,199	2,133	2,119	2,199	2,133	2,119	2,199	2,133	2,119	2,199	2,133	2,119
Number of Countries	56	56	45	56	56	45	56	56	45	56	56	45
Number of Ethnic Groups	465	449	448	465	449	448	465	449	448	465	449	448

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A43: Coefficients on herding when controlling for different measures of transhumant pastoralism (THP) in Africa

	Dependent variable: Number of conflict events											
	All conflicts			Civil conflicts			Non-civil conflicts			Localized conflicts		
	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count	Count (IHS)		Count
	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML	OLS	OLS	PPML
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
No THP controls	1.706 (0.512)*** [0.420]***	1.904 (0.591)*** [0.491]***	2.085 (0.739)*** [0.760]***	1.687 (0.456)*** [0.383]***	1.716 (0.519)*** [0.463]***	2.418 (0.794)*** [0.877]***	1.378 (0.436)*** [0.374]***	1.577 (0.494)*** [0.383]***	1.813 (0.794)** [0.760]**	1.113 (0.383)*** [0.367]***	1.276 (0.435)*** [0.354]***	2.300 (0.811)*** [0.841]***
Avg. neighbor THP (cont.)	1.458 (0.565)** [0.479]***	1.626 (0.637)** [0.535]***	2.843 (0.874)*** [1.030]***	1.464 (0.485)*** [0.398]***	1.503 (0.546)*** [0.499]***	3.313 (1.009)*** [1.173]***	1.186 (0.484)** [0.443]**	1.402 (0.536)*** [0.427]***	2.529 (0.958)*** [1.028]**	0.971 (0.418)** [0.426]**	1.151 (0.468)** [0.393]***	2.917 (0.991)*** [0.905]***
Max. neighbor THP (cont.)	0.943 (0.553)* [0.500]*	1.109 (0.601)* [0.544]**	1.709 (0.892)* [0.824]**	1.039 (0.469)** [0.388]***	1.085 (0.516)** [0.476]**	2.021 (1.077)* [1.040]*	0.731 (0.465) [0.464]	0.932 (0.501)* [0.435]**	1.457 (0.865)* [0.750]*	0.568 (0.392) [0.433]	0.739 (0.437)* [0.408]*	1.554 (0.838)* [0.760]**
Any neighbor THP > 0.15	1.591 (0.474)*** [0.453]***	1.612 (0.549)*** [0.536]***	2.434 (0.854)*** [0.901]***	1.586 (0.426)*** [0.383]***	1.497 (0.489)*** [0.489]***	2.761 (1.017)*** [1.022]***	1.311 (0.406)*** [0.454]***	1.372 (0.454)*** [0.430]***	2.070 (0.879)** [0.923]**	0.997 (0.362)*** [0.435]**	1.076 (0.399)*** [0.395]***	2.265 (0.822)*** [0.937]**
Any neighbor THP > 0.35	1.033 (0.502)** [0.532]*	1.184 (0.578)** [0.567]**	1.785 (0.871)** [0.907]**	1.051 (0.441)** [0.437]**	1.092 (0.507)** [0.492]**	2.112 (1.032)** [1.094]*	0.846 (0.425)** [0.478]*	1.011 (0.485)** [0.453]**	1.588 (0.805)** [0.843]*	0.621 (0.370)* [0.424]	0.751 (0.425)* [0.422]*	1.271 (0.783) [0.968]
Any neighbor THP > 0.55	1.178 (0.534)** [0.476]**	1.385 (0.608)** [0.573]**	1.828 (0.847)** [0.848]**	1.243 (0.464)*** [0.412]***	1.284 (0.524)** [0.523]**	2.124 (0.940)** [0.920]**	0.894 (0.452)** [0.416]**	1.111 (0.508)** [0.437]**	1.581 (0.830)* [0.878]*	0.740 (0.392)* [0.389]*	0.921 (0.444)** [0.388]**	1.852 (0.807)** [0.957]*
Any neighbor THP > 0.75	1.390 (0.517)*** [0.412]***	1.557 (0.600)*** [0.492]***	1.429 (0.789)* [0.678]**	1.401 (0.462)*** [0.380]***	1.413 (0.530)*** [0.461]***	1.514 (0.885)* [0.725]**	1.125 (0.435)*** [0.358]***	1.293 (0.499)*** [0.384]***	1.350 (0.791)* [0.714]*	0.897 (0.378)** [0.333]***	1.038 (0.437)** [0.351]***	1.759 (0.808)** [0.842]**
Baseline Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. The model controls for agricultural suitability within the polygon, the level of THP in neighboring polygons, the interaction between these two factors, and other baseline controls. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level.

Table A44: Traditional herding and contemporary conflict in grow and non-grow seasons globally

	<i>Dependent variable: Number of conflict events (IHS)</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	Grow	Non-grow	Grow	Non-grow	Grow	Non-grow
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	0.835 (0.229)*** [0.291]***	0.805 (0.232)*** [0.284]***	0.641 (0.199)*** [0.247]**	0.684 (0.202)*** [0.232]***	0.697 (0.173)*** [0.215]***	0.599 (0.169)*** [0.207]***
<i>Historical ethnicity-level covariates:</i>						
Settlement complexity	-0.007 (0.015) [0.018]	-0.007 (0.015) [0.018]	-0.006 (0.014) [0.016]	-0.005 (0.014) [0.016]	-0.005 (0.011) [0.012]	-0.006 (0.011) [0.013]
Jurisdictional hierarchy	0.055 (0.021)** [0.027]**	0.057 (0.022)*** [0.030]*	0.049 (0.019)*** [0.020]**	0.054 (0.019)*** [0.021]**	0.053 (0.017)*** [0.017]***	0.048 (0.017)*** [0.021]**
Historical latitude (abs.)	6.441 (6.096) [6.444]	6.785 (6.385) [6.150]	4.718 (5.325) [5.534]	5.365 (5.594) [5.352]	5.534 (4.240) [4.505]	5.227 (4.259) [4.329]
<i>Contemporary language-level covariates:</i>						
Population density (ln)	0.101 (0.014)*** [0.029]***	0.094 (0.013)*** [0.027]***	0.087 (0.013)*** [0.027]***	0.080 (0.012)*** [0.025]***	0.050 (0.009)*** [0.016]***	0.052 (0.008)*** [0.016]***
Ethnic fractionalization	-0.197 (0.147) [0.200]	-0.113 (0.147) [0.217]	-0.220 (0.134) [0.182]	-0.212 (0.133) [0.186]	-0.070 (0.102) [0.136]	0.081 (0.105) [0.159]
Ethnic polarization	0.178 (0.585) [0.693]	0.050 (0.588) [0.742]	0.433 (0.509) [0.634]	0.482 (0.507) [0.625]	-0.499 (0.406) [0.488]	-0.803 (0.416)* [0.567]
Nighttime lights	-0.017 (0.006)*** [0.008]**	-0.016 (0.005)*** [0.008]**	-0.019 (0.005)*** [0.008]**	-0.016 (0.005)*** [0.007]**	-0.006 (0.005) [0.005]	-0.008 (0.004)** [0.005]
Share of land for herding	0.126 (0.104) [0.128]	0.103 (0.102) [0.125]	0.070 (0.092) [0.113]	0.058 (0.091) [0.109]	0.063 (0.072) [0.079]	0.042 (0.072) [0.080]
Ruggedness	-0.534 (0.176)*** [0.238]**	-0.605 (0.169)*** [0.232]***	-0.486 (0.157)*** [0.218]**	-0.526 (0.154)*** [0.212]**	-0.455 (0.129)*** [0.180]**	-0.473 (0.119)*** [0.173]***
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.10	0.10	0.089	0.096	0.12	0.11
Oster δ	4.30	4.63	4.50	4.47	4.14	5.39
Mean of dependent var	0.48	0.48	0.38	0.38	0.26	0.27
SD of dependent var	1.24	1.24	1.11	1.11	0.87	0.87
Adj. R-squared	0.30	0.29	0.31	0.28	0.23	0.23
Number of Obs.	6,239	6,239	6,239	6,239	6,239	6,239
Number of Countries	211	211	211	211	211	211
Number of Ethnic Groups	985	985	985	985	985	985

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. They are measured as the inverse hyperbolic sine (IHS) of the value. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A45: Traditional herding and contemporary conflict in grow and non-grow seasons globally: PPML estimates

	<i>Dependent variable: Number of conflict events</i>					
	All conflicts		Civil conflicts		Non-civil conflicts	
	Grow	Non-grow	Grow	Non-grow	Grow	Non-grow
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	2.851 (1.142)** [1.059]***	2.918 (1.059)*** [1.022]***	2.788 (1.390)** [1.245]**	2.973 (1.259)** [1.155]**	3.031 (1.135)*** [1.195]**	2.773 (1.251)** [1.300]**
<i>Historical ethnicity-level covariates:</i>						
Settlement complexity	0.201 (0.167) [0.151]	0.206 (0.147) [0.131]	0.206 (0.186) [0.167]	0.221 (0.166) [0.144]	0.141 (0.104) [0.096]	0.121 (0.104) [0.101]
Jurisdictional hierarchy	0.203 (0.168) [0.189]	0.225 (0.154) [0.163]	0.145 (0.199) [0.231]	0.172 (0.186) [0.201]	0.378 (0.113)*** [0.099]***	0.379 (0.109)*** [0.094]***
Historical latitude (abs.)	78.866 (27.798)*** [28.827]***	76.324 (25.357)*** [26.874]***	78.606 (33.943)** [35.837]**	81.194 (29.981)*** [32.283]**	80.313 (28.063)*** [30.966]***	66.226 (29.541)** [32.743]**
<i>Contemporary language-level covariates:</i>						
Population density (ln)	0.606 (0.090)*** [0.064]***	0.594 (0.081)*** [0.065]***	0.618 (0.102)*** [0.068]***	0.605 (0.096)*** [0.069]***	0.639 (0.094)*** [0.088]***	0.620 (0.089)*** [0.077]***
Ethnic fractionalization	-1.844 (1.420) [1.607]	-2.031 (1.115)* [1.366]	-1.663 (1.612) [1.857]	-2.020 (1.269) [1.587]	-2.014 (0.902)** [0.979]**	-1.680 (0.786)** [0.960]*
Ethnic polarization	5.756 (3.785) [5.099]	4.373 (3.661) [4.773]	9.389 (4.069)** [5.709]	9.069 (3.778)** [5.152]*	-4.593 (3.216) [4.189]	-6.906 (3.382)** [4.479]
Nighttime lights	-0.158 (0.040)*** [0.036]***	-0.143 (0.030)*** [0.026]***	-0.205 (0.049)*** [0.044]***	-0.179 (0.038)*** [0.032]***	-0.087 (0.023)*** [0.018]***	-0.085 (0.023)*** [0.017]***
Share of land for herding	1.680 (0.541)*** [0.461]***	1.441 (0.492)*** [0.439]***	1.792 (0.612)*** [0.551]***	1.456 (0.546)*** [0.492]***	1.601 (0.452)*** [0.595]***	1.479 (0.497)*** [0.644]**
Ruggedness	-2.271 (0.821)*** [0.924]**	-2.160 (0.715)*** [0.791]***	-2.385 (0.853)*** [0.984]**	-2.117 (0.720)*** [0.811]***	-1.714 (0.616)*** [0.598]***	-1.951 (0.601)*** [0.600]***
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	11.3	9.98	8.98	8.12	2.76	2.70
SD of dependent var	147.3	114.3	135.6	105.1	27.0	25.8
Pseudo R-squared	0.59	0.58	0.62	0.60	0.54	0.53
Number of Obs.	5,461	5,501	5,297	5,121	5,061	4,941
Number of Countries	99	99	95	91	74	74
Number of Ethnic Groups	922	929	873	767	892	863

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Uppsala Conflict Data Program* (UCDP) for the period 1989-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A46: Traditional herding and contemporary conflict in grow and non-grow seasons in Africa

	Dependent variable: Number of conflict events (IHS)							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	Grow	Non-grow	Grow	Non-grow	Grow	Non-grow	Grow	Non-grow
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	1.721 (0.504)*** [0.415]***	1.732 (0.531)*** [0.431]***	1.556 (0.440)*** [0.375]***	1.480 (0.464)*** [0.423]***	1.397 (0.417)*** [0.340]***	1.391 (0.429)*** [0.317]***	1.112 (0.368)*** [0.315]***	1.123 (0.373)*** [0.300]***
<i>Historical ethnicity-level covariates:</i>								
Settlement complexity	-0.006 (0.044) [0.043]	-0.017 (0.045) [0.043]	-0.009 (0.039) [0.043]	-0.022 (0.040) [0.042]	-0.017 (0.037) [0.034]	-0.022 (0.038) [0.040]	-0.033 (0.032) [0.033]	-0.033 (0.032) [0.035]
Jurisdictional hierarchy	0.144 (0.053)*** [0.045]***	0.146 (0.057)** [0.053]***	0.147 (0.045)*** [0.047]***	0.133 (0.050)*** [0.045]***	0.133 (0.045)*** [0.040]***	0.142 (0.048)*** [0.049]***	0.114 (0.039)*** [0.037]***	0.119 (0.041)*** [0.041]***
Historical latitude (abs.)	12.963 (14.311) [15.102]	-2.663 (12.874) [12.859]	11.133 (12.443) [13.789]	4.021 (12.491) [13.959]	4.185 (11.807) [13.237]	-7.560 (10.621) [11.957]	-1.920 (10.227) [12.715]	-10.378 (9.853) [10.743]
<i>Contemporary language-level covariates:</i>								
Population density (ln)	0.313 (0.040)*** [0.061]***	0.341 (0.040)*** [0.063]***	0.229 (0.035)*** [0.054]***	0.245 (0.037)*** [0.056]***	0.264 (0.034)*** [0.049]***	0.285 (0.032)*** [0.054]***	0.233 (0.029)*** [0.042]***	0.248 (0.028)*** [0.043]***
Ethnic fractionalization	0.173 (0.918) [0.709]	0.472 (1.026) [0.885]	-0.004 (0.806) [0.671]	0.280 (0.889) [0.886]	0.130 (0.897) [0.586]	0.275 (0.901) [0.649]	-0.120 (0.831) [0.572]	0.301 (0.834) [0.607]
Ethnic polarization	0.903 (2.321) [3.164]	2.387 (2.461) [3.114]	1.568 (2.028) [2.733]	2.101 (2.127) [2.439]	-0.048 (2.099) [2.670]	1.544 (2.144) [2.760]	-0.789 (1.855) [2.317]	0.945 (1.906) [2.232]
Nighttime lights	0.007 (0.019) [0.034]	-0.002 (0.018) [0.033]	-0.005 (0.016) [0.026]	-0.011 (0.017) [0.028]	0.009 (0.017) [0.030]	-0.001 (0.015) [0.027]	0.005 (0.014) [0.024]	-0.006 (0.014) [0.024]
Share of land for herding	0.342 (0.227) [0.279]	0.329 (0.225) [0.257]	0.255 (0.200) [0.246]	0.253 (0.195) [0.223]	0.373 (0.182)** [0.215]*	0.324 (0.182)* [0.234]	0.360 (0.158)** [0.189]*	0.400 (0.157)** [0.189]**
Ruggedness	-0.633 (0.500) [0.602]	-0.489 (0.490) [0.619]	-0.208 (0.461) [0.530]	0.128 (0.451) [0.536]	-0.530 (0.395) [0.642]	-0.641 (0.391) [0.681]	-0.705 (0.323)** [0.519]	-0.731 (0.315)** [0.507]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.14	0.14	0.14	0.14	0.13	0.13	0.12	0.12
Oster δ	5.98	7.73	4.06	4.36	6.10	8.19	5.74	7.32
Mean of dependent var	1.27	1.29	0.93	0.94	0.90	0.92	0.68	0.69
SD of dependent var	1.82	1.85	1.58	1.60	1.53	1.55	1.34	1.36
Adj. R-squared	0.32	0.32	0.31	0.30	0.29	0.30	0.29	0.30
Number of Obs.	2,133	2,133	2,133	2,133	2,133	2,133	2,133	2,133
Number of Countries	56	56	56	56	56	56	56	56
Number of Ethnic Groups	449	449	449	449	449	449	449	449

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. They are measured as the inverse hyperbolic sine (IHS) of the value. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A47: Traditional herding and contemporary conflict in grow and non-grow seasons in Africa: PPML estimates

	<i>Dependent variable: Number of conflict events</i>							
	All conflicts		Civil conflicts		Non-civil conflicts		Localized conflicts	
	Grow	Non-grow	Grow	Non-grow	Grow	Non-grow	Grow	Non-grow
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependence on herding	2.100 (0.745)*** [0.783]***	2.072 (0.736)*** [0.744]***	2.424 (0.780)*** [0.875]***	2.415 (0.813)*** [0.887]***	1.841 (0.808)** [0.785]**	1.785 (0.787)** [0.743]**	2.323 (0.806)*** [0.838]***	2.276 (0.831)*** [0.862]***
<i>Historical ethnicity-level covariates:</i>								
Settlement complexity	-0.037 (0.082) [0.080]	-0.048 (0.080) [0.075]	-0.016 (0.091) [0.094]	-0.020 (0.092) [0.089]	-0.065 (0.086) [0.083]	-0.080 (0.080) [0.079]	-0.014 (0.081) [0.092]	-0.015 (0.076) [0.091]
Jurisdictional hierarchy	0.651 (0.137)*** [0.144]***	0.640 (0.137)*** [0.146]***	0.601 (0.139)*** [0.139]***	0.623 (0.145)*** [0.144]***	0.699 (0.156)*** [0.172]***	0.657 (0.146)*** [0.168]***	0.669 (0.124)*** [0.133]***	0.635 (0.124)*** [0.140]***
Historical latitude (abs.)	-10.550 (23.623) [27.842]	-10.856 (22.436) [28.110]	-34.099 (30.350) [34.864]	-30.504 (27.882) [33.297]	-2.761 (19.098) [21.455]	-4.604 (18.935) [22.877]	15.578 (19.932) [23.533]	16.357 (20.339) [25.069]
<i>Contemporary language-level covariates:</i>								
Population density (ln)	0.638 (0.087)*** [0.064]***	0.655 (0.083)*** [0.061]***	0.485 (0.091)*** [0.080]***	0.512 (0.087)*** [0.070]***	0.825 (0.097)*** [0.080]***	0.834 (0.096)*** [0.082]***	0.890 (0.096)*** [0.088]***	0.876 (0.099)*** [0.090]***
Ethnic fractionalization	-1.938 (1.046)* [1.179]	-1.662 (1.011)* [1.182]	-1.727 (1.163) [1.179]	-1.325 (1.072) [1.204]	-2.508 (1.133)** [1.246]**	-2.316 (1.133)** [1.225]*	-2.331 (1.060)** [0.968]**	-2.012 (1.151)* [1.018]**
Ethnic polarization	7.934 (3.478)** [3.526]**	8.702 (3.426)** [3.407]**	7.023 (3.823)* [3.998]*	8.029 (3.705)** [3.625]**	9.736 (3.631)*** [3.556]***	10.246 (3.510)*** [3.571]***	8.276 (3.064)*** [3.791]**	9.254 (3.047)*** [3.801]**
Nighttime lights	-0.052 (0.015)*** [0.012]***	-0.053 (0.015)*** [0.011]***	-0.053 (0.017)*** [0.013]***	-0.053 (0.017)*** [0.012]***	-0.057 (0.014)*** [0.013]***	-0.059 (0.014)*** [0.013]***	-0.064 (0.014)*** [0.014]***	-0.070 (0.013)*** [0.013]***
Share of land for herding	1.460 (0.855)* [0.882]*	1.293 (0.833) [0.843]	1.574 (0.927)* [0.949]*	1.421 (0.901) [0.913]	1.451 (0.775)* [0.862]*	1.255 (0.754)* [0.817]	1.851 (0.615)*** [0.760]**	1.612 (0.637)** [0.713]**
Ruggedness	0.500 (1.119) [1.617]	0.254 (1.077) [1.505]	0.730 (1.069) [1.615]	0.341 (1.021) [1.465]	0.264 (1.308) [1.712]	0.095 (1.255) [1.638]	-1.131 (1.220) [1.458]	-1.464 (1.172) [1.333]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent var	27.4	28.5	14.3	14.6	13.1	13.9	7.76	8.47
SD of dependent var	211.0	209.3	116.7	111.4	102.7	106.6	57.2	65.0
Pseudo R-squared	0.61	0.61	0.60	0.59	0.63	0.63	0.66	0.66
Number of Obs.	2,119	2,119	2,119	2,119	2,119	2,119	2,119	2,119
Number of Countries	45	45	45	45	45	45	45	45
Number of Ethnic Groups	448	448	448	448	448	448	448	448

Note. The unit of observation is a language group, which is defined as a language from *Ethnologue* spoken in a country. The dependent variables are the number of conflict events reported in the *Armed Conflict Location and Event Data Project* (ACLED) about conflict events in Africa for the period 1997-2016. Coefficients are reported with standard errors in parentheses clustered at the ethnicity level. Standard errors in square brackets are clustered at the country level. The coefficients for historical latitude (abs.) and ruggedness have been scaled up by 1000.

Table A48: The historical origins of a psychology of punishment: Additional individual-level controls

	<i>Dependent variable:</i>			
	Negative reciprocity	Negative reciprocity components, 0–10 index		
		Punish if ... treated unfairly		Willingness to
		Self	Others	take revenge
	Summary index			
	(1)	(2)	(3)	(4)
Dependence on herding	0.435 (0.216)** [0.265]	1.309 (0.627)** [0.715]*	1.230 (0.531)** [0.663]*	0.840 (0.565) [0.685]
<i>Historical ethnicity-level covariates:</i>				
Settlement complexity	0.010 (0.019) [0.019]	0.030 (0.057) [0.058]	0.006 (0.048) [0.044]	0.038 (0.051) [0.053]
Jurisdictional hierarchy	0.019 (0.024) [0.029]	0.053 (0.066) [0.079]	0.014 (0.066) [0.078]	0.069 (0.064) [0.075]
Historical latitude (abs.)	-0.003 (0.004) [0.006]	-0.007 (0.010) [0.013]	-0.016 (0.012) [0.022]	-0.004 (0.011) [0.012]
<i>Individual-level covariates:</i>				
Age	-0.004 (0.001)*** [0.002]**	-0.006 (0.004) [0.006]	-0.003 (0.004) [0.005]	-0.020 (0.004)*** [0.005]***
Age squared	-0.000 (0.000)*** [0.000]*	-0.000 (0.000)*** [0.000]***	-0.000 (0.000)*** [0.000]***	0.000 (0.000) [0.000]
Female indicator	-0.129 (0.009)*** [0.012]***	-0.346 (0.028)*** [0.038]***	-0.290 (0.026)*** [0.032]***	-0.345 (0.026)*** [0.036]***
Subj. cognitive skills	0.040 (0.003)*** [0.004]***	0.097 (0.007)*** [0.010]***	0.101 (0.008)*** [0.011]***	0.106 (0.008)*** [0.012]***
Log [Household income p/c]	0.013 (0.007)** [0.009]	0.036 (0.019)* [0.025]	0.045 (0.018)** [0.024]*	0.021 (0.018) [0.023]
Education level (1-3)	-0.006 (0.008) [0.010]	0.044 (0.023)* [0.035]	0.108 (0.024)*** [0.034]***	-0.163 (0.024)*** [0.025]***
Religion FE	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.064	0.063	0.060	0.041
Oster δ	23.9	26.4	14.2	98.7
Mean of dependent var	-0.00053	4.21	4.36	3.64
SD of dependent var	1.00	3.04	3.04	3.00
Adj. R-squared	0.11	0.078	0.088	0.11
Number of Obs.	72,538	72,601	72,596	73,300
Number of Countries	75	75	75	75
Number of Subnational Regions	951	951	951	951

Note. The unit of observation is an individual from the *Global Preferences Survey* (GPS). The dependent variable in columns 1 is the negative reciprocity index from the GPS, constructed as the weighted average of three self-assessments in GPS that measure people's propensity for different types of punishment and revenge taking. The dependent variables in columns 2-4 correspond to the individual survey questions. Coefficients are reported with standard errors in parentheses clustered at the subnational region level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A49: The historical origins of a psychology of punishment: Placebo outcomes

	<i>Dependent variable:</i>									
	Trust		Patience		Risk-taking		Pos. reciprocity		Altruism	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Dependence on herding	0.453 (0.137)*** [0.159]***	0.472 (0.148)*** [0.168]***	-0.018 (0.112) [0.108]	-0.047 (0.124) [0.095]	0.232 (0.165) [0.188]	0.213 (0.183) [0.208]	0.020 (0.166) [0.190]	-0.076 (0.185) [0.222]	0.098 (0.149) [0.145]	0.025 (0.166) [0.154]
<i>Historical ethnicity-level covariates:</i>										
Settlement complexity		0.006 (0.011) [0.013]		-0.004 (0.010) [0.010]		0.002 (0.015) [0.014]		-0.023 (0.014) [0.015]		-0.017 (0.014) [0.016]
Jurisdictional hierarchy		-0.013 (0.015) [0.016]		0.020 (0.014) [0.016]		0.026 (0.017) [0.021]		-0.020 (0.019) [0.025]		-0.009 (0.015) [0.013]
Historical latitude (abs.)		-0.008 (0.003)*** [0.005]		-0.008 (0.004)* [0.004]*		-0.009 (0.003)*** [0.004]**		0.004 (0.003) [0.006]		-0.002 (0.004) [0.005]
<i>Individual-level covariates:</i>										
Age		0.002 (0.001)** [0.002]		0.007 (0.001)*** [0.002]***		-0.002 (0.001) [0.002]		0.010 (0.001)*** [0.002]***		-0.001 (0.001) [0.001]
Age squared		0.000 (0.000) [0.000]		-0.000 (0.000)*** [0.000]***		-0.000 (0.000)*** [0.000]***		-0.000 (0.000)*** [0.000]***		-0.000 (0.000) [0.000]
Female indicator		0.027 (0.010)*** [0.015]*		-0.073 (0.009)*** [0.014]***		-0.204 (0.010)*** [0.015]***		0.022 (0.010)** [0.011]*		0.065 (0.010)*** [0.012]***
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.067	0.070	-0.0026	-0.0069	0.034	0.031	0.0029	-0.011	0.014	0.0037
Oster δ		34.6		-5.65		17.7		-2.51		1.00
Mean of dependent var	-0.00080	-0.00079	0.000068	-0.00089	0.015	0.015	-0.0045	-0.0039	0.0066	0.0067
SD of dependent var	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00
Adj. R-squared	0.088	0.092	0.14	0.16	0.095	0.15	0.12	0.12	0.12	0.12
Number of Obs.	74,333	74,095	75,158	74,922	75,141	74,899	75,594	75,344	75,346	75,098
Number of Countries	75	75	75	75	75	75	75	75	75	75
Number of Subnational Regions	951	951	951	951	951	951	951	951	951	951

Note. The unit of observation is an individual from the *Global Preferences Survey* (GPS). The dependent variables are from the GPS. Coefficients are reported with standard errors in parentheses clustered at the subnational region level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A50: The historical origins of a psychology of punishment: Controlling for trust

	<i>Dependent variable:</i>			
	Negative reciprocity Summary index	Negative reciprocity components, 0–10 index		
		Punish if ... treated unfairly		Willingness to take revenge
		Self	Others	
	(1)	(2)	(3)	(4)
Dependence on herding	0.453 (0.218)** [0.272]	1.362 (0.637)** [0.737]*	1.267 (0.541)** [0.687]*	0.876 (0.569) [0.694]
Trust	0.083 (0.009)*** [0.015]***	0.156 (0.025)*** [0.042]***	0.270 (0.025)*** [0.041]***	0.209 (0.027)*** [0.042]***
<i>Historical ethnicity-level covariates:</i>				
Settlement complexity	0.014 (0.020) [0.020]	0.040 (0.058) [0.059]	0.017 (0.050) [0.045]	0.046 (0.050) [0.053]
Jurisdictional hierarchy	0.022 (0.024) [0.030]	0.062 (0.068) [0.082]	0.022 (0.067) [0.081]	0.076 (0.063) [0.075]
Historical latitude (abs.)	-0.003 (0.004) [0.006]	-0.007 (0.011) [0.013]	-0.016 (0.012) [0.022]	-0.003 (0.011) [0.012]
<i>Individual-level covariates:</i>				
Age	-0.005 (0.001)*** [0.002]**	-0.006 (0.004) [0.006]	-0.004 (0.004) [0.005]	-0.023 (0.004)*** [0.005]***
Age squared	-0.000 (0.000)*** [0.000]**	-0.000 (0.000)*** [0.000]***	-0.000 (0.000)*** [0.000]***	0.000 (0.000) [0.000]
Female indicator	-0.161 (0.009)*** [0.012]***	-0.430 (0.028)*** [0.038]***	-0.382 (0.025)*** [0.031]***	-0.420 (0.026)*** [0.037]***
Country FE	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.067	0.066	0.062	0.043
Oster δ	164.5	163.2	28.7	-38.5
Mean of dependent var	0.0019	4.22	4.37	3.65
SD of dependent var	1.00	3.04	3.03	3.00
Adj. R-squared	0.10	0.072	0.085	0.10
Number of Obs.	72,990	73,045	73,045	73,725
Number of Countries	75	75	75	75
Number of Subnational Regions	951	951	951	951

Note. The unit of observation is an individual from the *Global Preferences Survey* (GPS). The dependent variable in column 1 is the negative reciprocity index from the GPS, constructed as the weighted average of three self-assessments in GPS that measure people's propensity for different types of punishment and revenge taking. The dependent variables in columns 2-4 correspond to the individual survey questions. Coefficients are reported with standard errors in parentheses clustered at the subnational region level. Standard errors in square brackets are clustered at the country level. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A51: Culture-of-honor related folklore in Ethnographic Atlas societies with at least 40 motifs

	<i>Dependent variable:</i>					
	Share of words related to ... that appear in group's folklore					
	Violence/Deterrence/ Punishment/Revenge		Violence/Deterrence		Punishment/Revenge	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	0.151 (0.046)*** [0.065]**	0.171 (0.048)*** [0.068]**	0.144 (0.049)*** [0.070]**	0.169 (0.050)*** [0.073]**	0.197 (0.057)*** [0.064]***	0.184 (0.082)** [0.088]**
ln(number of publications)	0.077 (0.010)*** [0.013]***	0.078 (0.011)*** [0.012]***	0.073 (0.009)*** [0.013]***	0.073 (0.010)*** [0.012]***	0.101 (0.023)*** [0.032]***	0.106 (0.026)*** [0.034]***
ln(year of first publication)	1.304 (0.501)** [0.486]***	1.381 (0.641)** [0.553]**	1.225 (0.492)** [0.542]**	1.300 (0.626)** [0.631]**	1.809 (1.080)* [0.861]**	1.902 (1.411) [1.055]*
Settlement complexity		0.003 (0.007) [0.004]		0.004 (0.006) [0.004]		-0.006 (0.012) [0.008]
Jurisdictional hierarchy		0.002 (0.005) [0.006]		0.002 (0.005) [0.005]		-0.000 (0.010) [0.012]
Historical latitude (abs.)		0.002 (0.001) [0.002]		0.002 (0.001) [0.002]		0.004 (0.003)* [0.003]
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.19	0.21	0.18	0.21	0.14	0.13
Oster δ		3.23		4.51		3.77
Mean of dependent var	0.59	0.59	0.57	0.57	0.72	0.72
SD of dependent var	0.16	0.15	0.16	0.15	0.28	0.28
Adj. R-squared	0.54	0.54	0.53	0.53	0.33	0.33
Number of Obs.	852	830	852	830	852	830
Number of Countries	131	130	131	130	131	130
Number of Linguistic Phyla	106	105	106	105	106	105

Note. The unit of observation is a society from the *Ethnographic Atlas*. The dependent variable is the share of culture-of-honor-related terms tagged in a society's folklore by Michalopoulos and Xue (2021). The dependent variable in columns (1) and (2) is the share of all terms related to violence/deterrence or punishment/revenge. In columns (3) and (4), the dependent variable is the share of terms related to violence/deterrence and in columns (5) and (6), it is the share of terms related to punishment/revenge. Standard errors in parentheses are clustered at the country level. Standard errors in square brackets are clustered at the language phylum level (i.e., largest language family) as defined in the *Ethnographic Atlas*. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

Table A52: Culture-of-honor related folklore in Ethnographic Atlas societies weighted by log number of motifs

	<i>Dependent variable:</i>					
	Share of words related to ... that appear in group's folklore					
	Violence/Deterrence/ Punishment/Revenge		Violence/Deterrence		Punishment/Revenge	
	(1)	(2)	(3)	(4)	(5)	(6)
Dependence on herding	0.174 (0.036)*** [0.052]***	0.199 (0.040)*** [0.055]***	0.170 (0.037)*** [0.054]***	0.198 (0.041)*** [0.056]***	0.206 (0.057)*** [0.062]***	0.209 (0.071)*** [0.078]***
ln(number of publications)	0.141 (0.010)*** [0.010]***	0.141 (0.010)*** [0.010]***	0.136 (0.010)*** [0.011]***	0.136 (0.010)*** [0.010]***	0.172 (0.019)*** [0.023]***	0.172 (0.020)*** [0.024]***
ln(year of first publication)	1.620 (0.676)** [0.449]***	1.680 (0.805)** [0.501]***	1.662 (0.697)** [0.491]***	1.748 (0.836)** [0.555]***	1.346 (0.991) [0.783]*	1.243 (1.175) [0.889]
Settlement complexity		0.003 (0.005) [0.004]		0.004 (0.005) [0.003]		-0.004 (0.009) [0.008]
Jurisdictional hierarchy		0.000 (0.005) [0.006]		0.001 (0.005) [0.006]		-0.003 (0.010) [0.009]
Historical latitude (abs.)		0.002 (0.001)** [0.002]		0.002 (0.001) [0.002]		0.004 (0.002)* [0.002]*
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.15	0.17	0.15	0.18	0.12	0.12
Oster δ		3.04		3.65		3.71
Mean of dependent var	0.51	0.50	0.49	0.49	0.62	0.62
SD of dependent var	0.22	0.22	0.21	0.21	0.34	0.34
Adj. R-squared	0.60	0.61	0.60	0.60	0.40	0.41
Number of Obs.	1,135	1,107	1,135	1,107	1,135	1,107
Number of Countries	149	148	149	148	149	148
Number of Linguistic Phyla	115	114	115	114	115	114

Note. The unit of observation is a society from the *Ethnographic Atlas*. The dependent variable is the share of culture-of-honor-related terms tagged in a society's folklore by Michalopoulos and Xue (2021). The dependent variable in columns (1) and (2) is the share of all terms related to violence/deterrence or punishment/venge. In columns (3) and (4), the dependent variable is the share of terms related to violence/deterrence and in columns (5) and (6), it is the share of terms related to punishment/venge. Standard errors in parentheses are clustered at the country level. Standard errors in square brackets are clustered at the language phylum level (i.e., largest language family) as defined in the *Ethnographic Atlas*. The regressions are weighted by the log of the number of motifs in the society.

Table A53: Culture-of-honor related folklore in Ethnographic Atlas societies: Alternative text analysis methods

	Dependent variable: Share of folklore motifs related to ...					
	Violence/Deterrence/ Punishment/Revenge		Violence/Deterrence		Punishment/Revenge	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Logistic regression classifier with TF-IDF						
Dependence on herding	0.140 (0.036)*** [0.038]***	0.161 (0.041)*** [0.039]***	0.136 (0.037)*** [0.037]***	0.157 (0.041)*** [0.039]***	0.041 (0.015)*** [0.017]**	0.045 (0.017)*** [0.017]***
ln(number of publications)	0.032 (0.010)*** [0.010]***	0.027 (0.010)*** [0.010]***	0.029 (0.010)*** [0.009]***	0.024 (0.009)** [0.009]**	0.006 (0.004) [0.003]*	0.005 (0.004) [0.004]
ln(year of first publication)	0.053 (0.568) [0.596]	-0.314 (0.582) [0.663]	0.065 (0.557) [0.594]	-0.318 (0.568) [0.660]	-0.187 (0.199) [0.216]	-0.305 (0.206) [0.240]
Historical controls	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.21	0.24	0.21	0.24	0.17	0.19
Oster δ		-3.83		-2.49		4.11
Mean of dependent var	0.31	0.31	0.30	0.30	0.079	0.079
SD of dependent var	0.12	0.12	0.12	0.12	0.045	0.045
Adj. R-squared	0.42	0.42	0.40	0.41	0.27	0.28
Number of Obs.	1,135	1,107	1,135	1,107	1,135	1,107
Number of Countries	149	148	149	148	149	148
Number of Linguistic Phyla	115	114	115	114	115	114
Panel B: Logistic regression classifier with sentence embeddings						
Dependence on herding	0.088 (0.037)** [0.034]**	0.103 (0.042)** [0.034]***	0.072 (0.038)* [0.029]**	0.086 (0.041)** [0.030]***	0.084 (0.022)*** [0.023]***	0.090 (0.026)*** [0.023]***
ln(number of publications)	0.024 (0.010)** [0.009]**	0.019 (0.010)* [0.009]**	0.020 (0.010)** [0.008]**	0.016 (0.010) [0.009]*	0.010 (0.007) [0.006]	0.007 (0.007) [0.006]
ln(year of first publication)	-0.103 (0.514) [0.623]	-0.481 (0.501) [0.689]	-0.095 (0.529) [0.628]	-0.478 (0.525) [0.702]	-0.066 (0.367) [0.407]	-0.231 (0.379) [0.459]
Historical controls	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.14	0.16	0.12	0.14	0.18	0.19
Oster δ		-4.22		-1.48		3.03
Mean of dependent var	0.30	0.30	0.28	0.28	0.17	0.17
SD of dependent var	0.12	0.12	0.11	0.11	0.088	0.087
Adj. R-squared	0.41	0.41	0.41	0.42	0.43	0.43
Number of Obs.	1,135	1,107	1,135	1,107	1,135	1,107
Number of Countries	149	148	149	148	149	148
Number of Linguistic Phyla	115	114	115	114	115	114
Panel C: Classification by GPT						
Dependence on herding	0.082 (0.027)*** [0.034]**	0.089 (0.028)*** [0.034]***	0.078 (0.036)** [0.033]**	0.082 (0.038)** [0.034]**	0.051 (0.026)* [0.027]*	0.055 (0.024)** [0.027]**
ln(number of publications)	0.017 (0.008)** [0.007]**	0.014 (0.009) [0.007]**	0.019 (0.010)* [0.009]**	0.016 (0.010) [0.009]	0.003 (0.006) [0.005]	0.002 (0.006) [0.006]
ln(year of first publication)	-0.409 (0.517) [0.631]	-0.730 (0.570) [0.744]	-0.224 (0.548) [0.634]	-0.582 (0.585) [0.726]	-0.338 (0.315) [0.346]	-0.485 (0.365) [0.417]
Historical controls	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Beta coef. for Herding	0.14	0.15	0.14	0.14	0.15	0.17
Oster δ		7.16		3.42		2.72
Mean of dependent var	0.38	0.38	0.33	0.33	0.13	0.13
SD of dependent var	0.11	0.11	0.11	0.11	0.063	0.063
Adj. R-squared	0.35	0.36	0.28	0.29	0.29	0.30
Number of Obs.	1,135	1,107	1,135	1,107	1,135	1,107
Number of Countries	149	148	149	148	149	148
Number of Linguistic Phyla	115	114	115	114	115	114

Note. The unit of observation is a society from the *Ethnographic Atlas*. The dependent variables are the share of motifs in a society's folklore that are related to culture-of-honor concepts. For Panel A, the motifs are identified using a logistic regression classifier with TF-IDF features. For Panel B, the motifs are identified using a logistic regression classifier with sentence embedding features. For Panel C, the motifs are identified using gpt-4o-2024-05-13. Standard errors in parentheses are clustered at the country level. Standard errors in square brackets are clustered at the language phylum level (i.e., largest language family) as defined in the *Ethnographic Atlas*. The Oster (2019) tests are with reference to a baseline specification that only includes country fixed effects.

B. Data description for the conflict analysis

a. Contemporary conflict data

Our measures of contemporary conflict are from two sources of geocoded data: the *Uppsala Conflict Data Program* (UCDP) and the *Armed Conflict Location and Event Data Project* (ACLED). The UCDP data has global coverage and spans a longer time period. The ACLED data is limited to Africa for a shorter period, but it is more comprehensive in terms of small-scale conflicts and provides textual descriptions of the events.

Global conflict data from UCDP We use data from the *Uppsala Conflict Data Program* (UCDP) to construct measures of contemporary conflict at the *Ethnologue* language group level. We use the *UCDP Georeferenced Event Dataset (GED) Global version 17.1*, which covers the whole world (with the exception of Syria) for the period 1989–2016. This dataset is UCDP’s most disaggregated dataset, covering individual events of organized violence (phenomena of lethal violence occurring at a given time and place). These events are sufficiently fine-grained to be geo-coded down to the level of individual villages, with temporal durations disaggregated to single, individual days. The dataset provides information on the names of the two actors involved in each conflict event in variables *side_a* and *side_b*. We code an event as a civil conflict if one of the actors involves the government of a given state (identified by the keyword “*Government*”). We code an event as a non-civil conflict if neither of the two actors is the government of a given state. We exclude events in which both actors are state governments.⁴⁶

We provide the definition and examples of entries in the dataset for each type of conflict.

- **All conflict:** This includes both civil and non-civil conflicts.
- **Civil conflict:** either the variable *side_a* or *side_b* contains the word “*Government*”.
 - Side A: the Government of Somalia; Side B: Somali National Movement (SNM); Starting date: January 6, 1991; Ending date: January 6, 1991; Location: Woqooyi Galbeed region; Country: Somalia; Deaths (side A): 100; Deaths (side B): 0.
 - Side A: the Government of Azerbaijan; Side B: Republic of Nagorno-Karabakh; Starting date: January 25, 1994; Ending date: January 25, 1994; Location: Kelbajar rayon; Country: Azerbaijan; Deaths (side A): 70; Deaths (side B): 4.
 - Side A: the Government of Sudan; Side B: Darfur Joint Resistance Forces; Starting date: June 27, 2014; Ending date: June 28, 2014; Location: Kutum district; Country: Sudan; Deaths (side A): 10; Deaths (side B): 13.
 - Side A: the Government of Georgia; Side B: Republic of Abkhazia; Starting date: September 1, 1992; Ending date: September 1, 1992; Location: Sukhumi town; Country: Georgia; Deaths (side A): 15; Deaths (side B): 60.
- **Non-civil conflict:** neither the variable *side_a* nor *side_b* contains the word “*Government*”.
 - Side A: Fulani; Side B: Tiv; Starting date: December 19, 2016; Ending date: December 19, 2016; Location: Gassol lga; Country: Nigeria; Deaths (civilians): 11.
 - Side A: Afar; Side B: Issa; Starting date: November 15, 2002; Ending date: December 10, 2002; Location: Gewane; Country: Ethiopia; Deaths (civilians): 30.
 - Side A: Al-Maraziq; Side B: Al-Saida; Starting date: November 23, 2004; Ending date: December 3, 2004; Location: al-Jawf governorate; Country: Yemen; Deaths (side A): 6; Deaths (side B): 22.
 - Side A: the National Democratic Front of Boroland (NDFB); Side B: Civilians; Starting date: July 24, 1994; Ending date: July 24, 1994; Location: Kokrajhar district; Country: India; Deaths (civilians): 69.
- **Interstate conflict:** both the variables *side_a* and *side_b* contain the word “*Government*.” This category is excluded from the main analysis.

⁴⁶The dataset also encodes each individual conflict event into one of the three types (variable *type_of_violence*): (i) state-based conflict, (ii) non-state conflict, and (iii) one-sided violence.

- Side A: the Government of Iraq; Side B: the Government of Kuwait; Starting date: February 25, 1991; Ending date: February 25, 1991; Location: Al Khubar town; Country: Saudi Arabia; Deaths (side A): 27.
- Side A: the Government of South Sudan; Side B: the Government of Sudan; Starting date: April 17, 2012; Ending date: April 17, 2012; Location: Southern Darfur state; Country: Sudan; Deaths (side A): 15; Deaths (side B): 7.
- Side A: the Government of Afghanistan; Side B: the Government of the United Kingdom and the Government of the United States of America; Starting date: October 10, 2001; Ending date: October 10, 2001; Location: Karam village; Country: Afghanistan; Deaths (civilians): 160.

In order to construct language group-level measures of contemporary conflict, we first use a spatial join algorithm to match the geographic location (using the latitude and longitude) of each conflict event to shapefile polygons of the language groups in *Ethnologue*. In the second step, we aggregate all conflict events matched to each language group to calculate the total number of conflict events that took place within the boundary of a language group during the period 1989–2016. We also aggregate the total number of conflict events separately for each of the three types of conflicts.

In addition to the number of conflict events, we also construct the number of conflict deaths as an additional measure of the intensity or severity of the conflict. The number of conflict deaths is reported in the variable *best*, which gives the best (most likely) estimate of total fatalities resulting from an event. It is calculated as the sum of deaths sustained by each side of the conflict, dead civilians in the event, and deaths of persons of unknown status. We aggregate the number of deaths at the language group level for all conflict events as well as for each of the three types.

African conflict data from ACLED For groups within the African continent, we are able to use data from the *Armed Conflict Location and Event Data Project* (ACLED) as an alternative source of conflict data. The database includes information on the location, date, and other characteristics of all known conflict events in Africa for the period 1997–2016. Compared to the UCDP database, ACLED data is more comprehensive when it comes to small-scale conflicts, yet it has less spatial and temporal coverage. We use the “*Interaction*” variable to group conflicts by the following six sub-types:

- **All Conflict** if an event is either a civil conflict, a non-civil conflict, or a within-group conflict.
- **Civil Conflict** if the *Interaction* variable takes a value between 10–17, or 20–27. These are all conflict events that involve the government military or rebels (who are seeking to replace the central government) as one of the actors.
- **Non-Civil Conflict** if the *Interaction* variable takes a value between 30–37, 40–47, 50–57, or 60–67. These are all conflict events that are not civil conflicts.
- **Within-Group or Localized Conflict** if the *Interaction* variable takes a value between 40–47, 50–57, or 60–67. These are all conflict events for which both actors in the conflict are geographically local and/or ethnically local groups. They are a subset of non-civil conflicts.
- **External Conflict** if the *Interaction* variable takes a value of 18, 28, 38, 48, 58, 68, 78, 80, and 88. These are all conflict events between civilians or involving external actors (e.g., international organizations, state forces active outside of their main country of operation, private security firms and their armed employees, and hired mercenaries acting independently). This category is excluded from the main analysis.
- **Non-conflicts** if the *Interaction* variable takes a value of 70 and 77. These are events that are peaceful interactions between civilians. This category is excluded from the main analysis.

Another benefit of the ACLED data is that they provide a description of most conflict events, which allows us to perform a textual analysis to test for a relationship between herding and revenge-taking conflict. We obtain a bag-of-words that proxies whether a conflict event involves herding or revenge-taking actions. To minimize our degree of discretion with regard to the construction of the bag-of-words, we take two steps. First, we select a set of seed words that describe the key concepts. For herding concepts, we use *herding* and *herder*; for punishment and revenge concepts, we use the same set of seed words that we have used in the folklore analysis: *punish*, *retaliation*, *revenge*. Then, we follow the methodology proposed

by Michalopoulos and Xue (2021) to retrieve the top-50 list of related terms from *ConceptNet*, which gives us the following terms after dropping duplicates:⁴⁷

- [illegible]

We constructed two indicators at the conflict event level: one for the mentioning of herding concepts and the other for the mentioning of revenge concepts. From the 137,140 African events in the ACLED database, we identified 2,979 events that involved herders and 1,936 events that involved revenge-taking actions. We aggregate the frequency of revenge-taking conflict at the language group level for *all*, *civil*, *non-civil*, and *localized* conflicts.

GPT revenge variable construction

We create an alternative revenge identifier for conflicts in ACLED using the most recent (at the time of writing) GPT-4o model to identify whether a conflict event description explicitly referenced acts of revenge. Each conflict event was individually processed using a specifically crafted prompt to detect its relevance to these concepts. The prompt, which was carefully tested and refined to ensure that it would only label an event as true when the reference to the concept was explicit and direct, thereby enhancing the model’s performance, was as follows: *“The following is a description of a conflict event. Description: {description}. Please determine whether the event clearly references the concept of retaliation/revenge. Only label as True if there is a direct and explicit reference to the concept within the description. Retaliation/Revenge: (True/False). Only return 1 result bool value.”* To ensure replicability and minimize the influence of randomness in the labeling process, we set a zero temperature with a fixed random seed.

GPT herding variable construction

We also create an alternative herding identifier for conflicts in ACLED using the GPT-4o model to identify whether a conflict event description explicitly referenced herding activities. Each conflict event was individually processed using the following prompt: *“The following is a description of a conflict event. Description: {description}. Please determine whether the event clearly references the concept of herding/herders. Only label as True if there is a direct and explicit reference to the concept within the description. Herding/Herders: (True/False). Only return 1 result bool value.”* Consistent with the previous process, we set a zero temperature with a fixed random seed.

⁴⁷There are more terms than in our folklore analysis because here we do not restrict the list to concepts that also appear in the folklore catalogue.

Synthetic text approach for event classification

We follow Halterman (2025) and classify conflict events using a synthetic-text approach that combines LLM-generated training data with machine-learning classification. This approach is useful in our setting because the events of interest are rare, making it difficult to assemble a sufficiently balanced hand-labeled training set of real ACLED events. Specifically, we use GPT to generate 500 short synthetic event descriptions in the style of ACLED event descriptions, with prompts designed to produce a balanced set of positive and negative examples for revenge-taking and herding activities. We then train logistic-regression classifiers on these synthetic texts using both TF-IDF and sentence embeddings as features, and apply the trained classifiers to the full set of ACLED events. The LLM is used only to generate the synthetic training data; no real ACLED event is labeled by GPT or by a human in this exercise.⁴⁸

b. Dependence on herding

Historical dependence on herding is constructed based on the *Ethnographic Atlas* (EA) data. We first create an index based on variable “v4,” which measures an ethnic group’s traditional dependence on animal husbandry for subsistence. The original variable is a 10-point scale representing intervals from “0–5% dependence” at the lower end to “86–100% dependence” at the upper end. We take the midpoint of each range and divide by 100. We then use variable “v40” to create an indicator for whether or not the predominant type of animal was suitable for herding. This is equal to 1 for “sheep and/or goats,” “equine animals,” “camels, alpacas, or llamas,” and “bovine animals.” It is equal to 0 for the absence or near absence of large domesticated animals and for “pigs the only large animal.” Historical dependence on herding is then calculated as the product of the two variables, yielding an 11-point scale ranging from 0 to 0.92.

c. Baseline covariates

- **Settlement complexity** The ordinal variable ranging from 1 to 8 is from “v30” in the EA. Each ethnic group is categorized into one of the following categories describing their pattern of settlement: (1) nomadic or fully migratory, (2) semi-nomadic, (3) semi-sedentary, (4) compact but not permanent settlements, (5) neighborhoods of dispersed family homesteads, (6) separate hamlets forming a single community, (7) compact and relatively permanent settlements, and (8) complex settlements.
- **Jurisdictional hierarchy** We use the number of jurisdictional hierarchies beyond the local community to quantify the pre-industrial political sophistication of an ethnic group. The original measure, taken from the variable “v33” of the *Ethnographic Atlas*, takes on the values of 1 to 5, with 1 indicating no levels of hierarchy beyond the local community and 5 indicating four levels.
- **Historical latitude (abs.)** The historical latitude variable is from “v104” in the EA. We take the absolute value of the latitude to capture the distance from the equator.
- **Population density (ln)** Population density is computed for 2006 using the grid-cell level population estimates from *Landscan*.
- **Ethnic fractionalization** The measure of ethnic fractionalization is constructed in the following way: For each language group, we first generate a 500 km radius circle centered at its centroid. Letting n_i be the share of the i -th language group (in terms of population) within the circle such that $\sum_i n_i = 1$, fractionalization is defined as $F = 1 - \sum_i n_i^2$. It captures the degree to which the circle is split into distinct groups.
- **Ethnic polarization** The measure of ethnic polarization is constructed in the following way: For each language group, we first generate a 500 km radius circle centered at its centroid. Letting n_i be the share of the i -th language group (in terms of population) within the circle such that $\sum_i n_i = 1$, polarization is defined as $\sum_i n_i^2(1 - n_i)$ and it captures how far the distribution of ethnic groups is away from the bipolar (1/2, 0, ..., 0, 1/2) distribution.

⁴⁸Because we do not have a hand-labeled held-out set for validation and threshold calibration, we cannot align the classification threshold to the true prevalence of positive cases (which is unknown) in the ACLED data. As a result, this approach yields substantially higher predicted positive rates than the bag-of-words and GPT-based approaches, since the classifiers are trained on a balanced synthetic dataset.

- **Nighttime lights** Nighttime light density is the average luminosity across pixels that fall within a language group's polygon. We use the values from 2006. Source: Available at <https://ngdc.noaa.gov/eog/dmsp/downloadV4composites.html>
- **Share of land for herding** The variable is constructed based on data from the ESA CCI/C3S Land Cover map for the year 1992. The share of land for herding is the percentage of land within a language group's polygon that is covered by shrubland (class code: 120, 121, 122), herbgrass (class code: 110, 130), or sparse land (class code: 150, 151, 152, 153). Source: Harper, Lamarche, Hartley, Peylin, Ottlé, Bastrikov, San Martín, Bohnenstengel, Kirches, Boettcher, Shevchuk, Brockmann and Defourny (2023), available at <https://maps.elie.ucl.ac.be/CCI/viewer/download.php>
- **Ruggedness** We construct a measure of average land ruggedness at the country-language group level following the procedure suggested by Nunn and Puga (2012). We first compute the ruggedness index at the grid cell level, which is defined as “the square root of the sum of the squared differences in elevation between one central grid cell and the eight adjacent cells” (Riley, DeGloria and Elliot, 1999). The data for elevation (meters) are from GTOPO30, a “global digital elevation model (DEM) with a horizontal grid spacing of 30 arc seconds,” which can be accessed at: <https://1ta.cr.usgs.gov/GTOPO30>. We then take the grid-cell level ruggedness index and aggregate the average land ruggedness within each of the country-language group polygons in the *Ethnologue* shapefile.

d. *Additional controls (for Tables A9–A11 and Tables A18–A21)*

- **Distance to capital (ln)** Distance to the capital city of the country is calculated as the minimum distance in kilometers from the boundary of the language group to the capital city of the country.
- **Distance to country border (ln)** Distance to the country border is calculated as the minimum distance in meters from the boundary of the language group to the country border. The shapefile of the country borders is from the World Administrative Boundaries database.
- **Distance to coastline (ln)** Distance to the coastline is calculated as the minimum distance in meters from the boundary of the language group to the coastline. The shapefile of the coastline is from the Global Self-consistent Hierarchical High-resolution Geography (GSHHG), version 2.3.7., which can be accessed at: <https://www.ngdc.noaa.gov/mgg/shorelines/>.
- **On or cross a major river** The indicator variable takes the value of 1 if the language group's polygon intersects with a permanent major river. The shapefile of permanent major rivers is from the Global Self-consistent Hierarchical High-resolution Geography (GSHHG), version 2.3.7., which can be accessed at: <https://www.ngdc.noaa.gov/mgg/shorelines/>.
- **Tsetse suitability index** This variable is from (Alsan, 2015), who models the steady-state population of tsetse flies based on their physiology as a function of historical temperature and humidity data. The variable is constructed at the ethnic group level.
- **Temperature (mean and SD)** The mean and standard deviation of temperature across the 1989–2016 period for the globe (and the 1997–2016 period for Africa) are calculated using data from GHCN CAMS, accessed at <https://psl.noaa.gov/data/gridded/data.ghcncams.html>. The data are at 0.5×0.5 degree resolution. We first calculate the average temperature within each group's polygon and then take the mean and standard deviation across time.
- **Precipitation (mean and SD)** The mean and standard deviation of precipitation across the 1989–2016 period for the globe (and the 1997–2016 period for Africa) are calculated using data from GPCC, accessed at <https://psl.noaa.gov/data/gridded/data.gpcc.html>. The data are at 0.25×0.25 degree resolution. We first calculate the average precipitation within each group's polygon and then take the mean and standard deviation across time.
- **Nomadic indicator** The indicator variable takes the value of 1 if the EA variable *settlement pattern* (“v30”) is nomadic (1) or seminomadic (2).
- **Semi-sedentary indicator** The indicator variable takes the value of 1 if the EA variable *settlement pattern* (“v30”) is semisedentary (3) or compact but not permanent settlement (4).
- **Former slavery** The indicator variable takes the value of 1 if the EA variable *former presence of slavery* (“v71”) is present in the past but not current (2) or present currently and in the past (3).
- **Excluded from state power** The indicator variable measuring whether an ethnic group is excluded from state power is coded from the Ethnic Power Relations (EPR) Core Dataset Family 2021 (Vogt,

Bormann, Rüegger, Cederman, Hunziker and Girardin, 2015), accessed at <https://icr.ethz.ch/data/epi/>. It takes the value of 1 if the “status” variable of the ethnic group is “POWERLESS,” “DISCRIMINATED,” or “SELF_EXCLUSION” in any of the years 1989–2016.

- **Ethnic segregation** For each language group, we first generate a 500km radius circle centered at its centroid. We define regions of the circle as the 1×1 degree cells contained within the circle. Let T be the total population of the circle, t_j be the population of region j , J be the total number of regions in the circle, M be the total number of language groups in the circle, π_{jm} be the fraction of language group m in region j , and π_m be the fraction of language group m in the circle. The segregation index for the centered language group i is defined as: $S = \frac{1}{M-1} \sum_m \sum_j \frac{t_j}{T} \frac{\pi_{jm} - \pi_m}{\pi_m}$.
- **Share of cropland** The variable is constructed based on data from the ESA CCI/C3S Land Cover map for the year 1992. It measures the percentage of land within a language group’s polygon that is covered by crops (class code: 10, 11, 12, 20, 30). Source: Harper et al. (2023), available at <https://maps.elie.ucl.ac.be/CCI/viewer/download.php>
- **Share of urban areas** The variable is constructed based on data from the ESA CCI/C3S Land Cover map for the year 1992. It measures the percentage of land within a language group’s polygon that is urban (class code: 190). Source: Harper et al. (2023), available at <https://maps.elie.ucl.ac.be/CCI/viewer/download.php>
- **Share of main religious denominations** The variables are coded from the World Religion Database, accessed at <https://hub.arcgis.com/datasets/nga::religion/about>. This database provides information on the main religious denominations for each subnational region. For each language group’s geographic area, we calculate the share of the population adhering to each of the following religious denominations for the global sample: Christians, Muslims, Hindus, Buddhists, Jews, and agnostics, with the excluded group comprising all other religious denominations. For the African sample, we calculate the share of the population adhering to Christians and Muslims only because the other religious denominations have no within-country variation.
- **Religious fractionalization** The variable is coded from the World Religion Database, available at <https://hub.arcgis.com/datasets/nga::religion/about>. Letting n_i be the share of the i -th religious denomination (in terms of population) within a language group’s polygon such that $\sum_i n_i = 1$, fractionalization is defined as $F = 1 - \sum_i n_i^2$. It captures the degree to which the language group is split into distinct religious denominations.
- **Religious polarization** The variable is coded from the World Religion Database, available at <https://hub.arcgis.com/datasets/nga::religion/about>. Letting n_i be the share of the i -th religious denomination (in terms of population) within a language group’s polygon such that $\sum_i n_i = 1$, polarization is defined as $\sum_i n_i^2 (1 - n_i)$ and it captures how far the distribution of religious denominations is away from the bipolar (1/2, 0, ..., 0, 1/2) distribution.

e. *Other Ethnographic Atlas Characteristics (for Tables A15 and A26)*

- **Jurisdictional hierarchy beyond local, 1-5** The variable is from the EA variable *Jurisdictional hierarchy beyond local community* (“v33”). It takes the value of 1–5 if the jurisdictional hierarchy beyond the local community is absent, one level, two levels, three levels, or four levels.
- **Political centralization, 0/1** The indicator variable takes the value of 1 if the EA variable *Jurisdictional hierarchy beyond local community* (“v33”) is larger chiefdoms (3), states (4), or large states (5).
- **Jurisdictional hierarchy local, 1-3** The variable is from the EA variable *Jurisdictional hierarchy of local community* (“v32”). It takes the value of 1 if the jurisdictional hierarchy of the local community is two levels, 2 if it is three levels, and 3 if it is four levels.
- **Headman elected, 0/1** The indicator variable takes the value of 1 if the EA variable *political succession* (v72) is through election (6).
- **Property rights in land, 0/1** The indicator variable takes the value of 1 if the EA variable *real property inheritance rule* (v74) is not absent (2–7).
- **Single inheritor for land, 0/1** The indicator variable takes the value of 1 if the EA variable *real property (land) inheritance distribution* (v75) is *Exclusively or predominantly to the one adjudged best qualified* (2), *Ultimogeniture (to the junior individual)* (3), or *Primogeniture (to the senior individual)* (4).

- **Institutional characteristics (PCA)** We construct a principal component index of institutional characteristics based on each category of the following EA variables: Jurisdictional hierarchy of local community (*v32*), Jurisdictional hierarchy beyond local community (*v33*), political succession (*v72*), real property inheritance rule (*v74*), and real property inheritance distribution (*v75*).
- **Patrilineality, 0/1** The indicator variable takes the value of 1 if the EA variable *major descent type* (*v43*) is patrilineal (1).
- **Matrilineality, 0/1** The indicator variable takes the value of 1 if the EA variable *major descent type* (*v43*) is matrilineal (3).
- **Patrilocality, 0/1** The indicator variable takes the value of 1 if the EA variable *marital residence* (*v12*) is patrilocal (8).
- **Matrilocality, 0/1** The indicator variable takes the value of 1 if the EA variable *marital residence* (*v12*) is matrilocal (5).
- **Polygyny, 0/1** The indicator variable takes the value of 1 if the EA variable *marital composition* (*v9*) is not independent nuclear (3–7).
- **Clan communities, 0/1** The indicator variable takes the value of 1 if the EA variable *Community marriage organization* (*v15*) is *clan communities* (6).
- **Kinship tightness, 0–1** The continuous variable ranging from 0 to 1 is coded following the procedure described in Enke (2019). It is defined as the average of four indicator variables: extended family (*v8==1,2*), joint residence (*v11==1,3*), unilineal descent (*v43==6*), and clans (*v15==2, 5, 6*).
- **Cousin marriage, 0/1** The indicator variable takes the value of 1 if the EA variable *cousin marriage* (*v24*) is present (1–7).
- **Bride price, 0/1** The indicator variable takes the value of 1 if the EA variable *Mode of marriage* (*v6*) is bride price (1).
- **Female participation in agriculture, 1–6** The ordinal variable ranging from 1–6 is from the EA variable *Sex differences in agriculture* (*v54*).
- **Settlement complexity, 1–8** The ordinal variable ranging from 1 to 8 is from “*v30*” in the EA.
- **Historical latitude (abs.)** The historical latitude variable is from “*v104*” in the EA. We take the absolute value of the latitude to capture the distance from the equator.
- **Nomadic, 0/1** The indicator variable takes the value of 1 if the EA variable *settlement pattern* (“*v30*”) is nomadic (1) or seminomadic (2).
- **Semi-sedentary, 0/1** The indicator variable takes the value of 1 if the EA variable *settlement complexity* (“*v30*”) is semisedentary (3) or compact but not permanent settlement (4).
- **Former slavery, 0/1** The indicator variable takes the value of 1 if the EA variable *former presence of slavery* (“*v71*”) is present in the past but not current (2) or present currently and in the past (3).
- **High Gods Moral, 0/1** The indicator variable takes the value of 1 if the EA variable *High Gods* (*v34*) is *Supportive of human morality* (4).

f. *Transhumant pastoralism variables (for Tables 8, 9, A42, and A43)*

- **Transhumant and non-transhumant herding** The measure of transhumance is derived from the EA variable *v30*, which describes settlement patterns. It takes the value of one if settlement patterns are “*nomadic or fully migratory* (1)” or “*seminomadic* (2).” Transhumant herding is defined as the product of transhumance and historical dependence on herding, while non-transhumant herding is defined as the product of non-transhumance and historical dependence on herding.
- **Neighbor’s Transhumant Pastoralism** We measure transhumance based on the EA variable *v30*, which describes settlement patterns. It takes the value of one if settlement patterns are “(1) *nomadic or fully migratory*” or “(2) *seminomadic*.” Transhumant pastoralism (THP) is defined as the product of (i) the transhumance indicator and (ii) the historical dependence on herding variable. We create six different measures of neighbors’ transhumant pastoralism: (1) the average THP of all neighbors; (2) the maximum THP of all neighbors; (3) an indicator that takes the value of one if any neighbor’s THP exceeds 0.15; (4) an indicator that takes the value of one if any neighbor’s THP exceeds 0.35; (5) an indicator that takes the value of one if any neighbor’s THP exceeds 0.55; and (6) an indicator that takes the value of one if any neighbor’s THP exceeds 0.75.

C. Data description for GPS analysis

a. Data construction

For the analysis, we link the contemporary individual-level GPS data to the historical ethnographic data using the region of residence of the respondent in the GPS and district-level measures of the ethnographic data which are taken from the *Ancestral Characteristics Database* (ACD) (Giuliano and Nunn, 2018).⁴⁹ To construct the ACD, Giuliano and Nunn (2018) first combine the grid-cell level population estimates from *Landscan* and the shapefile of the language groups in *Ethnologue* to associate each grid cell with a specific language group in *Ethnologue*. Next, they calculate the average ancestral characteristics of populations in each subnational region using the shapefile of global administrative boundaries provided by ESRI. The ancestral traits are taken from *Ethnographic Atlas*, and Giuliano and Nunn (2018) manually matched them to the language groups in *Ethnologue*.

However, the regions reported in ACD do not overlap exactly with those in the GPS data. Because the GPS does not include shapefiles at the subnational level, we manually link regions in the GPS data to regions in the ACD by combining multiple sources of information, accounting for potential name changes, merges, and splits. The GPS data report 1,146 distinct regions. Of these, 823 regions in the ACD (72%) match exactly. For 246 GPS regions (21%), the GPS regions are smaller than the ACD regions. For these, the measures from the larger ACD region are used. For 44 of the GPS regions (3.8%), the GPS region is larger than the ACD region. For these, we use the same methodology as in Giuliano and Nunn (2018) to construct ethnographic measures at the larger GPS region level. Lastly, for 33 GPS regions (2.9%), the GPS regions did not nest the ACD regions or vice versa, so a clean match was not possible. We omit these regions from the analysis. In doing so, we obtain 951 subnational regions over which the dependence on herding is cleanly defined.

b. Definitions of variables

- **Psychology of punishment.** We use the individual-level data from the Global Preference Survey (GPS) to measure a psychology of punishment. The measure is constructed by Falk et al. (2018) as a weighted average of three questions that elicit people’s propensity for altruistic punishment and for second-party punishment, each rated on a scale of 0 to 10. The questions are: (i) how willing are you to punish someone who treats others unfairly, even if there may be costs for you? (ii) how willing are you to punish someone who treats you unfairly, even if there may be costs for you? (iii) if I am treated very unjustly, I will take revenge at the first occasion, even if there is a cost to do so. The measure is normalized to have a mean of 0 and a standard deviation of 1. For robustness, we also use the responses to these three questions separately.
- **Age.** The age of the respondent is measured in years and is from the GPS individual-level dataset. This variable is included as a control in our individual-level analysis of the psychology of punishment.
- **Female indicator.** An indicator for female respondents is included in the GPS individual-level dataset. This variable is included as a control in our individual-level analysis of the psychology of punishment.
- **Subjective cognitive skills.** We measure subjective cognitive skills using the respondent’s self-assessment of math skills, which is included in the GPS individual-level dataset. The question is: “How well do the following statements describe you as a person? — I am good at math.” The measure takes values from 0 to 10, with 0 meaning “does not describe me at all” and 10 meaning “describes me perfectly”. This variable is included as a control in our individual-level analysis of the psychology of punishment.
- **Education level.** The measure of the respondent’s education level is taken from *Gallup World Poll 2012*, which can be linked to the GPS individual-level dataset using the personal identifiers contained in both datasets. The measure takes three values: (1) completed elementary education or less (up to 8 years of basic education), (2) Secondary – 3-year tertiary education and some education beyond secondary education (9–15 years of education), and (3) completed four years of education beyond

⁴⁹The version we use is the extension that includes Easternmost Europe, Siberia and the World Ethnographic Sample. The results are similar using any other version of their data.

high school and/or received a 4-year college degree. This variable is included as a control in our individual-level analysis of the psychology of punishment.

- **Household income.** The measure of household income per capita is taken from *Gallup World Poll 2012*, which can be linked to the GPS individual-level dataset using the personal identifiers contained in both datasets. The respondents are asked to report their household income in local currency. The measure is constructed by converting local currency to international dollars (ID) using purchasing power parity (PPP) ratios. Log household income is computed as $\ln(1 + \text{household income})$. This variable is included as a control in our individual-level analysis of the psychology of punishment.

D. Data description for Folklore analysis

a. Data construction

We follow Michalopoulos and Xue (2019, 2021) in quantifying ethnic groups' cultural beliefs and practices using textual data on folklore. The anthropologist and folklorist Yuri Berezkin assembled a dataset that codes the presence of 2,564 motifs across approximately 1,100 ethnolinguistic groups. A motif reflects a combination of images, episodes, or structural elements found in two or more texts.⁵⁰ The data are designed to capture a society's traditional beliefs, customs, and culture as they are transmitted from generation to generation through word-of-mouth, often in the form of folktales and narratives.⁵¹ Based on this catalog of motifs, Michalopoulos and Xue (2019, 2021) use text analyses to construct a folklore dataset. For a large number of economic, psychological, and cultural concepts, this dataset codes whether a given concept appears in a given motif.⁵² In these text analyses, a concept is said to appear in a motif if the text mentions either the seed word itself or one of the 50 most closely related terms according to the knowledge representation project ConceptNet.⁵³ Based on this approach, the authors construct the intensity of each concept in the folklore of a given group.

Most importantly for our purposes, the data contain many concepts that are related to the culture of honor hypothesis. Michalopoulos and Xue (2019) study the association between herding and a culture of honor by examining words associated with 'anger' and 'retaliation'. Following the same basic logic, we first selected all seed words that Nisbett and Cohen (1996) used to introduce the idea of a culture of honor. These are:

1. Violence and deterrence concepts: violence, perpetrator, strength, toughness, predation, predator, aggressiveness, affront, deterrence, defend, mayhem, guard.
2. Punishment and revenge concepts: punish, retaliation, revenge.

For each seed word, we retrieve the top-50 list of related terms from ConceptNet. We then select concepts from the folklore catalogue by Michalopoulos and Xue (2019) that appear in the top-50 list of our seed words, finding the following terms:

1. Violence and deterrence concepts: power, strong, crime, tough, violence, victim, threat, conflict, strength, violent, aggressive, hunter, habitat, intensity, courage, weakness, chaos, aggression, offender, predator, insult, riot, thief, prey, offend, outrage, aggressively, grit, endurance, coyote,

⁵⁰As described in detail in Michalopoulos and Xue (2019, 2021), Berezkin constructed this dataset by consulting a large number of books and journal articles. These primary sources were written by anthropologists, adventurers, and missionaries who had visited an ethnolinguistic group. Berezkin systematized these accounts into a consistent catalog. Each motif in Berezkin's catalogue is associated with a title and a short description of an image or an episode. These can be analyzed using text analysis. The median group in Berezkin's data has 62 motifs, and there is large variation across groups in which types of motifs appear in the records.

⁵¹A potential concern is that the data are more reflective of the biases of the individual who coded the primary sources rather than of the genuine folklore of a group. To address this concern, Michalopoulos and Xue (2019, 2021) extensively validate the catalog by documenting that the presence of objectively verifiable motifs is strongly correlated with real circumstances. For example, the presence of earthquake-related motifs is significantly higher in earthquake regions. Similar associations are found for other environmentally-determined variables, such as the presence of storms and lightning, or information about different modes of economic production.

⁵²The data are available at: <https://sites.google.com/site/steliosecon/folklore-catalogue?authuser=0>

⁵³ConceptNet originated from the MIT Media Lab. To construct a ConceptNet-based list of related terms Michalopoulos and Xue (2019, 2021) retrieve the top-50 list for each seed word.

perpetrator, attacker, vitality, brutality, unrest, culprit, victimization, humiliate, robber, vigor, rapist, resilience, nonviolent, abuser, predatory, disgrace, defense, security, protect, guard, protection, defend, disorder, mess, strategic, defensive, assert, confusion, prevention, protective, discourage, defender, uphold, guardian, disturbance, protected, madness, safeguard, turmoil, disruption, deter, preventive, frenzy, chaotic, bodyguard, lineman, warden, fend, upheaval, persuasion, havoc, protector, deterrent, militarily.

2. Punishment and revenge concepts: retaliate, retaliation, discipline, penalty, punishment, punish, revenge, disciplinary, backlash, vengeance, grievance, punitive, scold.

For each of the concepts, we generate a binary indicator that equals one if the concept appears in the folklore of an ethnic group. We then average across all concepts within a given domain (violence/deterrence and psychology of punishment/ revenge) to arrive at a summary measure that captures the fraction of concepts in the domain that are present in a society's folklore. In addition to measures for both domains, we also compute an overall summary measure of a culture of honor by taking the average across all concepts.

b. GPT folklore analysis

Specifically, the prompt is: *"The following is a description of a motif from folklore. Title: {title}. Description: {description}. Please determine whether the motif references any of the following concepts: violence/deterrence, revenge/punishment. Note that not all motifs relate to these concepts, and some may relate to multiple concepts. Violence/Deterrence: (True/False). Revenge/Punishment: (True/False). Only return 2 result bool values separated by a space."* To ensure replicability and minimize the stochasticity of GPT's output, we set a zero temperature with a fixed random seed.

c. Examples of revenge-related folklores

- **a38d The Sun ruins cloak:** Because the Sun does damage to a person or animal (spoils, burns his cloak, skin, etc.), the person or animal kills him or catches him in a snare
- **b117 The dogs' certificate:** The animals (usually dogs) got a certificate which was lost because of the cat (is swallowed by the cat, burned, eaten by mice). Since then dogs and cats are enemies, usually also cats and mice
- **b22 Offended person destroys the tree:** People who have climbed up a tree do not share food with a person who remained below. He or she revenges on them destroying the tree. The events have cosmic dimensions
- **b28a Pierced to the ground:** Person who have been transforming people into animals or stones is pierced to the ground with a pole
- **c11 Father avenges his son:** Person uses his supernatural powers to destroy a large group of people who have offended or insulted his son
- **e21 Avenged victim of water creatures:** Child fishes with poison or his or her body produces fish poison. Fish or water snake kills him or her. The perished child is avenged
- **i4d Thunders instrument stolen:** Thunders instrument is stolen from him. He or his helper comes unrecognized to the thief, gets his instrument and kills the enemies
- **k77a Small objects and animals defeat the ogre:** Small objects and animals (rare: animals alone but including harmless and passive ones, e.g. on the dry land fishes) revenge on a powerful enemy making attack on him in succession (usually they hide in his or her house); the enemy is badly injured, runs away or dies
- **m39e1: The eaten up iron and the kidnapped child:** A man appropriates money or property trusted to him by another. The latter has no proves but gets his property back after he or his helper puts the theft in such a position when the best choice for him becomes to return the money (usually the first man kidnaps a child of the second one)
- **m57d Beat, cudgel!:** Person first gets magic objects that bring food or treasure and after this he receives a cudgel (a whip, etc.) that beats people, usually those who have stolen the first objects. (In Greek text the stealer of magic objects is punished otherwise)

d. Traditional machine-learning classification of folklore

To perform the traditional machine-learning classification of folklore, we manually label a random sample of 500 motifs as either related to violence/deterrence, revenge/punishment, both, or neither. Of these, 400 motifs are used as a training set, and 100 motifs are used as a test set for validation.⁵⁴ We use the training set to train a logistic regression classifier using two different text representation approaches: (i) TF-IDF (term frequency-inverse document frequency), which weights terms based on their importance within each motif relative to the entire corpus, and (ii) sentence embeddings, which represent each motif as a learned vector that captures semantic meaning and similarity to other motifs. The models perform well on the test set, yielding an area under the curve (AUC) of 0.85–0.89 for violence/deterrence and 0.88–0.91 for revenge/punishment, depending on the text representation approach. We then apply the trained models to the full catalogue of motifs and convert the predicted probabilities into binary indicators using a classification threshold tuned on the held-out set to balance precision and recall. We classify all motifs in the folklore catalogue and then construct the same summary measures of a culture of honor as in our main analysis.

⁵⁴The validation set reflects the overall distribution of motifs, whereas the training set oversamples positive cases, as is standard when positive cases are relatively infrequent.